The Pilot-Proxy Method

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Abstract

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North American broadcast television occupies 470–608 MHz, a 138-MHz interval that maps to $1.34 \leq z \leq 2.02$ for the redshifted 21 cm line and therefore covers roughly one third of CHIME's cosmological observing band. The received 8-VSB payload can lie below thermal noise in an individual detector frame yet remain correlated long enough to bias a multi-year intensity-mapping survey. The resulting problem is not simply whether television can be detected, but whether a frame-level mask can remove enough contamination without discarding so much exposure that the baryon acoustic oscillation (BAO) measurement becomes impractical.

This dissertation develops a pilot-proxy detector around the dominant continuously present, spectrally concentrated feature of an ATSC A/53 transmission: the suppressed-carrier pilot. Under the declared known-tone, locally Gaussian model, the pilot projection is a matched-filter statistic; the complete proposed runtime decision also uses local references, a robust rank baseline, and per-epoch calibration measured from telescope data. The implementation is an exact fixed-point pipeline whose projections, frozen Fourier transform, power sums, and mask decision are bit-reproducible against independent references. A restartable survey processes triggered archival baseband captures and retains the decision statistics and provenance needed to revisit thresholds without reprocessing the raw archive. Because the archive is trigger-selected rather than a uniform time sample, the dissertation states its selection limits explicitly and reserves final operating points for blocked, per-epoch validation.

A Fisher-forecast framework converts masked fraction and surviving residual into integration-time cost and parameter-bias tolerance. The archive survey completed in August 2026: all 23 allocations, channels 14–36, carry products and residual-chain evaluations, spanning 7.6 years of captures. Within the present pilot-to-shelf transfer model and per-redshift-bin Fisher estimator, channels 32, 33, and the early epoch of channel 35 are forecast-feasible for the BAO dilation measurement; six allocations with live carriers (17, 22, 24, 30, 31, 35) are excision candidates; the measured transmitter sign-offs (19, 20, 26, 27, 32) convert most of the band into clean, per-epoch-recoverable observing time at a tolerance-priced working threshold; and the remaining verdicts are controlled by coherence measurements the snapshot archive cannot supply. Recovery of the growth-rate parameter is conditional on booking the same BAO-preserving delay filter in both the residual and Fisher models. These are deliberately conditional verdicts: the closing campaign must validate the pilot proxy across each 6-MHz allocation, calibrate its transfer to visibility-domain residuals, reproduce the combined-bin forecast estimator, and pass the blocked per-epoch holdout. The principal result is therefore both an instrument and an auditable decision framework: exact where the digital contract permits exactness, measured where the archive supplies evidence, and explicitly bounded where it does not.

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List of Acronyms and Abbreviations

8-VSB	eight-level vestigial sideband
ATSC	Advanced Television Systems Committee
BAO	baryon acoustic oscillation
CANFAR	Canadian Advanced Network for Astronomical Research
CFAR	constant false-alarm rate
CHIME	Canadian Hydrogen Intensity Mapping Experiment
CUDA	Compute Unified Device Architecture
DFT	discrete Fourier transform
DP4A	four-way packed 8-bit integer dot-product instruction
DTV	digital television
ENBW	equivalent noise bandwidth
FFT	fast Fourier transform
FRB	fast radio burst
GPU	graphics processing unit
HPC	high-performance computing
NTSC	National Television System Committee
OFDM	orthogonal frequency-division multiplexing
PFB	polyphase filter bank
PNR	pilot-to-noise ratio
PSD	power spectral density
Q15/Q16	signed fixed-point formats with 15/16 fractional bits
RFI	radio-frequency interference
RRC	root-raised cosine
SNR	signal-to-noise ratio

Chapter 1

A Ruler, a Broadcast, & an Error Budget

This section is deliberately pedagogical. It assumes no radio astronomy, no cosmology, and no detection theory, and it exists so that a reader from outside every one of those fields can follow the argument of the whole dissertation before meeting any of its mathematics. Later chapters distinguish what is specified by a standard, what is proved under an analytic model, what is measured from the archive, and what remains conditional on a calibration transfer. Figure 1.1 gives that distinction a visual form at the outset.

1.1 A Ruler Older Than the Stars

Just after the Big Bang, the universe was a hot plasma of light and matter, so hot that atoms could not hold together. Light and matter were locked to each other — photons scattering off free electrons, electrons bound by electric forces to the protons — and the mixture behaved as a single fluid with enormous pressure. Wherever matter happened to be slightly overdense, gravity pulled inward and pressure pushed back out, and the contest launched sound waves: pressure waves in the plasma, physically the same phenomenon as sound in air, moving at more than half the speed of light. This much was worked out on paper decades before it was observed [22, 23].

Figure 1.2 follows what happened next to a single overdense spot. Its sound wave spread outward as a growing spherical shell, carrying matter with it. Then, about 380,000 years after the Big Bang, the universe cooled through the temperature at which electrons and protons combine into neutral hydrogen. Recombination sharply reduced the free-electron density and therefore the Thomson opacity of the plasma, so the photons decoupled and streamed away — we see them today as the cosmic microwave background — and with them went the pressure that drove the wave. The shell stopped. It froze at exactly the distance sound had managed to travel in those 380,000 years: the **sound horizon**, about 150 megaparsecs once expanded to today's scale, roughly half a billion light-years. Every overdense region in the universe launched such a wave at the same time and froze it at the same radius, so the universe was left with a faint statistical preference for pairs of overdense regions separated by exactly that distance. Galaxies later formed where matter was dense — both at the original centers and on the frozen shells — and inherited the preference. It was detected directly in 2005, in the distribution of 46,748 SDSS luminous red galaxies [26] (and contemporaneously in $\sim$221,000 2dFGRS galaxies), thirty-five years after it was predicted.

This is the baryon acoustic oscillation (BAO) scale, and it is the most trustworthy ruler cosmology owns. Its length is set by well-understood physics of the early universe — the sound speed of a

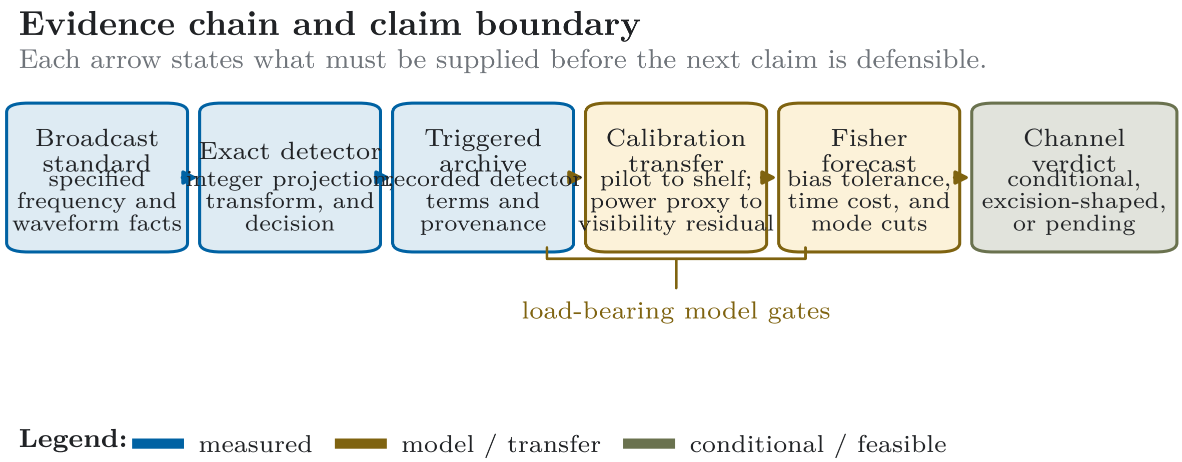

Figure 1.1: Evidence chain and claim boundary for the dissertation. The broadcast standard, exact detector, and recorded archive products are independently auditable. The transfers from a pilot statistic to the full DTV shelf, from a scalar power proxy to complex visibility residuals, and from a residual template to a cosmological verdict are model-dependent gates. Exact arithmetic secures the detector block; it does not by itself validate the scientific transfers downstream.

photon-baryon plasma and the age of the universe at recombination, both pinned tightly by the microwave background — and by nothing that happened since. A ruler of known length is a way to measure distance: if the 150-Mpc separation *appears* compressed or stretched at some epoch, that tells us exactly how much the universe expanded between that epoch and now. Measure the apparent size of the ruler at several epochs and the expansion history follows — and the expansion history is where dark energy, whatever it is, shows itself.

The redshifts CHIME reaches are not an arbitrary stretch of that history. Somewhere around seven billion years ago the expansion of the universe stopped slowing down — as gravity acting on matter says it must — and began to speed up, which is the single observation the phrase "dark energy" exists to name. A ruler measurement made only nearby sees the acceleration but not the approach to it; a ruler measurement at $z \approx 1.3$–2.0, where CHIME's television-contested third of the band sits, watches the era when matter still governed the expansion and any departure from the simplest model of dark energy would first become visible. The transverse dilation at each redshift traces out the distance-redshift relation, the radial dilation traces the expansion rate itself, and together, epoch by epoch, they are the record dark-energy models must reproduce. This is why the redshifts under the television band are worth a dissertation-length fight: they are not spare bandwidth, they are the leverage arm of the whole measurement.

1.2 The Wiggle

What does a "statistical preference for one separation" actually look like in data? Cosmologists compress a map of matter into a **correlation function**, written $\xi(r)$: the excess probability, relative to pure chance, that two regions separated by a distance r are both overdense. If matter were scattered at random, ξ would be zero at every separation. Gravity makes nearby regions strongly correlated, so ξ is large at small r and falls smoothly — except at one special separation. At

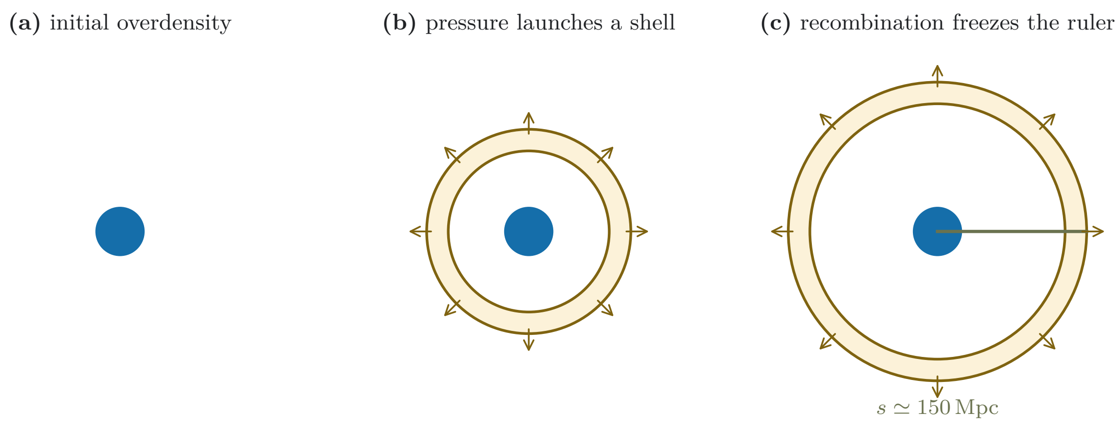

Figure 1.2: Where the ruler comes from (schematic). Follow one initially overdense spot. During the plasma era its pressure launches a spherical sound wave, which carries matter outward at over half the speed of light. When the universe cools enough for atoms to form — about 380,000 years in — the light that supplied the pressure escapes, the wave has nothing left to push it, and it simply stops where it is: a faint spherical shell of extra matter at the distance sound had traveled by that moment, the *sound horizon* $s \approx 150$ Mpc. Every overdense spot in the universe did this simultaneously, so the pattern is imprinted statistically everywhere.

$r \approx 150$ Mpc the frozen shells contribute a small localized excess: the **bump** in the left panel of Figure 1.3. That bump *is* the ruler, as it appears in a measurement.

The same information can be organized by scale instead of separation. Any map can be decomposed into waves of different wavelengths, and the **power spectrum** $P(k)$ records how much the map fluctuates at each wavenumber k (long wavelengths at small k, short at large k). A single preferred separation in $\xi(r)$ becomes a series of *oscillations* in $P(k)$ — the right panel of Figure 1.3 — for the same reason a bell struck once rings with a fundamental and overtones: a feature localized at one scale in one description is a harmonic series in the other. These are the "wiggles," and colloquially the whole feature is *the wiggle.* The two descriptions are mathematically equivalent, and the forecast machinery of Chapter 9 works in $P(k)$; the intuition is often easier in $\xi(r)$, and this section will move between them freely. One more reason panel (b) is the operative picture: when Chapter 9 quotes a *BAO detection significance*, the quantity being detected is precisely the amplitude of the curve in panel (b) — the forecast writes $P(k) = P_{\text{smooth}}(k)\,[1 + A\, f_{\text{BAO}}(k)]$ with f_{BAO} the wiggle-only template, and "detecting BAO" means measuring A away from zero, while the dilations are read from the *phase* of the same oscillations. Detection reads the wiggle's amplitude; measurement reads where its nodes sit.

Three numbers are read off the wiggle, and Figure 1.4 shows what each one does to it. The first two ask *where the bump sits.* Cosmologists predict its apparent position from a reference cosmology; the data are then fit for two **dilation** parameters, $\alpha_\perp$ and $\alpha_\parallel$, the ratios of the observed to the predicted position across and along the line of sight. (The two directions carry different information: across the sky the ruler's apparent size measures the distance to that epoch, while along the line of sight — where distance is inferred from redshift — it measures how fast the universe was expanding right then.) If the reference cosmology is right, both dilations come out 1.000; any departure is a discovery about the expansion history. These are *feature-position* measurements, the BAO measurement proper.

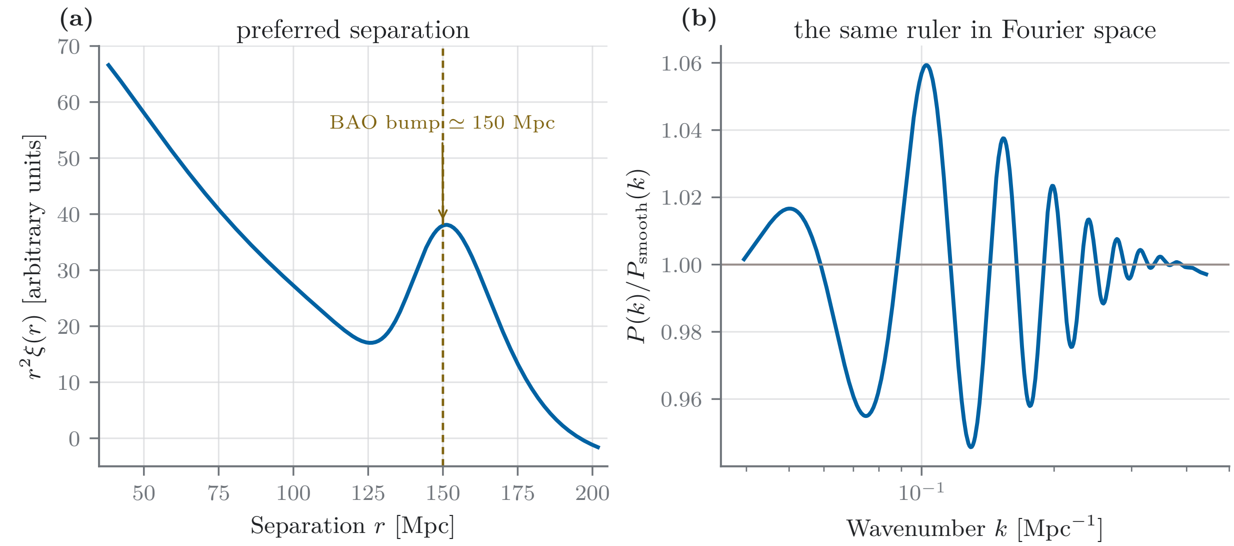

Figure 1.3: The ruler as it appears in data — computed from the identical matter power spectrum used inside every forecast of Chapter 9 (CAMB, Planck-2018 fiducial). *(a)* The correlation function: the excess tendency of two regions separated by r to both be overdense, scaled by r^2 so the feature is visible against the smooth decline. The bump at 150 Mpc is the frozen shell of Figure 1.2, seen statistically. *(b)* Exactly the same information organized by scale instead of separation: dividing the power spectrum $P(k)$ by a smooth, wiggle-free counterpart [24] reveals the oscillations that give the phenomenon its name. A single preferred length produces a harmonic series, spaced $\Delta k = 2\pi/s$, damped toward small scales.

The third number rides along in the same data. Galaxies are not merely carried by expansion; they also fall toward concentrations of matter, and those infall velocities add to the cosmological redshift. Along the line of sight, this compresses the apparent pattern by an amount proportional to how fast structure is growing [25] — the phenomenon is called **redshift-space distortion**. Its strength is quoted as $f\sigma_8$: the growth rate f times the overall clumpiness amplitude σ_8. Unlike the dilations, $f\sigma_8$ is an *amplitude* measurement — how strong the pattern is, not where its feature sits — and that distinction, drawn in Figure 1.4, turns out to govern the results of this dissertation. Contamination that adds smooth power to a map mimics a change of amplitude far more readily than it mimics a shift in a localized feature's position, so the growth rate will prove roughly an order of magnitude more fragile against interference than the ruler itself (Chapter 9).

1.3 Watching Hydrogen Instead of Galaxies

The classical way to measure the wiggle is to catalog a few million galaxies, one by one, with their positions and redshifts. The Canadian Hydrogen Intensity Mapping Experiment (CHIME) takes a different route: it never resolves a single galaxy. Neutral hydrogen — the most abundant element, present in and between galaxies everywhere — emits at a precise radio wavelength of 21 cm (a frequency of 1420.4 MHz). Cosmic expansion stretches that emission in transit, by exactly the factor the universe expanded, so the frequency at which the line *arrives* says how far back in time it *departed*. A radio telescope tuned to one frequency is therefore looking at one definite slice of cosmic history. Frequency is distance. Tune across a band and you sweep out a three-dimensional volume, no catalog required.

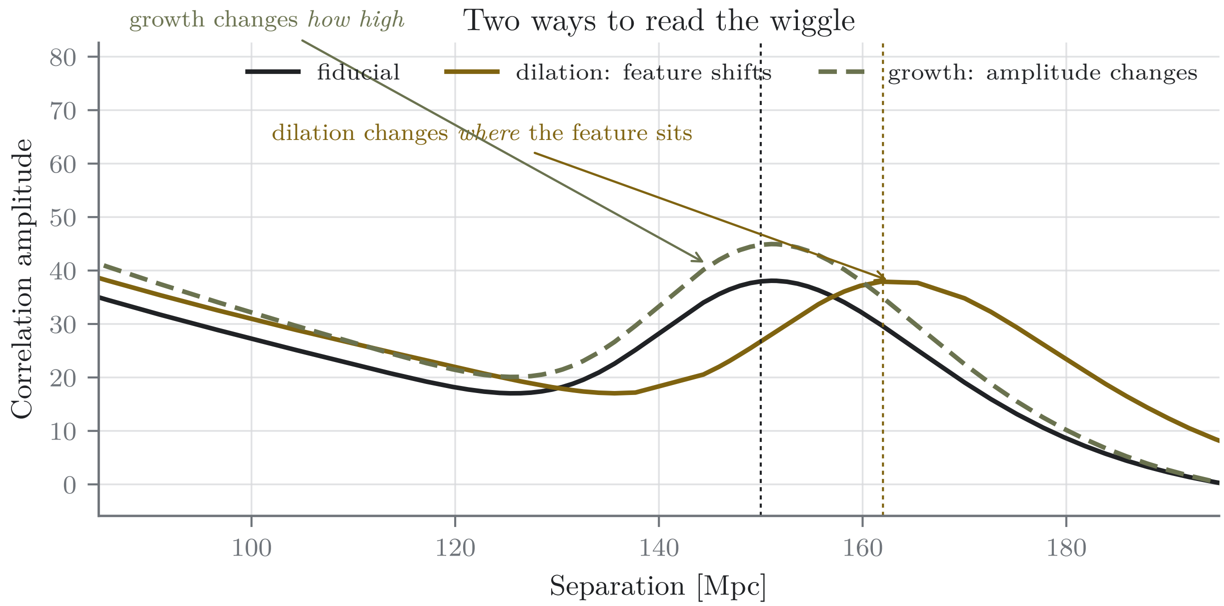

Figure 1.4: Two ways to read the wiggle, drawn from the same forecast spectrum (offsets exaggerated for visibility). A *dilation* slides the bump in separation without changing its shape: that is what the BAO parameters $\alpha_\perp$ and $\alpha_\parallel$ measure, and it is a geometry measurement — where the feature sits. *Growth* raises the whole curve where it stands: that is what $f\sigma_8$ measures, an amplitude. The distinction governs this dissertation's results: smoothly distributed contamination power mimics a change of amplitude far more readily than a shift of a feature's position, and Chapter 9 finds the measured tolerance for $f\sigma_8$ to be ten to twenty-five times tighter than for the dilations.

This strategy is called **intensity mapping**: map the total 21 cm brightness in broad pixels — each containing many unresolved galaxies — and measure the statistical pattern, $\xi(r)$ or $P(k)$, directly from the map. The BAO-scale information is a property of that large-scale pattern and does not require resolving each galaxy, although intensity mapping carries its own tracer bias, calibration, foreground, and shot-noise limitations. What it gains is survey speed and volume. CHIME observes 400–800 MHz, which maps to redshifts $0.8 \leq z \leq 2.5$ — light that left the hydrogen roughly seven to eleven billion years ago, spanning the approach to the epoch when the expansion of the universe stopped decelerating and began to accelerate [1]. Modern spectroscopic surveys, especially DESI, now overlap much of this range with galaxies, quasars, and the Lyα forest [30, 31]; CHIME contributes a complementary tracer, enormous instantaneous volume, and a different set of systematics rather than an uncontested redshift domain.

The instrument itself matters to this dissertation's story, so one paragraph on it. CHIME is four fixed cylindrical reflectors, each 20 m by 100 m, like half-pipes laid side by side, with 1,024 dual-polarization receivers along their focal lines [1]. It has no moving parts. It points nowhere, and therefore everywhere: the sky drifts overhead as the Earth turns, and the correlator reconstructs where each signal came from. Two consequences follow. First, enormous survey speed — the whole northern sky, every day. Second, a property that will quietly become one of this work's main quantitative assets: an instrument that never points is differential in time, and emission fixed to the ground can project strongly onto modes removed by the sidereal analysis. Chapter 9 measures a 5–23 dB reduction in the scalar pilot-power proxy under a per-sidereal-day subtraction model. That reduction is useful, but it is not automatically identical to the attenuation of complex,

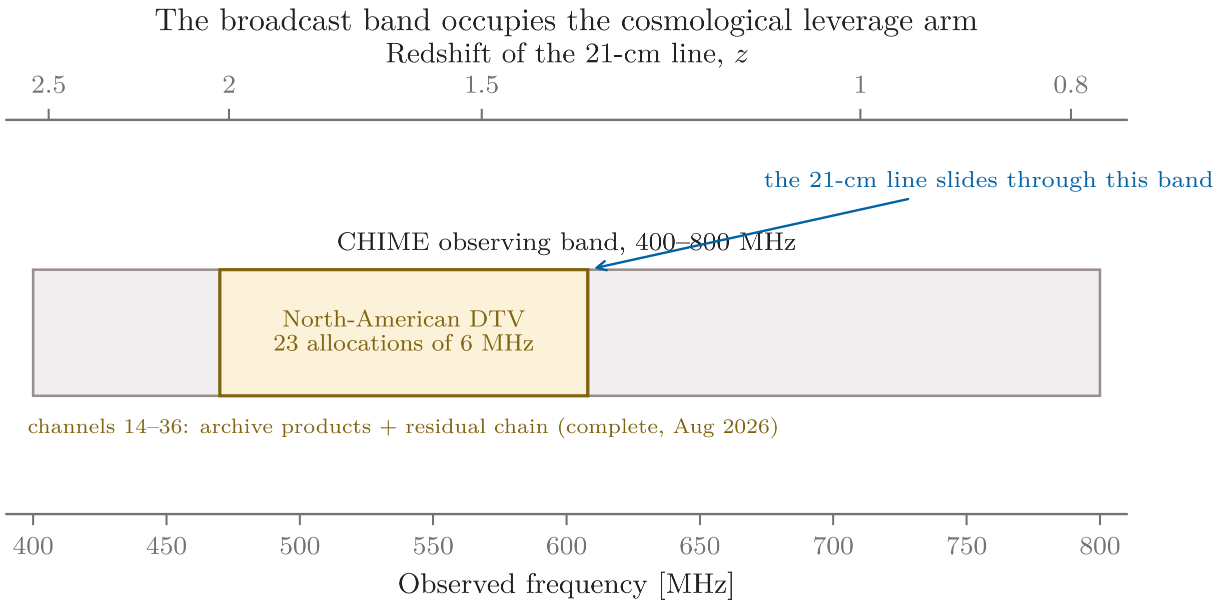

Figure 1.5: The contested ground. CHIME observes 400–800 MHz; North American digital television occupies 470–608 MHz, which the 21 cm line maps to redshifts $1.34 \leq z \leq 2.02$ — a third of the band, sitting on the deepest redshifts CHIME reaches. The 138 MHz of television spectrum is organized as 23 allocations of 6 MHz each (ATSC physical channels 14–36). The figure marks the completed August 2026 coverage: all 23 allocations (channels 14–36, 470–608 MHz) carry archive products and the residual-chain analysis of Chapter 9.

baseline-dependent visibility contamination; the required transfer is made explicit there rather than treated as a free theorem.

The price of intensity mapping is faintness and foregrounds. The cosmological signal is of order a tenth of a millikelvin against a system temperature of tens of kelvin — parts in a hundred thousand — so the measurement is statistical from the outset: observe for years, let thermal noise average down, and the wiggle emerges. Worse, the same band carries **foregrounds**, chiefly the synchrotron glow of our own galaxy, three to four orders of magnitude brighter than the signal. The saving grace is that foregrounds are spectrally *smooth* — their brightness drifts gently with frequency while the cosmological signal fluctuates rapidly (each frequency is a different distance slice) — so smooth-in-frequency modes can be filtered out, at the cost of some signal [1, 19]. Foreground removal is a discipline of its own and is emphatically *not* this dissertation's subject; following the convention of the CHIME Overview forecast, everything here assumes the rest of the pipeline does its job perfectly, claims no credit for any help foreground filtering might incidentally give against interference, and prices only what remains (Sections 1.7 and 9). CHIME has already detected the cosmological 21 cm signal in cross-correlation with galaxy catalogs [18]; the BAO measurement is the long game.

1.4 Where the Measurement Lives, and What Else Lives There

Figure 1.5 shows the problem in one strip. North American over-the-air television occupies 470–608 MHz, dead center in CHIME's band. For the 21 cm line those frequencies are redshifts 1.34 to 2.02 — not an incidental corner of the measurement but a third of the band, and the third that reaches deepest into cosmic history. Each television station holds a 6 MHz allocation; twenty-three of them tile the range. There is no moving the telescope away from this: the spectrum is allocated by regulators, the transmitters are licensed to run at high power from elevated sites, and the telescope's own interference surveys identify television bands among the strongest persistent features in its spectrum [1].

It is worth dispatching the obvious remedies before reaching for a subtle one, because each fails in an instructive way. *Point somewhere else*: a transit telescope has no pointing to change, and the contested frequencies are the science — there is no elsewhere. *Filter the band out in hardware*: an analog notch per allocation would excise all 138 MHz permanently, which is excision of the very redshifts the measurement wants, purchased in copper instead of software; and a hardware filter cannot be turned off during the hours a transmitter happens to be silent, which Chapter 9 will show is where most of the recoverable science lives. *Move the telescope*: it is already in a protected radio-quiet valley, and the television power arrives anyway — diffracted over the horizon and scattered off aircraft and meteor trails [1]. Distance and terrain attenuate broadcast television; they do not remove it, and this measurement's tolerance is not "attenuated," it is a number. What remains is the subtle remedy: decide, moment by moment and in software, when each channel is contaminated, and discard exactly those moments. The rest of this dissertation is the theory, construction, and pricing of that decision.

Here is the fact that makes this a dissertation rather than a nuisance: *in any single data frame, the television signal is below the telescope's noise.* The detector built here decides in frames of 41.94 ms (Section 2.3), and in one frame the measured television shelf on these channels runs from system-noise level down to 44 dB below it — the loudest channel matches the noise; the faintest is twenty-five thousand times weaker (Chapter 9). Nothing about a single frame looks wrong. A sky map made from an afternoon of data would show nothing amiss. If the interference behaved like noise, it would average away with the noise and could be ignored entirely. The whole problem, and the whole of this work, comes from the fact that it does not.

1.5 Two Ways to Be Wrong

The vocabulary of the problem is worth pinning down, because every quantitative claim in this dissertation is stated in it.

Statistical error (the error bar). The scatter a measurement would show if it were repeated many times. It comes from randomness — here, thermal noise in the receivers — and it shrinks with more data, as $1/\sqrt{t}$ for observing time t. When an experiment quotes $x \pm \sigma$, the σ is this.

Systematic. Anything in the apparatus or environment that pushes the measurement consistently in one direction, rather than randomly in both. A bathroom scale that reads two pounds heavy is a systematic; weighing yourself every morning for a year does not fix it, it just gives you a very precise wrong weight.

Bias. The shift a systematic produces in the final answer: the distance between the average of many repeated measurements and the truth. Written b throughout.

Residual, r. The unit this dissertation prices contamination in: interference power that survives all filtering into the final statistical analysis, expressed as a fraction of the thermal noise power in the same data (Chapter 9). A residual of $r = 10^{-3}$ means the surviving contamination carries a thousandth of the noise power.

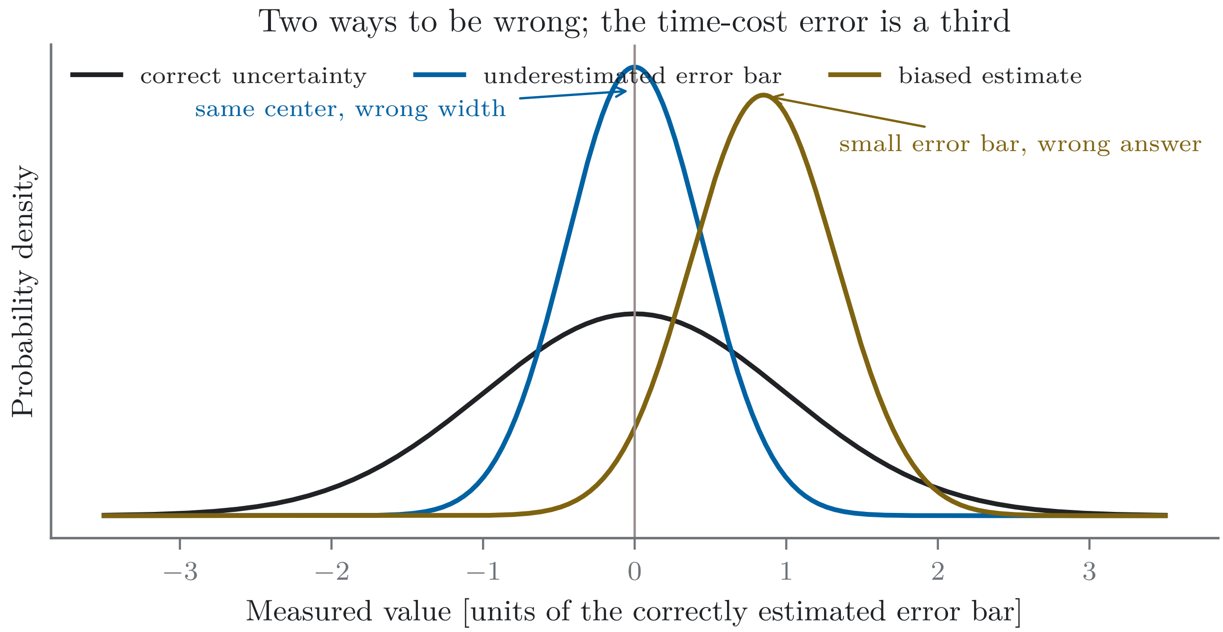

Figure 1.6: Two ways to be wrong (schematic). Each curve is the distribution of answers an experiment would give if repeated. More data narrows the distribution — the statistical error falls — but does nothing to the offset a systematic causes. The same bias that was hidden inside a wide early error bar becomes a confident, precise, wrong answer later: here $b = 0.6$ early error bars, invisible at first, is a 2.4σ discrepancy after sixteen times the data. The published tolerance criterion, $b \leq \sigma$ [6], is the statement that a measurement should never be allowed into that regime.

Figure 1.6 draws the distinction. Statistical error and bias are not two flavors of the same problem; they respond oppositely to the one resource a survey controls, which is time. Integration narrows the error bar and leaves the bias alone, so the *ratio* of bias to error bar grows as data accumulate. An experiment with an unaddressed systematic does not converge to the truth slowly. It converges, confidently, to the wrong answer — and every internal consistency check based on scatter will say things are going well.

When is a bias small enough to live with? The convention this work adopts is the one published for precision cosmology by Amara & Réfrégier [6]: a systematic is tolerable while the bias it induces does not exceed the statistical error, $|b| \leq \zeta\,\sigma$ with $\zeta = 1$. The logic is that an experiment's headline number is $x \pm \sigma$; if the truth lies within the quoted interval, the report is honest even in the presence of the systematic. Past that point the experiment is claiming precision it does not have. Nothing about $\zeta = 1$ is magic — a collaboration wanting margin can set $\zeta = 0.3$ and every tolerance in this work scales linearly — but $\zeta = 1$ is the published, citable line between an honest error bar and a false one, and Chapter 9 adopts it as the definition of "tolerable."

1.6 Why Time Does Not Fix This One

Why doesn't the television signal just average away with the noise it hides beneath? Because averaging removes fluctuations, and a broadcast transmitter, to first approximation, does not fluctuate. Picture a choir of a thousand people each humming a random note that changes every second, and one quiet singer holding a single steady pitch. In any one second the singer is inaudible.

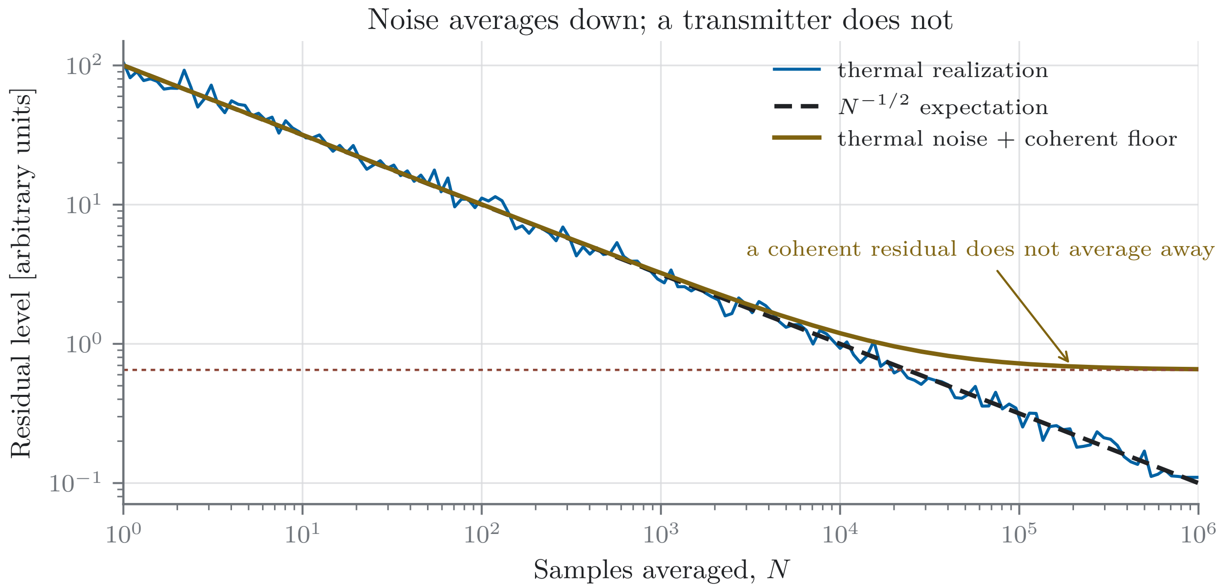

Figure 1.7: One seeded simulation of the mechanism at the heart of the problem. Blue: the running average of pure noise falls as $1/\sqrt{N}$ indefinitely. Orange: add a constant signal at 1/100 the noise amplitude — 40 dB weaker in power, invisible in any single sample — and the average tracks the noise until $N \sim 1/a^2$, then lands on the transmitter and stays there. Averaging is not a filter against something that does not fluctuate. The measured version of this figure, with both axes in survey units, is the convergence analysis of Chapter 9.

Record for an hour and average: the random hums cancel toward silence, and what remains is the held note. The telescope's thermal noise is the choir; the transmitter is the singer.

Figure 1.7 shows the arithmetic. Noise averages down as $1/\sqrt{N}$ without limit. A coherent signal of amplitude a (in noise units) does not average down at all, so after roughly $N \sim 1/a^2$ samples it owns the average. A signal 40 dB below the noise — thoroughly undetectable in any frame — dominates after 10^4 samples. At the 41.94-ms decision cadence, a continuously integrated channel contributes about 2.06×10^6 frames per day. A multi-year forecast target therefore makes even sub-noise coherent residuals dangerous: under the screening model of Chapter 9, the measured shelves — from system-noise level down to 44 dB under it — can exceed the adopted tolerance by factors from a few hundred (channel 33, the best-constrained case) to a few million (channel 30) if left untreated. Those ratios are forecast outputs, not counts inferred from the sparse trigger archive.

The boundary between "averages away" and "accumulates" has a name that will recur throughout this work: the **correlation time**, τ_c — how long the contamination holds steady before it changes character. Whatever varies faster than a frame averages down like noise; whatever holds steady accumulates coherently for as long as it holds, and is amplified relative to noise by the number of frames per correlation time. A residual with a 45-minute correlation time is some 64,000 frames' worth more dangerous than its single-frame power suggests, which is why Chapter 9 spends considerable effort measuring τ_c per channel — and declining to guess where the archive cannot support a measurement.

Reality is gentler than the cartoon in one important way, but the mitigation must be defined at the level where it is applied. The downstream analysis subtracts a per-sidereal-day mean from each *complex visibility*, suppressing contamination that is stationary in sidereal coordinates while retaining the portion that varies within a day through propagation flutter, aircraft reflections, or other changes. The current archive does not contain the corresponding before-and-after visibility products; it contains a scalar pilot-power proxy. In that proxy, the measured intra-day shares span roughly 0.5%–29%, equivalent to screening suppressions of about 5–23 dB. These are not direct measurements of complex-visibility attenuation. Chapter 9 therefore treats the proxy-to-visibility transfer as an explicit closing test rather than silently crediting the full scalar suppression.

One more twist separates this figure from the real analysis: the time scaling depends on what is held fixed. In the screening calculation of Chapter 9, the residual template is normalized to the survey noise at a declared target time, so the induced bias and statistical error can fall at similar rates and their ratio remains nearly flat. A fixed physical contaminant need not follow that family; as the thermal error shrinks, its significance can instead grow. The dissertation therefore does not infer safety from convergence alone. It defines the residual model, the target observing time, and the tolerance together, and it treats alternative time scalings as a validation gate.

1.7 How Much Is Too Much: From a Criterion to a Number

The criterion $|b| \leq \sigma$ is a principle, not a number. Turning it into a number an engineer can design against takes three steps, and the machinery for all three is a **Fisher forecast**. Fisher information formalizes a simple question: *how sharply do the data distinguish nearby universes?* Imagine generating the map CHIME will measure in two cosmologies that differ only slightly in one parameter — the bump shifted by a hair. If the two predicted maps differ by much more than the noise, the data carry a lot of information about that parameter; if the difference drowns in noise, they carry little. Fisher information is that ratio, made precise, and its reciprocal square root is the best statistical error any analysis of those data can achieve. A Fisher forecast assembles this for every parameter at once, given an instrument, a sky, and an observing time — before any data are taken. The forecast used throughout this work is built on the RadioFisher framework of Bull et al. [17], configured to CHIME following its Overview forecast conventions [1], and it predicts the statistical error $\sigma(\theta)$ on each parameter $\theta \in \{\alpha_\perp, \alpha_\parallel, f\sigma_8\}$ as a function of integration time.

The three steps. First, compute $\sigma(\theta; T_\star)$ for the clean survey at a declared target integration time $T_\star$ — that is the error bar the experiment is entitled to. Second, inject a small coherent residual with amplitude ratio $r_{\rm sys}$ and compute the first-order shift $\Delta\theta(r_{\rm sys}; T_\star)$ it drags into each parameter. Third, solve $|\Delta\theta| = \zeta\,\sigma$ for the allowed residual. The solution,

$$r_{\rm sys,\,tol}(\theta; T_\star) \;=\; \zeta\,\frac{\sigma(\theta; T_\star)}{|\partial\Delta\theta/\partial r_{\rm sys}|_{T_\star}},$$

is the **tolerance**: the largest residual admitted by that estimator, residual template, and target time before the quoted error bar on θ becomes misleading. In the current noise-normalized screening family the ratio is nearly flat with integration time, but Chapter 9 keeps the target time explicit and requires a separate fixed-physical-amplitude test. For the redshifts occupied by television, the screening tolerances are a few times 10^{-2} for the acoustic dilations and 1.5×10^{-3} for $f\sigma_8$: the growth rate is ten to twenty-five times more delicate than the ruler itself, for the position-versus-amplitude reason drawn in Figure 1.4.

It is worth being explicit about the three distinct jobs this forecast does in this dissertation, because together they are the reason the cosmological machinery belongs in an instrumentation thesis at all.

Feasibility. Before live integration, the forecast asks whether a channel has any defensible operating point under the declared model. The useful comparison is not merely whether a positive tolerance exists, but whether the measured proxy, transfer uncertainty, occupancy, and available suppression can place both stochastic variance and coherent bias inside their gates. The completed archive produces conditional recovery candidates on a subset of the 23 measured channels, negative screening results on others, and a class of cases whose outcome is controlled by measurements the snapshot cadence cannot supply. That mixture — rather than a blanket claim of recoverability — is why the detector and the decision framework are both needed.

Evaluation. In the archival analysis, the same forecast converts measured mask fractions and kept-frame pilot proxies into screening quantities that can be compared with the tolerance, channel by channel. Those comparisons become deployment verdicts only after the proxy is calibrated across each 6-MHz allocation and into visibility-domain residuals, the combined-bin estimator is reproduced, and the per-epoch policies pass blocked holdout tests. Chapter 9 carries the present calculation end to end and labels each conclusion by the evidence it actually has.

Optimization. Masking is not free. Every flagged frame is discarded observing time, so an aggressive threshold trades residual contamination for a longer survey. For the stochastic part, the equivalent-time cost is $(1 + r_{\mathrm{var}})/(1 - f)$, where f is the masked fraction and r_{var} is the visibility-domain excess-variance ratio. The coherent component r_{sys} is constrained separately by the Fisher-bias tolerance rather than folded into that variance penalty. Pricing the two effects with separate gates turns the threshold from a taste into an optimization variable — and identifies channels where no tested policy wins.

1.8 The Needle in the Shelf

CHIME already runs interference flaggers in production, and they work by the natural principle that contamination should look *different* — an outlier against its neighbors in time, or statistics inconsistent with the adopted noise null (the median-deviation and spectral-kurtosis families; [18, 21]). Against bursty, above-noise interference that principle is effective. A continuous sub-noise 8-VSB transmission is a harder case: the standard randomizes the payload to flatten its spectrum, and after channelization its narrow-cell statistics can lie close to the same null used by common power and variability tests [2]. It need not be mathematically Gaussian, nor is every flagger blind to it; the narrower claim tested here is that the incumbent archive flags are most responsive to transitions while long transmitter-on intervals can remain. The pilot supplies a deterministic feature against which that failure mode can be measured directly.

But the standard, having hidden the payload, could not hide everything. A receiver needs something deterministic to lock onto, so ATSC adds a **pilot**: a bare carrier at a frequency fixed by the standard to within transmitter tolerance, carrying about 7% of the station's power [2]. Figure 1.8 shows the anatomy. Seven percent sounds like a poor handle until you notice *where* that power sits: all of it at one known frequency, while the other 93% is smeared across 6 MHz. In the detector's own analysis bandwidth — the 3 kHz bin that detection sensitivity actually responds to — the pilot stands 21.6 dB *above* the shelf that carries it; per hertz of spectrum the contrast is larger still, about 56 dB (Chapter 3). The needle is brighter than the haystack, 146-fold, provided you know exactly where to look. And we do know: the broadcast standard is the calibration document.

The detector this dissertation builds is, at its core, one line of mathematics: correlate the data against a copy of what the pilot should be, and compare the power at the pilot's frequency to the power at two reference frequencies a few kilohertz away. That correlation is a matched filter — a single-term discrete Fourier transform, evaluated off the telescope's frequency grid so it lands exactly on the pilot, implemented as a dot product with precomputed integer weights (Chapter 4). Per 42 ms frame, per monitored channel, it yields one number and one decision: transmitter on, or off. Everything else in Chapters 4–6 — the weight synthesis, the reference placement, the exact integer

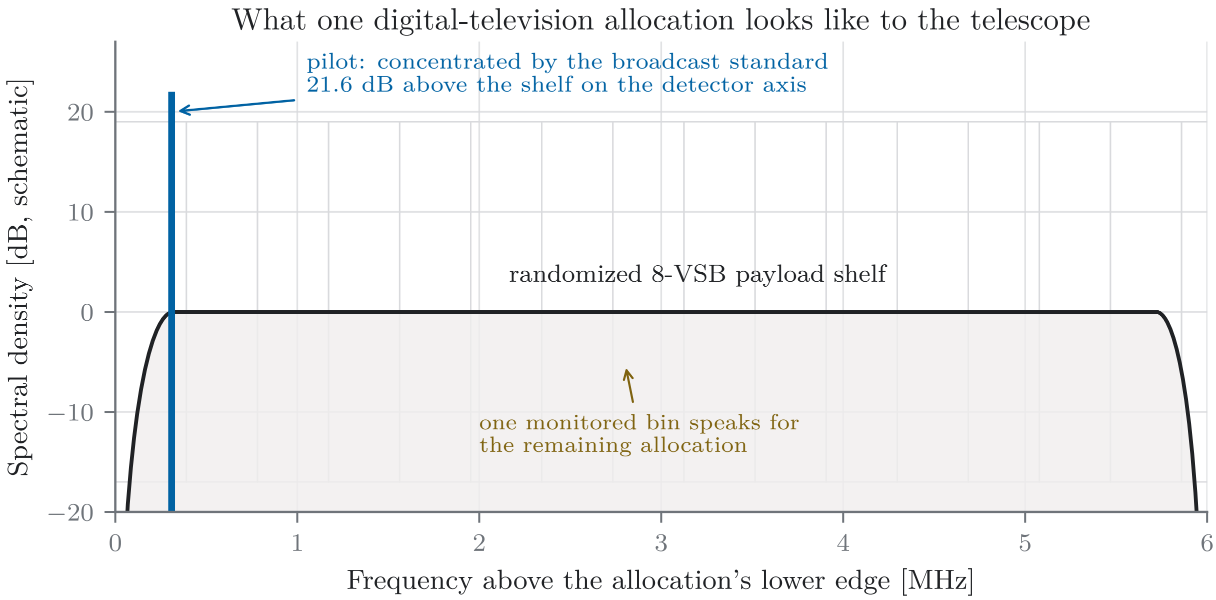

Figure 1.8: What one television allocation looks like to the telescope (schematic, drawn to the measured proportions). The 6 MHz shelf is the broadcast's randomized data payload; in narrow time–frequency cells it can approach the null assumed by power- and variability-based flaggers. The pilot carries about 7% of the station's power at a nominal standard-defined offset [2]: weak in total, but 21.6 dB *above* the shelf in spectral density under the transmitter-ratio model used here. CHIME's 390.625 kHz channel raster is drawn behind; the detector monitors the bin containing the pilot as a proxy for the 15.36 bins spanned by the allocation. Chapters 3 and 4 derive the geometry and state the transfer assumptions.

arithmetic that makes every implemented decision bit-reproducible — is engineering in service of that one dot product.

The economics of the method come from a geometric accident shown in the same figure. The pilot occupies one of CHIME's 390.625 kHz bins, but its allocation spans 15.36 of them: one monitored bin speaks for fifteen. Across the whole band, watching 23 pilot bins — 6.5% of the contested spectrum — decides the fate of all 138 MHz. The detector is designed and benchmarked for frame-continuous use, and its decision is per-frame, so a future live mask can remove transmitter-on time rather than entire channels.

1.9 What the Detector Must Be

The physics of the preceding subsections is a requirements document in disguise. Collecting the demands in one place shows why the detector of Chapters 4–6 has the shape it has, and gives the reader a checklist to carry into those sections.

R1 — See below the noise. On all but the loudest channel the contamination sits below thermal noise per frame — as much as 44 dB below — so a detector that inspects raw power is blind by construction. The pilot's 21.6 dB spectral concentration (Chapter 3) plus coherent integration across each frame (Chapter 4) must together close that gap.

R2 — Decide per frame. Transmitter occupancy changes on minutes to years, and every misclassified frame either leaks contamination or discards good sky. The decision cadence is one verdict per 42 ms frame per channel (Chapter 5), so the mask can remove transmitter-on *time* rather than defaulting to whole-channel excision whenever clean frames remain.

R3 — Fit the compute budget. The decision must run on every frame of every contested channel, indefinitely, inside a correlator that is already the busiest part of the instrument. The design therefore reduces the online work to one fixed-point dot-product pipeline per frame (Chapter 4) and treats measured runtime, memory traffic, and synchronization headroom as deployment gates rather than assuming the cost is negligible.

R4 — Reproduce to the bit. Eight years of archive products are calibrated against this detector's outputs, and the intended live stage uses the same arithmetic contract. Every emitted runtime decision is therefore required to be bit-identical to a reference implementation (Chapter 6): not close, identical, so that any decision can be re-derived and audited years later.

R5 — Record the statistic, not just the verdict. The survey products store the full decision statistic for every *processed* frame, not merely the mask bit (Chapter 5). This is what lets Chapter 9 re-decide every threshold offline, calibrate operating points against the forecast, and adapt to a broadcast environment that measurably changes over the archive — all without ever re-running the survey.

A sixth demand runs through the analysis rather than the hardware: *prefer "unmeasured" to a guess.* Where the archive cannot support a number — a correlation time on an episodic channel, a sensitivity floor on a channel that never falls quiet — the machinery of Chapter 9 is built to refuse and to carry the refusal forward as a labeled bound, because a budget assembled from optimistic guesses is precisely how a survey ends up confidently wrong, which is the failure mode this entire work exists to prevent.

1.10 Limits Within the Mask-and-Discard Paradigm

A masking method invites an obvious objection: perhaps a more sophisticated detector would save channels this one cannot. The right answer is narrower than "ours is optimal," because three distinct questions are easy to conflate: the optimal statistic for a declared signal model, the best complete detector after nuisance parameters and calibration are introduced, and the existence of clean frames for any mask to retain.

The model-conditional sensitivity result. For the decision "is a signal of known form present in Gaussian noise," the Neyman–Pearson lemma [27] establishes that the likelihood-ratio test maximizes detection probability at fixed false-alarm rate, and for a known signal in Gaussian noise that test is the matched filter [28, 29]. The pilot projection used here is therefore optimal for the known-tone component of the declared model. The complete operational detector is broader: it searches a finite residual-frequency window, estimates local background from displaced references, tolerates feed covariance and non-Gaussian tails through empirical calibration, and applies a rank statistic. Those additions are necessary because the telescope does not satisfy the ideal model exactly, and they prevent the matched-filter theorem from becoming a claim of global optimality for every possible detector.

The broadcast standard also contains deterministic data-segment and data-field synchronization sequences, so the pilot is not literally the only deterministic structure in an 8-VSB waveform [2]. It is, however, the dominant *continuously present, spectrally concentrated* feature available to a low-cost detector that does not decode the broadcast. The scrambled payload is noise-like at the time and bandwidth scales used here; the synchronization structures are distributed across the allocation; and the pilot places its coherent energy in one known spectral neighborhood. This is why a pilot matched filter is the natural sensitivity benchmark. It does not prove that every cyclostationary, decoding, multi-antenna, or subtraction method must perform worse. A fair successor comparison would require matched compute, identical frames, and out-of-sample receiver-operating curves.

The occupancy result. A different bound applies to channels whose transmitter is effectively always on. If a channel contains no demonstrable clean population, perfect detection within the mask-and-discard paradigm converges to excision: every contaminated frame is rejected and nothing remains. Imperfect detection retains contamination. This conclusion is detector-independent only in that restricted sense. It says that a better *mask* cannot manufacture clean exposure; it does not rule out waveform subtraction, spatial nulling, decoding-assisted cancellation, or a different observing strategy. Chapters 8 and 9 therefore use "excision candidate" rather than "impossible" whenever the archive cannot itself establish a clean population.

The boundary is scientifically useful. Sensitivity-limited channels motivate a better detector or a better reference model. Occupancy-pinned channels motivate a different paradigm. The two should not be given the same negative label, and the status matrix in Chapter 11 keeps them separate.

1.11 What This Dissertation Does

The argument of the work, compressed to a paragraph: CHIME's BAO measurement carries a computable tolerance for surviving contamination, and broadcast television — sub-noise per frame and potentially coherent across long integration — can exceed it while evading simple variability-based defenses. The astrophysics leads and the engineering follows: the forecast asks first whether a sufficiently clean mask could make these redshifts useful. The pilot then provides a continuously present, spectrally concentrated target whose projection is a matched filter under the declared model and is cheap enough to evaluate on every processed frame. This dissertation builds that detector to a bit-reproducible engineering contract, applies it to 7.6 years of triggered archival captures, and prices the resulting policies in the currency of the science. The final step is explicitly conditional: a channel verdict is only as strong as the pilot-to-shelf, visibility-domain, and forecast-estimator transfers that connect a detector statistic to cosmology.

The chapters follow that arc. Chapter 2 sets the instrument and the detection problem. Chapter 3 extracts the pilot from the broadcast standard and establishes every property the detector will rely on. Chapter 4 synthesizes the matched filter's weights under the telescope's integer arithmetic constraints. Chapter 5 builds the processing pipeline and its decision theory, and Chapter 6 verifies the implementation to the bit. Chapter 9 is the reckoning: the residual chain from a masked frame to a cosmological bias, the tolerance derivation sketched above in full, the four-way comparison against keeping everything, excising everything, and the incumbent flaggers, the ten-channel status map, and the August 2026 completion of the thirteen lower-band allocations. Chapter 10 then states the live-stage interface as an engineering contract rather than treating archive execution as telescope deployment, and Chapter 11 separates measured results, model-conditional forecasts, and the closing tests required before a final 23-channel verdict.

Chapter 2

Scientific Context, Instrument, & the Detection Problem

2.1 Scientific Context & the Interference Problem

The Canadian Hydrogen Intensity Mapping Experiment (CHIME) is designed to measure the expansion history of the universe by mapping redshifted 21 cm emission from neutral hydrogen over $0.8 \leq z \leq 2.5$, tracing the baryon acoustic oscillation scale through the epoch in which cosmic expansion transitioned from decelerating to accelerating [1]. The measurement is strongly foreground-limited: Galactic synchrotron emission exceeds the cosmological signal by several orders of magnitude, and foreground separation relies on the spectral smoothness of that emission. Radio-frequency interference directly undermines this separation when its spectral or temporal structure survives into averaged visibilities and overlaps modes used by the cosmological estimator. Interference excision therefore constitutes a science requirement for the experiment. False alarms carry the complementary cost: each incorrectly flagged frame discards integration time from a measurement whose statistical uncertainty falls only through accumulated exposure.

Within CHIME's measured interference environment, digital television broadcasting is of particular concern. The telescope's own RFI characterization identifies television station bands across 480–580 MHz among the strongest persistent features in its spectrum, together with intermittent few-second bursts of distant broadcast channels scattered into the beam by meteor ionization trails and aircraft [1]. DTV interference thus arrives in two modes — persistent carriers from line-of-sight and diffracted paths, and transient scattered events — and both must be caught. Chapter 3 develops the structural feature that makes this tractable: every DTV emission carries a standardized, spectrally concentrated pilot carrier whose frequency is known in advance to within transmitter tolerance. This dissertation develops the *pilot-proxy method*: a matched-filter projection operating on the channelized baseband, followed by local references and a robust calibrated decision. It produces one binary rejection decision per processed correlator frame per monitored channel and is implemented in exact integer arithmetic so that the candidate production path is bit-reproducible against an independent reference. Its calibration is derived from a survey of the DTV environment conducted on archived telescope baseband; live `kotekan` integration remains a separate interface milestone (Chapter 10).

Why the pilot is a targeted complement, not a universal replacement

Generic flaggers answer a different question. Spectral-kurtosis estimators are powerful when interference changes the distribution of power away from the Gaussian null [21]; a randomized, filtered 8-VSB payload can approach that null in a narrow analysis cell even while a coherent pilot remains present. Cyclostationary methods instead search for known periodic correlations and are natural candidates for digitally modulated signals [32]. Neither class is dismissed by theorem here. Their relative performance depends on observation length, bandwidth, nuisance uncertainty, and compute budget. The pilot method is justified by a narrower engineering claim: for a continuously present line with known nominal frequency in locally Gaussian noise, the matched projection is the likelihood-ratio statistic, and the line concentrates enough energy to be evaluated cheaply on every frame. The incumbent and alternative methods are therefore compared empirically on common data products in Chapter 9, rather than declared impossible in advance.

2.2 The Instrument & Signal Path

CHIME is a drift-scan transit telescope at the Dominion Radio Astrophysical Observatory near Penticton, British Columbia: four fixed cylindrical parabolic reflectors, each 20×100 m, oriented north–south, with no moving parts [1]. Each cylinder's focal line carries 256 dual-polarization feeds; four cylinders give 1024 antennas and $M = 2048$ independent analog chains. The observing band is 400–800 MHz: scientifically, this is the 21 cm redshift range $0.8 \leq z \leq 2.5$; from an engineering standpoint it is exactly one octave, so an 800 MSPS converter captures the band directly as its own alias (second-Nyquist-zone sampling), halving the conversion rate otherwise required [1].

Table 2.1 summarizes the telescope parameters used throughout this work, together with their governing constraints. The digital signal chain defines the environment in which the detector operates. Each analog input is digitized at 800 MSPS with 8-bit resolution — with dynamic range verified sufficient that site RFI does not drive the receiver into nonlinearity [1]. A polyphase filter bank (PFB) processes the samples in frames of 2048, applying a sinc-Hamming window across four consecutive frames and emitting one complex spectral value per $f_s = 400 \text{ MHz}/1024 = 390.625$ kHz channel every

$$\tau_{\text{PFB}} = \frac{2048}{800 \text{ MSPS}} = 2.56 \ \mu\text{s}; \tag{2.1}$$

the PFB is preferred over a direct FFT because its more compact spectral response localizes any disturbance — including RFI — to fewer channels [1]. Throughout this work, f_s denotes this per-channel complex sample rate, and one "sample" of Chapters 4–5 is one channelized PFB output. The 18+18-bit channelizer outputs are rounded to packed 4+4-bit complex values after per-channel complex gains chosen to load the 15-level quantizer optimally [1, 3]: this native 4+4-bit format defines the detector's input interface and motivates the `int4` arithmetic of Section 4.1.

The channelized data feed an FX correlator: a corner-turn network rearranges the F-engine outputs so that each of 256 GPU nodes of the X-engine receives all 2048 inputs for four frequency channels, where the kotekan real-time framework [5] forms the N_{feed}^2 visibility matrix, presently integrated over 12,288 channelized samples (≈31 ms) [1]. For the primary cosmological data product, a downstream receiver applies calibration and flagging, stacks redundant baselines, and integrates to 10 s cadence [1] ; this 10 s cadence is the granularity at which flagging decisions act on the science data, and it is adopted later in this work as the natural unit for quantifying flagging impact. The X-engine already hosts a general-purpose RFI stage (spectral-kurtosis excision at sub-millisecond cadence [1]); the detector of this dissertation complements it with a matched, narrowband, per-channel decision aligned one-to-one with correlator frames, targeting a specific known interferer at

Table 2.1: Telescope and signal-path parameters used throughout this work, with governing constraints. These are CHIME's values, and every numerical quantity quoted in this chapter follows from them; the constraint equations are receiver-independent, and Section 3.5 states which parameters a different receiver must re-derive. Instrument values from [1]; quantization from [1, 3].

Parameter	Value	Governing constraint
Observing band	400–800 MHz	21 cm at $0.8 \leq z \leq 2.5$; one octave (2nd Nyquist zone)
Analog inputs	$M = 2048$	4 cylinders $\times$ 256 dual-polarization feeds
ADC	800 MSPS, 8 bit	octave band; RFI headroom below nonlinearity
Channelizer	1024-channel PFB	sinc-Hamming, 4-tap; RFI localization
Channel rate	$f_s = 390.625$ kHz	400 MHz/1024
PFB cadence	$\tau_{\text{PFB}} = 2.56\ \mu$s	2048/800 MSPS
Channelized format	packed 4+4-bit complex	15-level mid-tread; gain-optimized loading
Visibility cadence	$\approx$31 ms (12,288 samples)	present X-engine integration
Cosmology product	10 s integrations	calibrated, stacked, flagged
Detector frame	$N = 16384$ samples = 41.94 ms	upgraded-engine design value (Sec. 4.1)

interference-to-noise ratios below the reach of broadband kurtosis statistics.

2.3 The Detection Problem & the Shape of This Work

The pilot-proxy detector taps the signal path at the channelized, quantized stage: its input is the packed 4+4-bit complex sample stream of one coarse channel across all $M = 2048$ inputs, and its output is one binary rejection decision per N-sample frame — $N = 16384$ channelized samples, i.e. $16384 \times \tau_{\text{PFB}} = 41.94$ ms, chosen to align one-to-one with the upgraded correlator's integration frames (Section 4.1). N is frozen in *samples*; its wall-clock duration and the resulting fine-axis span scale with the receiver's channel raster, so the millisecond and kilohertz figures quoted throughout are CHIME-cadence values rather than properties of the method. Everything between input and output is fixed-point integer arithmetic with widths chosen so that no intermediate value can overflow or round before the frozen transform's specified shifts: the GPU implementation and the offline reference are required to agree bit-for-bit, a property this dissertation treats as an engineering contract and exploits systematically for verification. Two execution surfaces share that datapath. The archive surface is exercised in this dissertation. The live surface is specified as a future `kotekan` stage ahead of visibility integration, with its per-frame mask consumed synchronously by the correlator path; the stage interface and failure contract are treated separately in Chapter 10. Offline, the identical arithmetic runs over archived full-array baseband captures — triggered recordings from the telescope's transient backend [4] — which constitute the survey data from which the per-channel calibration constants of Chapters 4 and 5 (measured anchors, null-bulk masks, CFAR ranks and multipliers) are derived. The remainder of this work is organized as follows: Chapter 3 derives the beacon from the broadcast standard; Chapter 4 constructs the detector's parameters and weights from explicit constraints; Chapter 5 develops the processing pipeline and its decision theory; Chapter 6 realizes the design in exact arithmetic and verifies it to the bit; Chapter 9 converts the detector's products — the mask and the per-frame contamination proxy — into conditional BAO science reach, pricing the archive rule against the alternatives a survey actually has; Chapters 7 and 8 state the evidence and calibration products required to turn those conditional results into a final operating bundle.

Notation. $\mathbf{X} \in \mathbb{C}^{M \times N}$ denotes one frame of channelized data for one coarse channel ($M = 2048$ inputs, $N = 16384$ samples); $K = 128$ the matched-filter tap length; $L = N/K = 128$ the sub-frames per frame; $L_F = 2L = 256$ the padded fine-spectrum length; $R = 3$ the weight profiles (target $r = 0$, references $r \in \{1, 2\}$); f_s the per-channel complex sample rate; ν_c the normalized pilot tone frequency of channel c; $S_r[f]$ and P_r the fine and coarse power statistics; $T[f]$ the per-bin detection ratio; $\mathcal{D}$ and $\mathcal{B}$ the designated window and usable null bulk; f_a, ρ, η the per-channel anchor bin, CFAR rank, and threshold multiplier; μ_0 the exact rational weight-norm correction; and β the sample dimension factor ($\beta = 2$ for complex data).

Chapter 3

Digital Television RF Anatomy & Parametric Beacon Characteristics

3.1 Physical Channel Allocations & the UHF/VHF Raster

In North American broadcast standards (ATSC A/53 [2]), Digital Television (DTV) signals are allocated across fixed $B_c = 6$ MHz terrestrial frequency channels — the allocation width inherited from the 1941 monochrome NTSC standard. The telescope observes 400–800 MHz, the octave band set by its 21 cm science goals (mapping neutral hydrogen across redshifts $0.8 \leq z \leq 2.5$, the epoch in which cosmic expansion transitioned from decelerating to accelerating) and by the electronics economy of second-Nyquist-zone sampling over exactly one octave (an 800 MSPS converter captures 400–800 MHz directly as its own alias, halving the sampling rate otherwise required) [1]. The intersection of this band with the UHF television allocation runs from the 470 MHz band edge — channel 14, the first UHF assignment, channels 2–13 occupying VHF allocations at 54–88 and 174–216 MHz — up to 608 MHz (channel 36), the top of the television band after the 2017–2020 incentive-auction repack, bounded above by the internationally protected radio-astronomy allocation at 608–614 MHz (channel 37). The monitored census is therefore $C = 36 - 14 + 1 = 23$ active or incumbent physical DTV channel allocations. Let $c \in \{1, 2, \ldots, 23\}$ index the monitored DTV channels. The lower RF boundary of the c-th allocation, denoted $f_{\text{edge}}^{(c)}$, is defined by the standard terrestrial UHF television raster: in the core UHF broadcast band (physical channels 14–36), the lower allocation edge for physical channel number n_{ch} is given analytically in MHz by

$$f_{\text{edge}}(n_{\text{ch}}) = 470 + 6(n_{\text{ch}} - 14) \text{ MHz}. \tag{3.1}$$

Because terrestrial broadcast transmitters emit high Effective Isotropic Radiated Power from elevated horizons, even distant or horizon-diffracted DTV emissions present severe, highly correlated RFI across high-gain astrophysical receiving arrays. This interference environment is directly measured at the site: the telescope's RFI characterization identifies television station bands across 480–580 MHz among the strongest persistent features in its spectrum, alongside intermittent few-second events — 6 MHz-wide bursts of distant broadcast channels scattered into the beam by meteor ionization trails and aircraft [1]. DTV energy therefore arrives through two distinct modes, persistent carriers on line-of-sight and diffracted paths, and transient scattered bursts; the flagging system of this work must detect both, which shapes the per-frame decision cadence of Chapter 5.

3.2 8-VSB Signal Anatomy & the Normative Clock Relations

An ATSC A/53 broadcast occupies a $B_c = 6$ MHz allocation using eight-level vestigial-sideband modulation. The quantities used by the detector come from two different kinds of statements in the standard, and the distinction matters: the clock relations are exact, whereas the root-raised-cosine roll-off is a separately specified nominal response. Treating those two categories as one exact "allocation-fill" identity would impose a relation the standard does not state.

The exact symbol rate is inherited from the 4.5-MHz picture–sound spacing and the NTSC line-rate relation. A/53 specifies

$$R_s = 684\left(\frac{4.5\text{ MHz}}{286}\right) = \frac{1539}{143}\text{ MBd} \approx 10.762238\text{ MBd}. \tag{3.2}$$

Equivalently, because 4.5 MHz $= 3B_c/4$ for a 6-MHz allocation,

$$R_s = \frac{513}{286}B_c, \qquad B_N = \frac{R_s}{2} = \frac{513}{572}B_c \approx 5.381119\text{ MHz}, \tag{3.3}$$

where B_N is the Nyquist bandwidth associated with the symbol clock. The standard's framing hierarchy is synchronized to the same clock: a data segment contains four segment-sync symbols and 828 data symbols, and the coded payload carried by those 828 symbols corresponds to one Reed–Solomon-protected transport packet [2]. These relations establish the exact timing used by the pilot-frequency calculation below; they do not determine the spectral roll-off.

A/53 separately specifies a nominal linear-phase transmitter RRC response with roll-off

$$\alpha = 0.1152. \tag{3.4}$$

The concatenated transmitter and receiver responses are therefore nominally a raised cosine. Using the exact B_N above gives $B_N(1+\alpha) \approx 6.001024$ MHz. The approximately 1-kHz difference from 6 MHz is not a physical excess to be explained by a new constraint; it is the consequence of combining an exact clock with a nominal roll-off quoted to four decimal places. The standard itself depicts the channel occupancy at rounded scales of 5.38 MHz and 0.31-MHz transition regions [2]. The transmission mask, not an inferred rational value of α, is the regulatory boundary. No later detector calculation depends on the skirts filling the allocation exactly.

The data stream is randomized, error-corrected, interleaved, and trellis-coded before mapping to levels $\{\pm1, \pm3, \pm5, \pm7\}$. In a narrow spectrometer cell and over the short frame used here, the resulting payload is therefore well approximated as broadband noise-like power, although it is not mathematically Gaussian and the standard also contains deterministic segment- and field-synchronization sequences. Those qualifications are important for claims about alternative detectors. They do not change the narrower design fact used here: the continuously present pilot concentrates coherent energy far more strongly in frequency than either the randomized payload or the distributed synchronization structure.

3.3 The Suppressed-Carrier Pilot as a Detection Beacon

A/53 inserts a small in-phase pilot at the suppressed-carrier frequency, describing its position nominally as 309 kHz above the lower band edge and providing an exact frequency relation in the accompanying note [2]. Expressed through the exact symbol rate, that relation is

$$\Delta f_{\text{pilot}} = \frac{B_c}{2} - \frac{R_s}{4} = B_c\left(\frac{1}{2} - \frac{513}{1144}\right) = \frac{59}{1144}B_c = \frac{177}{572}\text{ MHz} \approx 309.440559\text{ kHz}. \tag{3.5}$$

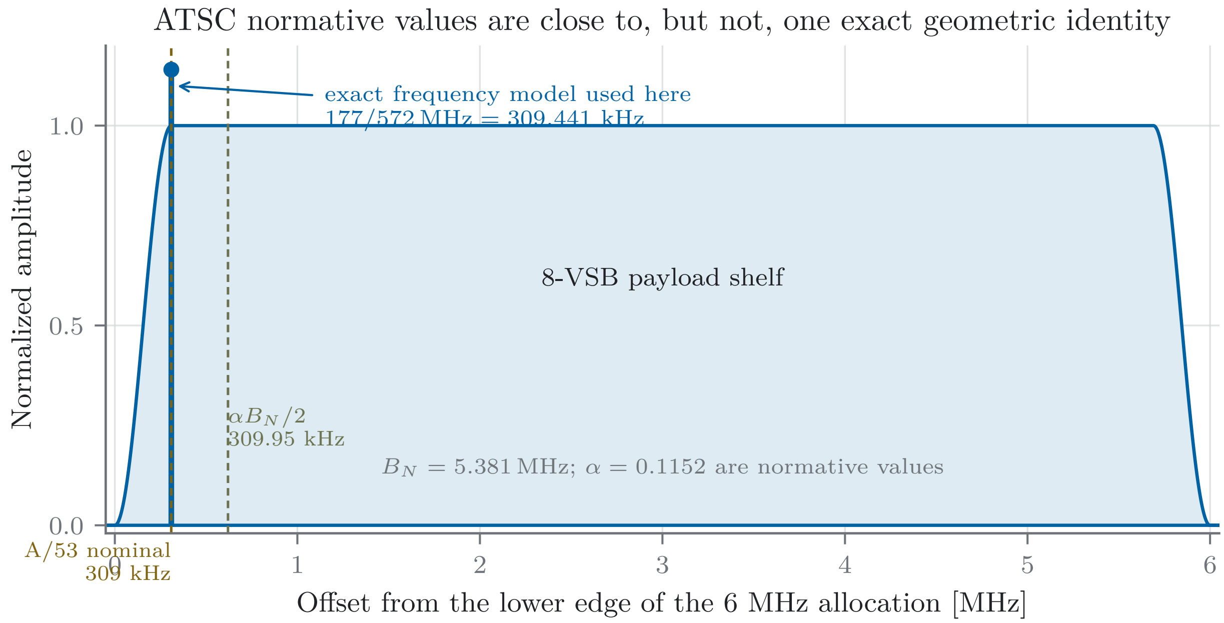

Figure 3.1: ATSC A/53 normative values and the near-coincidence that must not be promoted to an exact identity. The standard's exact pilot-frequency relation gives 309.441 kHz above the lower allocation edge (nominally 309 kHz), while the half-amplitude point implied by the separately specified nominal roll-off $\alpha = 0.1152$ lies near 309.95 kHz. The ∼0.51-kHz separation is negligible for the detector's measured-anchor strategy but invalidates the earlier claim that the pilot is exactly the −6-dB point because of a common integer factor. The spectral envelope is schematic; the regulatory transmission mask is the governing boundary.

For physical channel c,

$$f_{\text{pilot}}^{(c)} = f_{\text{edge}}^{(c)} + \frac{177}{572}\ \text{MHz}. \tag{3.6}$$

The weight generator stores this offset rounded to the hertz, 309.441 kHz. The resulting 0.44-Hz representation error is far below the 11.9-Hz fine-bin spacing and is absorbed by the per-channel anchor measured from the survey.

The exact frequency relation should not be confused with an exact roll-off identity. Under the independently specified nominal $\alpha = 0.1152$, the half-amplitude point of the lower composite raised-cosine transition lies at approximately 309.95 kHz. The pilot is therefore close to that point, as the standard's rounded 0.31-MHz drawing suggests, but not exactly fixed to it by the same rational integer. Figure 3.1 makes the distinction visible.

The pilot is synthesized by adding a DC level 1.25 to every composite symbol (data and synchronization) before suppressed-carrier modulation. For the nominal equiprobable eight-level data model, $\langle a_n^2 \rangle = 21$, so the pilot-to-data power ratio is

$$\frac{P_{\text{pilot}}}{P_{\text{data}}} = \frac{1.25^2}{21} = \frac{25}{336} \approx 0.0744, \tag{3.7}$$

consistent with the standard's statement that the pilot is approximately 11.3 dB below the average data signal power [2]. This is a nominal mean-power relation, not a statement that an arbitrary received 6-MHz spectrum must reproduce the ratio without scatter.

The corresponding transmitter-side proxy is

$$P_{\mathrm{data}} = \frac{336}{25} P_{\mathrm{pilot}}, \qquad S_{\mathrm{shelf}} \approx \frac{336}{25} \frac{P_{\mathrm{pilot}}}{B_N}, \tag{3.8}$$

where the second expression adopts a flat-shelf approximation over the Nyquist band. Concentrating the pilot power in a spectral line gives it roughly 56 dB more power spectral density than the payload per hertz. That concentration, continuous presence while the transmitter radiates, and nominal frequency standardization make it the appropriate low-cost beacon for this work.

Equation (3.8) is the starting model for the word "proxy," not its validation. Frequency-selective propagation, transmitter filtering, and the receiver bandpass can change the received pilot-to-shelf ratio across a 6-MHz allocation. The archive products used here observe only the pilot-containing coarse channel and therefore cannot measure that transfer. Chapter 9 carries it explicitly as a conditional model, and Chapters 7 and 8 specify the simultaneous across-allocation spectra required to turn it into a measured distribution. The matched-filter optimality statement is likewise limited to the declared known-tone Gaussian model; the complete calibrated detector and competing cyclostationary or decoding-assisted methods remain empirical comparisons.

3.4 The Transmitter Population & the Propagation Environment

Equation (3.6) fixes where a pilot *should* be; what stands behind the phrase "to within transmitter tolerance" is a population, and its structure turns out to matter as much as the standard itself. The population has two tiers, and the regulatory record explains why. Full-service ("primary") stations are high-power facilities whose analog-era rule held the visual carrier to ±1 kHz [12], and whose co-channel coordination practice — precise frequency offsets negotiated so that mutual interference between high-power neighbors interleaves rather than beats — holds them to tens of hertz in practice: the discipline is enforced less by rule than by the mutual protection of stations that can hurt each other. Translators, boosters, and low-power stations ("secondary" services under Part 74) are rebroadcast relays filling terrain shadows, historically licensed under tolerances orders of magnitude looser and typically built on aging or low-cost frequency synthesis; they protect full-service stations by displacement, not by precision, so nothing in their operating economics rewards a disciplined carrier. The current rules sharpen this asymmetry to its logical endpoint: today's §73.1545 assigns carrier departure tolerances to AM, FM, and international stations only — television is absent from the list — and the Part 74 tolerance section, §74.761, is *[Reserved]*: deleted [13, 14]. Frequency discipline in the modern DTV band is therefore a matter of coordination incentive rather than regulation, and the incentive binds exactly one tier. The expected offset population is a tight core of disciplined primaries surrounded by a broad skirt of undisciplined secondaries.

That expectation was tested directly, because the detector's window geometry depends on it. A regional station-list summary was assembled from the RabbitEars master compilation of the FCC's Licensing and Management System and ISED Canada's spectrum authorizations, with each entry's service class and distance to the site. The narrower summary contains 132 records across the DTV band within 500 miles: 49 full-service primaries and 83 secondaries — *sixty-three percent of that list is the undisciplined tier* — and Table 3.1 tabulates the first-measured ten channels. For mapping, the reproducible PilotProxy reduction applies the full on-air ATSC 1.0 and physical-channel normalization: it expands multi-channel facilities and merges shared physical carriers, yielding 499 emitter-channel records at 162 distinct source range–bearing sites. The Canadian side of that reduction is verified row-by-row against ISED's licence records and corrected where they disagree: matched rows take the licensed transmitter site — the compilation had anchored several entries on the community instead, and nearest to the telescope the Penticton channel-30 rebroadcaster CHKL-1 stands at 37.8 km, not the compiled 17.7 km, in agreement with the archived RabbitEars study printout — two relays with no current licence are dropped, nine analog-listed low-power stations holding digital licences enter, and every matched row carries the licensed effective radiated power. The two counts answer different questions and are not interchangeable. Figure 3.2 therefore shows the complete normalized 500-mile field while retaining the individual-record 120-mile detail. The population's spectral face is measured independently from the survey products themselves: each product stores the archive-averaged 16384-point spectrum of its coarse channel, and Figure 3.3 plots that average around each pilot. The channels sort into two clear morphologies. Where the local field is quiet, the average is a compact cluster — at most a resolved companion or shoulder inside the fine span and an isolated narrow line or two beyond it (channels 27, 28, 29, 34, 36; channel 29's cluster rides on a visibly tilted baseline, the odd-order structure the symmetric reference pair of Chapter 4 cancels exactly). Where the field is crowded, a *main lobe* at the dominant transmitter's offset stands over a resolved *forest of sidelobes* spanning several kilohertz (channels 30, 31, 32) — structure that is transmitter census, not propagation. The prior working model — a few stations per channel whose apparent frequency and power were pushed around by aircraft reflections — does not survive this measurement: the sidelobes are too narrow, too stable, and too numerous to be Doppler-smeared images of one carrier, and their count tracks the census.

The census and the spectra then cross-validate against every independent handle this work has. The survey-measured anchors of Section 5.5 concur — per-channel residuals of hundreds of hertz to 1.4 kHz, stable within eras, stepping when the dominant facility changes (Section 9.10). Occupancy tracks proximity: the five channels masked above 98% (27, 28, 30, 31, 34) all have a facility within 130 km — channel 27's is the set's nearest full-service primary, CHBC-DT at 60 km — and channel 30 — whose shelf sits at system noise, the loudest measured — hosts a rebroadcaster *in Penticton itself*, with six of its eight facilities secondary. Channel 33, the one coarse-threshold-feasible channel, has the sparsest field: three facilities, all disciplined primaries, none closer than 241 km great-circle — and its averaged spectrum is *dominated* by a second primary at its own offset, $\approx$3.6 kHz below the synthesized pilot and standing 16 dB over the lobe at the pilot position in the archive average, which is the measured origin of the spread-across-schedules fine structure that its wide-null diagnosis chases (Section 9.11). Channel 36 is the instructive exception that separates the two causes of occupancy: its nearest facility is 277 km out, yet it is masked 99.9% of the time and carries the largest intra-day share in the set — occupancy delivered by propagation rather than proximity, which is the bridge to the mechanisms below. It is this measured envelope — primary core, secondary skirt, and the Doppler smear below — that sizes the analysis window of Section 4.1.

Two propagation mechanisms then modulate what the census delivers, and they are worth distinguishing carefully because they act on different timescales and different terms of the residual budget. *Tropospheric ducting*: the radio refractive index of the lower atmosphere decreases with height, and when temperature inversions or sharp humidity gradients steepen that lapse past the critical value, a layer forms that refracts UHF energy back downward faster than the Earth curves away — a waveguide along the surface [15]. A ducted path delivers transmitters from hundreds of kilometers beyond the horizon at strengths that can rival line-of-sight, and the duct's formation and collapse follow the meteorology: evolution over tens of minutes to hours, organized diurnally and seasonally, and uncorrelated with the sidereal frame. *Aircraft scatter*: an airframe is a moving bistatic reflector, and the scattered path carries a Doppler shift bounded by $2v/\lambda$ — the same signals and Doppler scales that DTV-based passive radar exploits deliberately [16]. Table 3.2 evaluates the bound by traffic class at the band's midpoint: the civil envelope tops out near ± 1 kHz at jet cruise speed, inside the detector's fine span; military speeds exceed the span but are rare in this airspace; satellite forward scatter lands tens of kilohertz out — far outside the capture window — and transits

Table 3.1: The narrower 132-record regional station-list summary on the first-measured ten channels: records per physical channel, split by service tier (full-service primaries versus the translator/rebroadcaster/low-power secondary tier), with each channel's nearest listed facility. Compiled from the RabbitEars master station list (FCC LMS and ISED records); primary-station distances are great-circle to the verified licensed site. Across channels 14–36, this summary contains 49 primary and 83 secondary records. It is distinct from the normalized 499-row emitter-channel export used by Figure 3.2.

	facilities	primary	secondary	nearest facility	
ch 27	6	2	4	CHBC-DT (pri.)	60 km
ch 28	5	2	3	CH5533-DT (sec.)	130 km
ch 29	6	2	4	K29ED-D (sec.)	401 km
ch 30	8	2	6	CHKL-1 (sec.)	in Penticton
ch 31	7	1	6	CH4491-DT (sec.)	60 km
ch 32	7	3	4	CH5534-DT (sec.)	130 km
ch 33	3	3	0	CKVU-DT (pri.)	241 km
ch 34	6	2	4	CH4490-DT (sec.)	60 km
ch 35	5	2	3	CBUT-DT (pri.)	242 km
ch 36	4	1	3	KZJO (pri.)	277 km

Table 3.2: Maximum bistatic Doppler $2v/\lambda$ by reflector class, evaluated at the DTV band midpoint (581 MHz, $\lambda = 0.516$ m); values scale by ±5% across 554–608 MHz. The civil-aircraft envelope sits inside the ±1.526 kHz fine span; satellite scatter lands far outside it and transits the beam in seconds, so its contribution is confined to the fast (sub-acquisition) share of the timescale decomposition.

reflector class	typical speed	max Doppler
light general aviation	50 m/s	±0.19 kHz
turboprop / charter	130 m/s	±0.50 kHz
commercial jet, cruise	250 m/s	±0.97 kHz
military fast jet	600 m/s	±2.3 kHz
satellite, low Earth orbit	7.6 km/s	±29 kHz

in seconds. The observable signature is twofold: a frequency-smeared scattered component, and *amplitude flutter* — the beat between the direct and scattered paths at their Doppler difference, familiar from the analog era as "airplane flutter" — with fading on timescales of seconds.

Neither mechanism is directly proven from the survey products, and this subsection does not claim to have separated them. What can be said is that they are separable in principle by timescale, and that the decomposition of Section 9.4 already measures the relevant split: flutter's seconds-scale fading lands in the sub-acquisition (fast) share, which carries no coherence amplification and is benign; ducting's slow wander lands in the intra-day share, the term that carries n_{coh} and dominates every failing budget. The measured structure — intra-day shares that dominate the surviving power on most channels, and correlation times of 34 and 45 minutes where the estimator succeeds, timescales natural to duct evolution and impossible for flutter — is circumstantial evidence that ducting is the principal slow modulator and Doppler flutter a fast-share contributor. A definitive attribution — correlating shelf power against refractivity soundings, or the fine spectra against transponder-derived aircraft tracks — is specified for the closing campaign rather than claimed here.

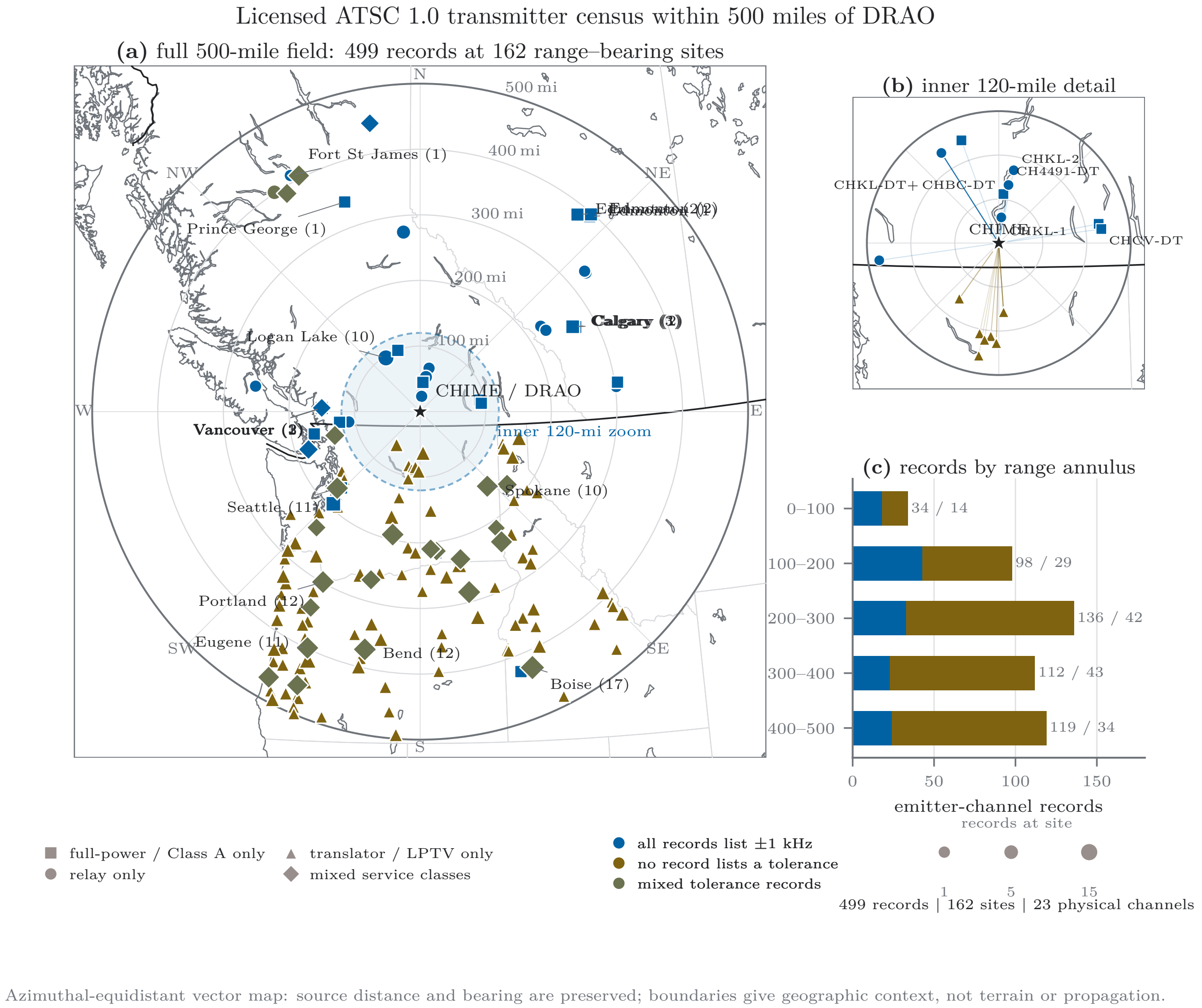

Figure 3.2: The normalized licensed-transmitter census within 500 miles of DRAO. **(a)** The complete export contains 499 emitter-channel records at 162 distinct source range–bearing sites; exactly coincident records are aggregated for legibility, with marker area retaining their multiplicity. Shape records the site's service-class composition and color records whether its entries list a ±1 kHz transmitter-frequency tolerance. **(b)** The inner 120-mile individual-record detail preserves the view used to diagnose the nearby field. **(c)** Emitter-channel records by 100-mile annulus; annotations give records/sites. The DRAO-centred azimuthal-equidistant vector map preserves the supplied distances and bearings. Coastlines and administrative boundaries provide geographic context but intentionally do not imply terrain screening or propagation.

3.5 Scope, Standard Evolution, & Generalization

The weight profiles of Chapter 4 are specific to the A/53 standard: the implemented single-tone method detects the 8-VSB suppressed-carrier pilot and nothing else. Two boundaries of that scope should be stated explicitly. First, the broadcast standard itself is evolving. ATSC 3.0 replaces 8-VSB with an OFDM physical layer whose synchronization features — a deterministic bootstrap preamble and embedded pilot subcarriers — are structured differently, and the implemented single-tone profiles do not track that transition. The per-channel census and epoch structure of the survey accommodate a gradual transition naturally (a converted channel leaves the A/53 census), but detection of ATSC 3.0 emissions would require new profiles. Second, the method does not transfer geographically merely because the backend hardware does. An instrument sharing this digital signal chain but operating in a different regulatory domain — HIRAX in South Africa, for example — faces a different broadcast standard (DVB-T2, an OFDM system) on a different channel raster, so the weight bank is regulatory-domain data in the same sense that the calibration constants are channel data.

Porting to another receiver. The constraint equations of Chapters 4 and 5 are receiver-independent; the numerical values are not, and it is worth being explicit about which is which, because the natural assumption — that a different telescope differs only by a spectral-sense convention — is wrong in a way that fails quietly. The sense *is* a convention, fixed per receiver profile, and a sense error announces itself: the synthesized tone lands at the mirror frequency, the pilot is not matched at all, and the channel simply never fires. What must be re-derived is subtler. The coarse-channel rate f_s sets the tap length through the census requirement, since what that census fixes is the analysis window's width in *frequency*: a receiver whose raster is half as wide needs half the taps to present the same window, and holding K fixed instead silently halves both the window and the unambiguous fine span. The frame length N follows the receiver's visibility-integration length, and with K determines the sub-frame count L and hence which member of the transform family of Section 6.3 is required. The channel raster sets each pilot's baseband offset δf_c and normalized tone frequency ν_c, so the weight bank is regenerated per profile rather than transferred. The input count M enters the degrees of freedom that set the null width. Only the beacon itself — its existence, its standardized offset $\Delta f_{\mathrm{pilot}}$, and the shelf proxy relation of Eq. (3.8) — is a property of the broadcast standard and carries over untouched.

The same observations define the method's generalization path. Nothing in the architecture is specific to a single sinusoidal beacon: the requirements on a detection feature are that it be standardized in frequency, spectrally concentrated or otherwise deterministic, and present whenever the transmitter radiates. OFDM broadcast standards satisfy these requirements through different structures — continual pilot subcarriers at fixed carrier indices with known modulation, and repeated deterministic preambles — and a generalized weight profile matched to a pilot comb rather than a single tone is the natural extension. This is not achievable in the current single-tone form, and it is left as future work; the framework of Chapters 4–6 (constraint-derived parameters, exact arithmetic, rank-based calibration against measured nulls) would carry over unchanged.

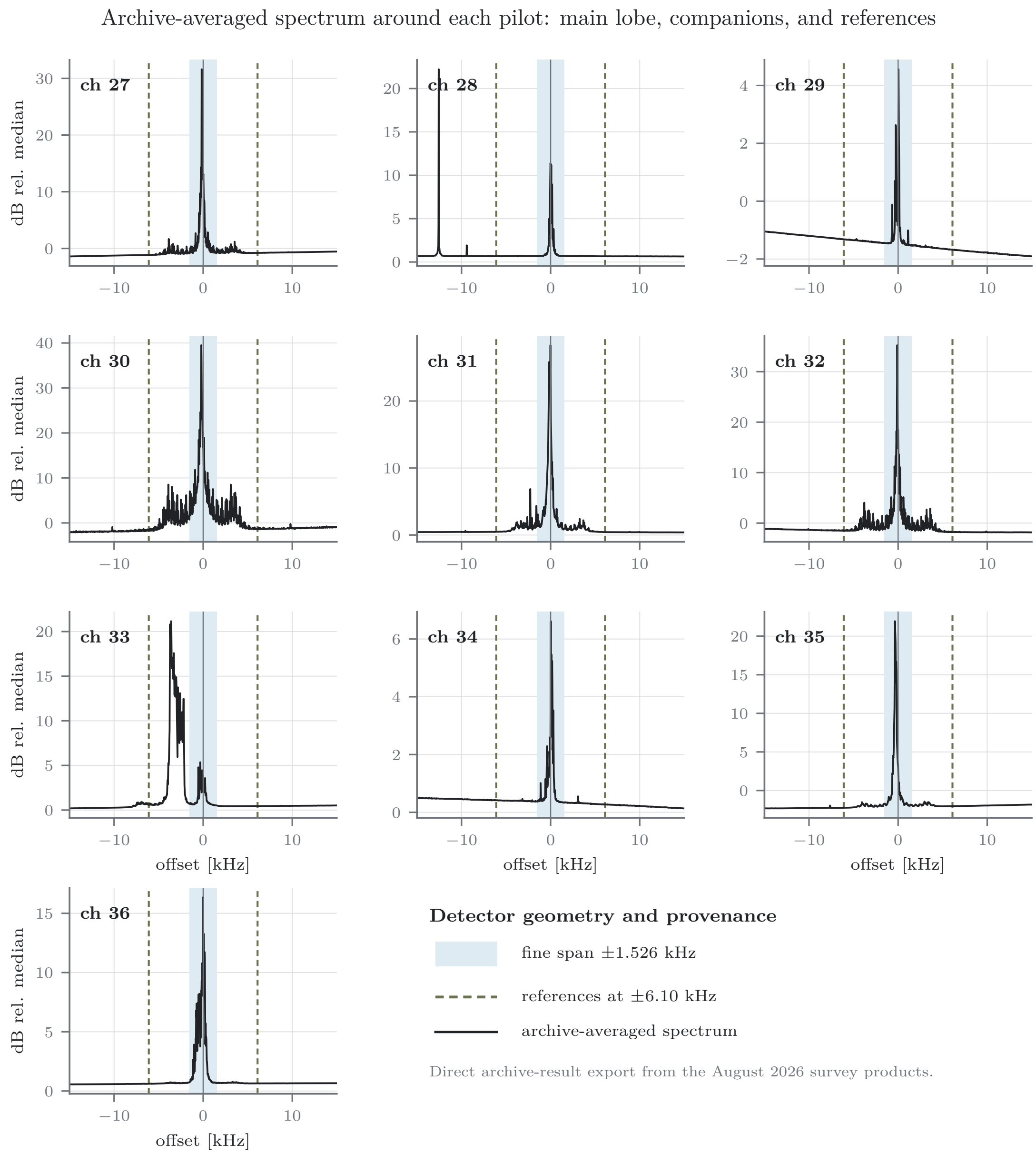

Figure 3.3: The transmitter population's spectral face: archive-averaged spectrum within ±15 kHz of each pilot, from the survey products' stored integrated spectra (dB relative to the channel median). Shaded: the $K = 128$ fine span (±1.526 kHz); dashed: the ±2-bin reference placements (±6.1 kHz). Panels are centred on each channel's *synthesized* (nominal) pilot in the transmitted-frequency sense, so a carrier's displacement from zero is its measured off-nominal offset. On eight of the ten channels the dominant carrier is a single compact lobe inside the fine span, within 0.4 kHz of synthesis; on the crowded-field channels (30, 31, 32) it stands over the resolved sidelobe forest discussed in the text. Channel 33 is the documented exception: its dominant carrier sits at −3.6 kHz, in the skipped guard between the fine span and the lower reference, with companion structure near −2.6 kHz — the off-nominal blind spot whose diagnostics Appendix B records. Channel 28's strongest in-window feature lies 12.6 kHz below synthesis, far outside the capture geometry, while its nominal pilot stands at 11 dB. The references sit in clean territory on every channel — the placement validation of Section 4.1 made visually. The thirteen lower-band channels appear in Figure 9.7.

Chapter 4

System Parameterization & Weight Profile Synthesis

Every dimension of the data frame $\mathbf{X} \in \mathbb{C}^{M \times N}$ and of the weight matrix $\mathbf{W} \in \mathbb{C}^{K \times R}$ is fixed by an explicit operational requirement — the array hardware, the correlator's framing contract, the fixed-point arithmetic budget, or the spectral structure of the interference being detected — rather than left as a free tuning parameter. This section derives each parameter from its governing constraint (Section 4.1), then constructs the weight profiles themselves: the target matched filter synthesized from the broadcast standard's pilot carrier (Section 4.2) and the displaced reference profiles that furnish the background estimate (Section 4.3). A recurring theme, developed fully in Chapter 6, is that the entire datapath downstream of these weights is exact integer arithmetic; several parameter choices below exist precisely to make that exactness possible.

4.1 Architectural Trade-Offs & Parameter Justifications

1. **Telescope Feed Count ($M = 2048$):** The spatial dimension $M = 2^{11} = 2048$ is strictly hardware-defined, corresponding to the total number of independent spatial inputs processed by the telescope array: four cylindrical reflectors, each instrumented with 256 dual-polarization feeds, giving 1024 antennas and $M = 2048$ analog chains [1]. The detector does not require per-feed phase calibration: as developed in Section 5.4, all cross-feed combining is non-coherent by design. The analytic null model initially treats the inputs as independent noise looks, but this is an approximation rather than a hardware identity; common sky signal, receiver coupling, and shared gain structure can reduce the effective degrees of freedom. The empirical null calibration therefore remains authoritative, and a release-quality calibration reports the effective-width inflation relative to the independent-feed model.

2. **Correlator Frame Synchronization ($N = 16384$):** The temporal frame length is fixed at $N = 2^{14} = 16384$ complex samples, the design value of the upgraded correlator engine's integration frame contract; at the receiver coarse-channel rate $f_s = 390.625$ kHz (the 400 MHz observing bandwidth divided into 1024 channelizer channels [1]) this corresponds to a frame cadence of 41.94 ms. Matching the correlator frame one-to-one is a structural decision, not a convenience: the detector's product is a per-frame rejection flag applied to the visibility integration of *the same samples*. Any mismatch between detection frames and correlator frames would force one of two failure modes — interpolating flags across partially overlapping integrations (contaminating clean integrations with flagged fractions, and vice versa), or

buffering and re-framing the baseband (adding latency and a second framing contract to keep synchronized). The contract that eliminates both is that a decision must span an *integer* number of correlator integrations: a flag either applies to an integration or it does not, with no partial-frame semantics. The target upgraded interface is the 1:1 case. A receiver whose visibility integration is shorter than its detector frame realizes an n:1 case, which is equally free of partial-frame semantics but coarsens the excision quantum by n — a real cost against intermittent interference, and the reason N is specified as an interface value rather than as a universal constant. The published correlator integrates visibilities over 12,288 channelized samples ($\approx$ 31 ms) [1], whereas $N = 16384$ is the upgraded-engine design used and benchmarked here. Chapter 10 therefore treats frame alignment as an integration acceptance gate, not as an already demonstrated live property; nothing in the detector construction depends on the specific value beyond the power-of-two tiling discussed next.

3. **Exact-Arithmetic and Geometric Weight Length ($K = 128$):** The tap length is set by exact-arithmetic and geometric constraints rather than by a sensitivity optimum, and it is *derived* rather than chosen: K is the power of two whose sub-frame rate f_s/K meets the analysis-window requirement set by the measured transmitter population of Section 3.4, subject to the fixed-point ceiling of item (c). At CHIME's $f_s = 390.625$ kHz that evaluates to $K = 2^7 = 128$; the same requirement on a receiver whose coarse raster is half as wide gives $K = 64$, since what the population fixes is the window *width in frequency*, not the tap count. Sensitivity does not constrain this parameter: because the coherent spectral integration of Section 5.3 restores coherence across the full frame regardless of how the frame is partitioned (the product $KL = N$ is fixed), the detector's deflection is governed by N to first order; K selects only the sub-frame geometry — where the boundary falls between the matched-filter projection and the cross-window transform. Within that freedom, $K = 128$ is pinned by four requirements:

   (a) *Exact frame tiling.* K must divide N exactly with both factors powers of two, giving $L = N/K = 128$ complete sub-frames with no residual samples and radix-2-friendly transform lengths at every stage.

   (b) *Fine-span coverage of the measured offset population.* The sub-frame rate $f_s/K \approx$ 3.052 kHz sets the unambiguous span of the fine spectral axis at $\pm f_s/(2K) \approx \pm 1.526$ kHz. Because the synthesized tone absorbs each pilot's nominal channelization offset exactly (Section 4.2), this span need only cover the *residual* offsets that actually occur — and that requirement is empirical, fixed by the census of Section 3.4: the disciplined primary core, the undisciplined secondary skirt, and the $\approx \pm$ 1 kHz aircraft-Doppler envelope on scattered components. $K = 128$ covers that population with margin, and the survey-measured anchors certify the sizing on archived telescope data: the largest measured residual, -1.37 kHz on channel 32, sits at 90% of the span edge — inside $K = 128$'s span, *outside* the ± 0.763 kHz span of the next radix-2 step $K = 256$, which would have clipped a measured primary outright (and which item (c)'s conservative bit bound excludes independently). The step down closes the bracket from the other side: $K = 64$ widens the span to ± 3.05 kHz, but the reference displacement is ± 2 bins *of the K-tap grid* — $\pm 2f_s/K$ in frequency — so halving K doubles the references' distance to ± 12.2 kHz, quadrupling the surviving curvature term of item 4's slope cancellation and moving the background samples away from the locally smooth neighborhood the ratio statistic requires. The radix-2 bracket around $K = 128$ is thus closed on both sides by measurement rather than preference: 256 clips the population the census maps, 64 degrades the background estimate the ratio needs, and 128 covers the one while preserving the other. Both disqualifications are statements at CHIME's f_s rather than about tap counts in isolation: on a receiver whose coarse raster is half as wide, $K = 64$ reproduces exactly the span and reference displacement in frequency that $K = 128$ realizes here, per the derivation above. One honest boundary of the coverage claim is visible in Figure 3.3: on the most crowded channels a minority of sidelobe structure falls *outside* the span (channel 33's second primary at ≈ -4 kHz is the clearest case), and no radix-2 value covers it without ruining the reference geometry — $K = 32$ would reach it at the price of references ± 24 kHz out. The window is sized to capture the *dominant* emitter on every channel; out-of-span structure aliases across the fine axis and is handled where it belongs, by the decision architecture (bulk, census exclusions, and the rank's contamination immunity, Section 5.5) rather than by the window. A final constraint completes the choice: K is a single band-wide constant, not a per-channel one. A per-channel tap length would fragment the kernel geometry, the weight-bank contract, the transform family, and the verification corpus into as many variants as channels — operationally undesirable for the single-geometry implementation evaluated here — so one value must serve the whole monitored band, and it is selected for the majority of channels with the weight placed on those whose potential science return is largest. That weighting is visible in the outcome: the channels $K = 128$ serves completely include every channel that is threshold-feasible under the current conditional forecast of Section 9.8, and the imperfectly-served minority structure lies on channels whose verdicts do not hinge on it.

(c) *Fixed-point bit-growth closure.* The exact integer datapath requires the complete dot product to fit fixed-width accumulators with no possibility of overflow. With 4-bit signed components, samples lie in $[-8, 7]$ and quantized weight components in $[-7, 7]$ (the receiver's own quantizer is in fact 15-level mid-tread, emitting only $[-7, 7]$ [3]; the datapath bound conservatively admits the full two's-complement range), so one complex-product component is bounded by $|x_r w_r| + |x_i w_i| \le 8 \cdot 7 + 8 \cdot 7 = 112$; accumulating $K = 128$ taps bounds the completed dot-product component at $128 \times 112 = 14336 < 2^{14}$. Equivalently: a 4×4-bit product pair contributes $4 + 4 + 1 = 9$ bits, and $\log_2 K$ accumulation guard bits bring the total to $9 + \log_2 K$ — exactly 16 signed bits at $K = 128$, the natural register width, with headroom. Each downstream stage carries its own bound derived from the same quantity: the coarse squared magnitude $2(128K)^2$ against the 32-bit accumulator, and the transform's per-stage growth against the same width. In the implemented kernel these are not prose but `static_assert`s evaluated per geometry, so the build — not a reader — answers whether a given K closes. The signed-32-bit squared-magnitude bound is the binding one, admitting $K \le 292$ under the true weight range and $K \le 128$ under the conservative bound used here; the guard-bit count is asserted equal to $\log_2 K$ so it cannot silently desynchronize from the window length. The 4-bit grid itself is not the detector's choice but the receiver's: the F-engine rounds its channelized output to packed 4+4-bit complex samples, applying per-channel complex gains beforehand to load that grid optimally [1, 3], and the detector adopts the same grid for its weight coefficients so that samples and weights share a single packed integer datapath.

(d) *Kernel geometry.* On the benchmark GPU, the packed `int4` dot product consumes two complex samples per fused instruction, so K presents $K/2$ packed tap-pairs. In the *staged* projection entry points these are distributed across the block, and $K = 128$ against a 64-thread block gives exactly one pair per thread with no inactive lanes. The implemented *fused* kernel distributes the window axis instead (Section 6.5): each thread owns whole windows and iterates the $K/2$ tap-pairs serially, so its block geometry is constrained by the window count L rather than by K, and K enters only as an inner trip count. This is the weakest of the four constraints and the only one that does not bind the fused form.

These four constraints concern the exactness and efficiency of the downstream arithmetic rather than detection statistics: sensitivity is set by the frame length and the coherent transform, while K is selected so that the implementation can be verified at the bit level.

4. **Symmetric Reference Count ($N_{\mathbf{ref}} = 2$) and Slope Cancellation:** The number of reference profiles is fixed at $N_{\text{ref}} = 2$, placed symmetrically about the target. Three independent justifications converge on this choice, and the alternatives they exclude are stated at the end of this item. The cancellation argument is local: it assumes that the two references sample the same sufficiently smooth response in a coordinate for which the placement is symmetric. A narrow instrumental line, an edge, or any other non-smooth or asymmetric feature inside the span is not cancelled by symmetry and must be handled empirically.

   First, *baseline cancellation.* In astrophysical and instrumental RF environments the background power spectral density across nearby bins is smooth but not flat; expanding it about the target bin as $P_{\text{bg}}(k) = P_0 + \alpha_1 k + \alpha_2 k^2 + \alpha_3 k^3 + \cdots$, a single reference displaced by Δk estimates P_0 with a first-order error $\alpha_1 \Delta k$. A symmetric pair at $\pm\Delta k$, averaged, cancels *every odd-order term identically*:

$$\frac{1}{2}\left(P_{\text{bg}}(+\Delta k) + P_{\text{bg}}(-\Delta k)\right) = P_0 + \alpha_2 \Delta k^2 + \alpha_4 \Delta k^4 + \cdots, \tag{4.1}$$

   leaving a leading smooth residual of second order, $\alpha_2 \Delta k^2 = 4\alpha_2$ at the implemented $\Delta k = 2$. For a locally smooth response sampled symmetrically, linear tilt and all other odd Taylor terms cancel algebraically; even-order curvature remains. Discrete or asymmetric structure is outside this Taylor model and is diagnosed from the recorded spectra rather than claimed to cancel.

   Second, *statistical sufficiency.* The reference estimate enters the test statistic's denominator with $\beta N_{\text{ref}} M$ degrees of freedom (Section 5.5); at $M = 2048$ the two-reference denominator already carries 8192 degrees of freedom, for a relative standard deviation of $\sqrt{2/8192} \approx 1.6\%$. Adding a third reference would narrow the idealized null from 2.71% to 2.55% — a 6% improvement — at the cost of a 33% increase in projection compute and weight-bank storage, moreover, it would provide no practical improvement in the measured operating regime: the *measured* null is inflated several-fold above the i.i.d. model by receiver systematics (Section 5.5), so idealized-variance refinements are not the binding constraint. $N_{\text{ref}} = 2$ is therefore the minimum count providing exact odd-order cancellation under the stated local-smoothness model; additional references yield statistically negligible improvement at material computational cost.

   Third, *the reference-window sizing law.* How many reference cells a CFAR detector should carry is a solved problem in the radar literature: the penalty for estimating one's own threshold (the *CFAR loss*) falls with reference count along a universal curve, and the window is sized at that curve's knee [8, 9]; the same law governs training-sample counts in adaptive detection through the Reed–Mallett–Brennan rule [10]. Applied to this geometry the curve has a closed form: the idealized null width of the ratio scales as $\sqrt{1 + 1/N_{\text{ref}}}$ relative to a perfect (infinite-reference) denominator — a detection-floor penalty of 0.9 dB at $N_{\text{ref}} = 2$, 0.6 at three, 0.5 at four, with 0.5 dB unrecoverable at *any* count because it belongs to the single target column, whose own fluctuation no reference reduces. The knee sits at two-to-three, and the reason it sits so low is the feed axis: a radar reference cell is a single look carrying $\beta = 2$ degrees of freedom, so classic windows need 16–32 cells, while each reference here is a full weight column pooled over $M = 2048$ inputs — $\beta M = 4096$ degrees of freedom per reference. The feed count does for this detector what the cell count does for radar, and $N_{\text{ref}} = 2$ sits where a radar designer's several-thousand-cell window would. Past the knee, the remaining idealized fraction of a decibel is not real in any case: the measured nulls are inflated 14–19× above the i.i.d. model by systematics (Section 5.5), so the operating width is carried by the contaminated tail, not by denominator degrees of freedom.

Three alternatives from the CFAR family were weighed and declined, each for a stated reason. *More references of the same kind* purchase the saturating curve above: +33% of the dominant projection cost per added column for at most 0.4 dB of idealized floor that the measured null structure erases. *Higher-order cancellation designs* — pairs at ±2 and ±4 bins combined with Richardson weights $(\frac{4}{3}, -\frac{1}{3})$, which null the quadratic term exactly and leave a fourth-order residual — pay the Savitzky–Golay tax: the curvature-free estimator's denominator carries 1.9× the variance of the plain pair, spending ≈0.6 dB of idealized floor against a term the ±2-bin locality already suppresses to second order; the measurement has since been made from the products' stored integrated spectra: fitting the smooth local background across a 5–12 kHz annulus around each pilot (median-filtered, sigma-clipped against discrete lines), the curvature bias at the reference distance is $|a_2\Delta^2| \approx 2 \times 10^{-3}$ of the background in the median — one fifth of the $\eta - 1 = 10^{-2}$ decision knee — and 3×10^{-3} or below on every threshold-feasible channel, while the slope terms the pair already cancels *exactly* run an order of magnitude larger (up to 4×10^{-2} on channel 29: the visible baseline tilt of Fig. 3.3). Where the pair's estimate is biased beyond this (the crowded-field channels, up to 1.4×10^{-2} on channel 32), the excess is discrete secondary structure, not smooth curvature, and the Richardson design removes essentially none of it — so the measurement closes the question in the plain pair's favor and redirects the residual concern to the censoring option below, which is built for exactly that failure mode. *Censored and ordered reference variants* (greatest-of, smallest-of, censored-mean, and adaptively selected windows [9, 11]) require $N_{\text{ref}} \geq 3$ to have anything to vote over, and are the one alternative whose value here is qualitative rather than statistical: with two references, a contaminant landing on either biases the background estimate upward without bound, whereas a median-of-three is evaluated by comparisons alone — fully compatible with the exact integer datapath — and would extend to the reference axis the same order-statistic philosophy the fine decision applies along the spectral axis (Section 5.5). It is declined for the archive-compatible configuration on contractual rather than statistical grounds: a new reference is a new weight bank, a new μ_0, and a new survey epoch, and the 7.6-yr archive is an $N_{\text{ref}} = 2$ archive. A censoring third reference is recorded as the natural forward-epoch upgrade should the evolving broadcast environment (Section 9.10) warrant it.

5. **Dirichlet Kernel Guard-Band Separation ($\Delta k = \pm 2$):** The displacement of the references is set by the spectral response of the finite-length projection. The K-tap conjugate-tone projection of Section 5.1 applies a rectangular temporal window, so its normalized response to a tone at fractional bin offset δ from the target, observed at integer bin displacement Δk, is the periodic-sinc (Dirichlet) kernel obtained by summing the geometric phase series:

$$D_K(\Delta k - \delta) = \frac{1}{K} \sum_{k=0}^{K-1} e^{j\frac{2\pi}{K}(\Delta k - \delta)k} = \frac{\sin\bigl(\pi(\Delta k - \delta)\bigr)}{K \sin\bigl(\pi(\Delta k - \delta)/K\bigr)} \, e^{j\pi(\Delta k - \delta)(K-1)/K}. \tag{4.2}$$

At exact bin centers ($\delta = 0$) integer displacements are orthogonal and leak nothing — the

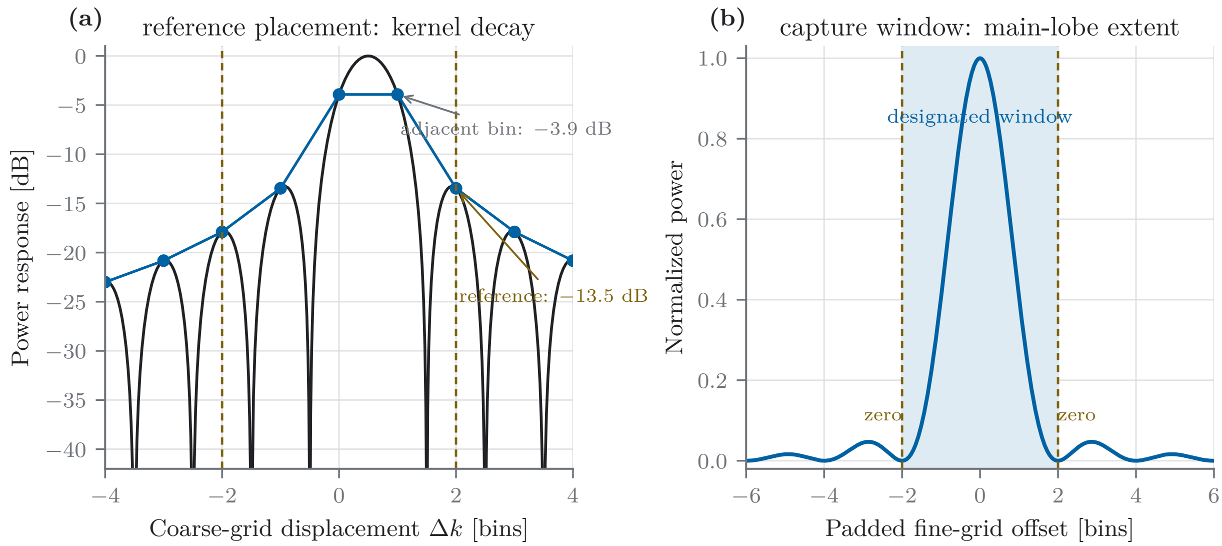

Figure 4.1: The two ± 2 displacements as dual applications of the Dirichlet response, Eq. (4.2). (a) On the coarse grid, the references are placed at the kernel's decayed skirts: a tone at worst-case half-bin offset leaks -3.9 dB into the adjacent bin but only -13.5 dB at ± 2, so the one-bin guard bounds self-contamination of the background estimate below 5% in power (Section 4.1). (b) On the padded fine grid, the window-axis kernel's main lobe spans exactly ± 2 bins, so the designated detection window of Section 5.5 is the main-lobe extent: an exclusion guard on one grid, a capture window on the other.

DFT basis property. The generic operating condition, however, is fractional: transmitter offset and drift place the pilot between bin centers. At the worst-case half-bin offset ($\delta = 1/2$), the immediately adjacent bin ($\Delta k = \pm 1$) receives $|D_K(1/2)| \approx 2/\pi$ of the tone amplitude (-3.92 dB, i.e. 40.5% of the power). References placed at ± 1 would therefore absorb up to 40% of the pilot's own power into the background estimate precisely when the pilot is present — inflating the denominator, deflating the detection ratio, and partially blanking the detector against exactly the signal it exists to find. At $\Delta k = \pm 2$ the worst-case leakage falls to $|D_K(3/2)| \approx 0.212$ (-13.5 dB, 4.5% of the power). Placing the references at ± 2 bins — leaving a one-bin guard — therefore bounds self-contamination of the background estimate below 5% in power under worst-case misalignment, while keeping the references close enough to preserve the local spectral covariance that the slope-cancellation argument requires. Figure 4.1(a) displays the kernel and both worst-case leakage points. The same kernel reappears in Section 5.3 as the fine-span edge rolloff of the sub-frame projection: Eq. (4.2) is the single spectral response function from which both the guard-band width and the fine-axis scalloping budget follow.

4.2 Target Profile Synthesis via DTV Pilot Matching

Section 3.3 derived the detection beacon from first principles: a stationary, coherent pilot line at the exact rational offset $\frac{177}{572}$ MHz ≈ 309.440559 kHz above each channel's lower allocation edge (Eq. (3.6); the implemented generator's hertz-rounded 309.441 kHz, as noted there, is absorbed by the measured anchors). What remains is to map each of the $C = 23$ pilot frequencies — UHF

Table 4.1: Pilot-to-channelizer mapping used by the evaluated configuration for the $C = 23$ monitored DTV channels: nominal pilot frequency, CHIME frequency-channel index (`freq_id`, the identifier carried by all data products in this work), RF-frame baseband offset $\delta f_c = f^{(c)}_{\text{pilot}} - f^{(c)}_{\text{center}}$ of the pilot from the channel center, and resolved reference-placement status (Section 4.3). Values are computed from the evaluated weight-generation configuration.

Ch.	Pilot (MHz)	`freq_id`	δf_c (kHz)	Placement
14	470.309441	844	−3.059	edge-wrapped (lower)
15	476.309441	829	+137.566	nominal
16	482.309441	813	−112.434	nominal
17	488.309441	798	+28.191	nominal
18	494.309441	783	+168.816	nominal
19	500.309441	767	−81.184	nominal
20	506.309441	752	+59.441	nominal
21	512.309441	736	−190.559	nominal
22	518.309441	721	−49.934	nominal
23	524.309441	706	+90.691	nominal
24	530.309441	690	−159.309	nominal
25	536.309441	675	−18.684	nominal
26	542.309441	660	+121.941	nominal
27	548.309441	644	−128.059	nominal
28	554.309441	629	+12.566	nominal
29	560.309441	614	+153.191	nominal
30	566.309441	598	−96.809	nominal
31	572.309441	583	+43.816	nominal
32	578.309441	568	+184.441	nominal
33	584.309441	552	−65.559	nominal
34	590.309441	537	+75.066	nominal
35	596.309441	521	−174.934	nominal
36	602.309441	506	−34.309	nominal

channels 14 through 36, on the raster of Section 3.1 — into the receiver's channelized baseband and synthesize the matched filter.

The receiver's channelizer partitions the band into coarse channels of complex bandwidth f_s; pilot c lands in a single coarse channel, and we define $\delta f_c = f^{(c)}_{\text{pilot}} - f^{(c)}_{\text{center}}$ as its baseband offset from that channel's center. The normalized tone frequency is

$$\nu_c = \frac{\delta f_c}{f_s} \pmod 1, \tag{4.3}$$

under the receiver's baseband frame convention. Table 4.1 lists the complete mapping for the evaluated receiver. Two features of the table merit note. First, the offsets span −190.6 to +184.4 kHz — nearly the full $\pm f_s/2$ — because the 6 MHz broadcast raster and the $f_s = 390.625$ kHz channelizer raster are incommensurate, so successive pilots precess through the coarse-channel span. (Offsets are quoted in the RF frame; the detector's spectrally inverted baseband convention negates them, per the sign remark below.) The synthesized tone absorbs this entire channelization offset *exactly*, which is what frees the fine spectral axis of Section 5.3 to resolve only residual transmitter error and slow drift within its ±1.5 kHz span. Second, the coarse-channel index *decreases* as RF frequency increases, reflecting the receiver's spectrally inverted channel ordering; this inversion is absorbed entirely by the weight sign convention noted below.

What the weights are. The synthesis follows from one observation, and it is worth stating before the algebra because it explains every property the profile has. A single bin of a discrete Fourier transform,

$$X[\kappa] = \sum_{k=0}^{K-1} x[k]\, e^{-j2\pi\kappa k/K}, \tag{4.4}$$

is an *inner product* between the data and a sampled complex exponential. Computing a whole transform means computing K of these against K different exponentials, which is what an FFT accelerates. But the detector does not need a whole transform: the broadcast standard fixes the pilot frequency exactly (Eq. (3.6)), so exactly one of those inner products carries the signal and the other $K-1$ are discarded work. Evaluating that one term directly is a single dot product of length K — and, because nothing forces the evaluation frequency onto the integer grid κ/K, it can be taken at the pilot's *actual* normalized frequency ν_c rather than at the nearest bin. The weight profile is nothing more than that one DFT basis vector, conjugated and sampled at the tap index.

Three consequences follow immediately, and they are the reasons the architecture works. The cost is one multiply-accumulate chain per window rather than a transform, which is what fits the detector inside a small fraction of the real-time budget. The evaluation is off-grid, so the channelizer offset δf_c of Table 4.1 is absorbed *exactly* instead of leaving a scalloping loss — and the fine axis downstream is then free to resolve only the residual transmitter error. And the basis vector is a compile-time constant: it depends on the standard and the receiver's raster, not on the data, so it can be quantized once, stored, hashed, and shared by every frame, which is what admits the exact integer datapath of Section 6.2.

This also fixes the two-stage structure of the pipeline. The single-term DFT is applied to each K-tap window, collapsing it to one complex number; the $L = N/K = 128$ resulting numbers, one per window, are then transformed *across* windows (Section 5.3) to resolve frequency finely. The first transform is one fixed term at a known frequency; the second is a full, short transform over the unknown residual. Neither does the other's work.

Let $\mathbf{w}^{(c)}_{\text{target}} \in \mathbb{C}^{K\times 1}$ denote the primary target weight profile ($r = 0$) synthesized for the c-th DTV pilot. Written out, it is the conjugated basis vector of Eq. (4.4) evaluated at ν_c and scaled to full fixed-point amplitude — a complex sinusoidal matched filter tuned to the pilot carrier rotation across the $K = 128$ tap sub-frame duration:

$$\mathbf{w}^{(c)}_{\text{target}} = \begin{bmatrix} w^{(c)}_{0,0} & w^{(c)}_{1,0} & \dots & w^{(c)}_{K-1,0} \end{bmatrix}^T, \qquad w^{(c)}_{k,0} = \gamma \, \exp\left(j2\pi\nu_c\, k\right), \quad k \in \{0, 1, \dots, K-1\}. \tag{4.5}$$

Here γ is the fixed-point full-scale amplitude: with b-bit signed components, $\gamma = 2^{b-1} - 1 = 7$ for the implemented `int4` representation. Driving every tap to full scale is the correct quantization policy for a constant-modulus profile — it maximizes the ratio of signal power to quantization-noise power in the stored coefficients, since any smaller amplitude wastes representable range without changing the profile's shape; optimal loading of few-level quantizers is analyzed in detail for this receiver in [3]. The continuous-domain Euclidean norm is $\|\mathbf{w}^{(c)}_{\text{target}}\|_2 = \gamma\sqrt{K}$, i.e. a squared norm of $\gamma^2 K = 6272$; after component-wise rounding to the `int4` grid, the measured squared norms across the 23 configured channels fall in $[6280, 6418]$ — scattered around the continuous value by up to 2.3% because each channel's tone samples the quantization grid at different phases. The consequences of this scatter for the detection statistic's zero point are quantified in Section 4.3. *Remark on sign convention:* the implementation adopts the conjugate (negative-exponent) form of the tone, absorbing the receiver's spectrally inverted channel ordering visible in Table 4.1; the coordinate convention is fixed *per receiver profile* — CHIME's second-Nyquist sampling inverts the baseband and orders `freq_id` descending in RF, whereas a first-Nyquist receiver is upright and ascending — and is recorded as explicit fields of the runtime bundle's detector contract (the conjugate tone form and the pre-kernel window time-reversal), detailed in Section 6.2. Because a sense error mis-synthesizes the tone rather than mis-labelling it, it manifests as a complete detection miss; the acceptance gate is therefore an injected pilot at a known *signed* offset, asserted to peak in the predicted bin, with the injector built from physical parameters independently of the detector's own convention path.

The reference profiles of Section 4.3 are the identical construction evaluated at neighboring frequencies: the same single-term DFT, displaced by $\pm\Delta k$ bins, which is why they share the target's norm structure and cancel against it in a ratio. By instantiating the detection pipeline across the $C = 23$ standardized pilot frequencies, the array evaluates 23 independent target weight profiles $\mathbf{w}^{(c)}_{\text{target}}$. Each target profile is accompanied by its own pair of frequency-shifted local reference profiles (nominally $\Delta k = \pm 2$ bins), as defined in Section 4.3, yielding 23 distinct $K \times 3$ weight matrices $\mathbf{W}^{(c)}$.

4.3 Reference Profile Generation via Unitary Shift

To construct background reference profiles ($r \in \{1, 2\}$) that share identical structural characteristics with the target profile while satisfying the guard-band constraint of Section 4.1, the target tone is displaced by a discrete frequency shift in the tap domain.

Let Δk_r denote the resolved integer spectral bin displacement of reference r, nominally $\Delta k_1 = +2$ and $\Delta k_2 = -2$. The reference profile vectors are generated by point-wise complex phase rotation of the target profile:

$$w_{k,r} = w_{k,0}\, \exp\left(j \frac{2\pi}{K} \Delta k_r\, k\right), \qquad k \in \{0, 1, \ldots, K-1\}. \tag{4.6}$$

In vector notation, let $\mathbf{E}(\Delta k) \in \mathbb{C}^{K \times K}$ define the diagonal phase-modulation matrix:

$$\mathbf{E}(\Delta k) = \operatorname{diag}\left(1,\, e^{j\frac{2\pi}{K}\Delta k},\, e^{j\frac{2\pi}{K}\Delta k\cdot 2},\, \ldots,\, e^{j\frac{2\pi}{K}\Delta k(K-1)}\right), \tag{4.7}$$

so that the complete weight matrix $\mathbf{W} \in \mathbb{C}^{K \times 3}$ is expressed compactly as:

$$\mathbf{W} = \begin{bmatrix} \mathbf{w}_{\text{target}} & \mathbf{E}(\Delta k_1)\mathbf{w}_{\text{target}} & \mathbf{E}(\Delta k_2)\mathbf{w}_{\text{target}} \end{bmatrix}. \tag{4.8}$$

This construction has two properties worth making explicit. Structurally, each reference is the target's response displaced in frequency — the same rectangular window, the same constant modulus, the same Dirichlet response of Eq. (4.2) recentered two bins away — so the reference channels measure the background *through the same instrument response* as the target channel measures the pilot, which is what makes their ratio a meaningful normalization. Algebraically, $\mathbf{E}(\Delta k)$ is unitary, so continuous-domain phase rotation preserves the Euclidean norm identically across all three columns ($\|\mathbf{w}_0\|_2 = \|\mathbf{w}_1\|_2 = \|\mathbf{w}_2\|_2$): in exact arithmetic the three channels have precisely equal gain, and the detection ratio's null center is exactly unity.

Adaptive placement. The ± 2-bin displacement is the nominal rule, not an invariant. Two per-channel hazards require resolution at weight-generation time. First, a nominal reference bin may fall outside the coarse channel's baseband span, in which case it wraps circularly within the channel (the projection is periodic in the tap domain, so the wrapped bin is the physically correct one); this is recorded as an *edge-wrapped* placement. Second, a nominal reference bin may coincide with the channelizer's forbidden DC bin — the bin onto which the receiver's DC offsets and quantizer bias concentrate after channelization, which therefore cannot serve as a background sample; in that case the reference is displaced outward bin-by-bin until clear (*DC-shifted*), while a *target* tone colliding with the DC bin is rejected outright rather than moved, since relocating the target would change what the detector measures. Every channel's resolved offsets, placement status, and warnings are recorded in the weight manifest and carried into the runtime calibration bundle, so the evaluated configuration is auditable after the fact. For the monitored array the resolution is almost entirely nominal: 22 of the 23 channels resolve at exactly ± 2 bins, and the single exception is channel 14 (Table 4.1), whose pilot lies only $+3.059$ kHz from the frame origin — less than the 2-bin displacement of 6.104 kHz — so its lower reference wraps across the channel edge. In implementation, each weight column is synthesized analytically at its *resolved* normalized frequency and quantized independently, rather than by modulating the quantized target; the unitary-shift formalism above is the exact description of the continuous-domain construction, and quantization is applied once, at the end, to each column.

Quantized norms and the statistic's zero point. Upon discretization to fixed-point `int4` representations, the three columns sample the quantization grid at different phase progressions, so their squared norms — exact integers — differ slightly. Across the 23 configured channels the measured norm-ratio factor $\mu_0 = 2\|\mathbf{w}_0\|^2/(\|\mathbf{w}_1\|^2 + \|\mathbf{w}_2\|^2)$ falls in [0.9853, 1.0111]: the null center of the detection ratio is displaced from unity by up to 1.5%. This bias is not negligible: the coarse survey axis operates with thresholds within a fraction of a percent of unity (Section 5.5), and an uncorrected 1.5% zero-point bias would exceed the available decision margin. The coarse recorded flag therefore applies μ_0 as an *exact rational* correction (the integer norms are known exactly, so the correction introduces no rounding); the fine CFAR rule of Section 5.5 requires no correction at all, because the norm factor is common to every fine bin and cancels identically between the designated bins and the null-bulk baseline.

Chapter 5

Multichannel Signal Processing & Detection Pipeline

5.1 Problem Formulation & System Definitions

Consider a multichannel receiving array consisting of $M = 2048$ spatial feeds operating over a discrete time frame of $N = 16384$ complex baseband samples. Let the received data frame be represented by the matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$, where the scalar element $x_{m,n}$ denotes the sample acquired from feed $m \in \{0, 1, \ldots, M-1\}$ at sample index $n \in \{0, 1, \ldots, N-1\}$:

$$\mathbf{X} = \begin{bmatrix} x_{0,0} & x_{0,1} & \cdots & x_{0,N-1} \\ x_{1,0} & x_{1,1} & \cdots & x_{1,N-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{M-1,0} & x_{M-1,1} & \cdots & x_{M-1,N-1} \end{bmatrix}. \tag{5.1}$$

The objective of the detection pipeline is to test, once per frame, whether a narrow pilot-like component is present in addition to the local background. In the analytical model,

$$H_0 : \quad x_{m,n} = n_{m,n}, \tag{5.2}$$

$$H_1 : \quad x_{m,n} = a_m e^{j(2\pi\nu n + \phi_m)} + n_{m,n}, \tag{5.3}$$

where a_m and ϕ_m are feed-dependent nuisance parameters, ν lies in the fine search window around the synthesized pilot frequency, and $n_{m,n}$ is initially modeled as circular complex Gaussian noise. This is the model under which the tone projection has the matched-filter interpretation; local references, rank normalization, frequency search, quantization, feed covariance, and empirical calibration are additional parts of the complete detector and are not covered by that simple optimality statement.

The background power is unknown and time-varying — receiver gain, sky temperature, and broadband interference all modulate it — so the test must be invariant to the absolute noise scale. This requirement shapes the entire pipeline: every statistic below is ultimately a *ratio* of target-matched power to reference-matched power. Under the idealized H_0 model the common scale σ^2 cancels. In operation, the thresholds are calibrated empirically rather than assumed to transfer perfectly across gain states; the ratio construction removes first-order scale dependence, while the measured null retains receiver correlations and non-Gaussian structure.

The test signal is defined by a length-K weight profile against background noise estimated from auxiliary reference profiles. The weight coefficient vector length is fixed at $K = 128$ taps. We define $R = 3$ distinct weight profiles: one target profile ($r = 0$) and $N_{\text{ref}} = 2$ reference profiles ($r \in \{1, 2\}$).

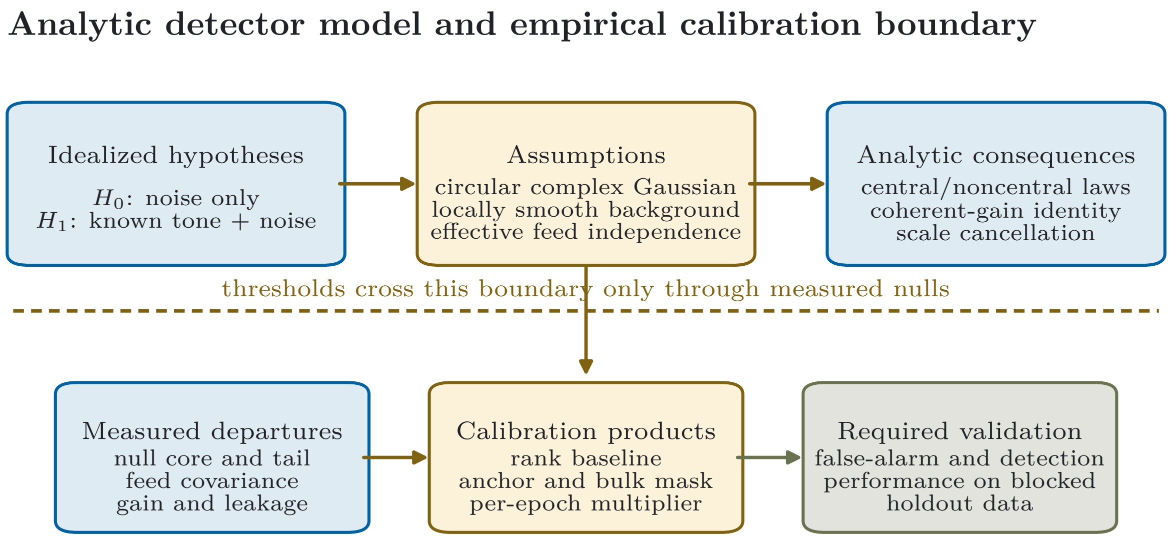

Figure 5.1: The detector has two evidence layers. The upper layer is the analytical known-tone, circular-Gaussian model used to derive the projection and idealized null. The lower layer contains the instrument-specific effects that set the actual operating point: quantization, feed covariance, local spectral structure, frequency search, and per-epoch null calibration. A theorem about the upper layer does not by itself establish global optimality of the complete lower-layer decision.

The reference profiles are the target profile displaced by ± 2 bins of the length-K filter's spectral grid (distinct from the fine bins of Section 5.3), leaving a one-bin guard separation between target and reference, thereby establishing a spectrally local background estimate; their construction and the justification of every dimension are given in Chapter 4. These profiles are assembled into the weight matrix $\mathbf{W} \in \mathbb{C}^{K \times R}$:

$$\mathbf{W} = \begin{bmatrix} \mathbf{w}_{\text{target}} & \mathbf{w}_{\text{ref},1} & \mathbf{w}_{\text{ref},2} \end{bmatrix}. \tag{5.4}$$

Remark on fixed-point arithmetic: In the physical architecture, input samples and weight coefficients are packed as complex integer `int4` values with exact integer accumulation and frozen fixed-point FFT arithmetic that is bit-identical between reference software and GPU execution. While this section formulates the pipeline using continuous complex variables for analytical clarity, complete quantization and fixed-point implementation details are deferred to Chapter 6. The formulation is nonetheless exact in structure: every sum written below is computed as written, in integer arithmetic, with accumulator widths chosen in Section 4.1 so that no rounding or saturation occurs before the final ratio. Figure 5.2 summarizes the complete processing chain developed in this section.

5.2 Sub-Frame Segmentation & Weight Projection

To apply the weight profiles across the temporal frame, each feed's time series of length N is partitioned into $L = N/K = 128$ non-overlapping sub-frames of length K. Let $\mathbf{s}_{m,\ell} \in \mathbb{C}^{K \times 1}$ denote

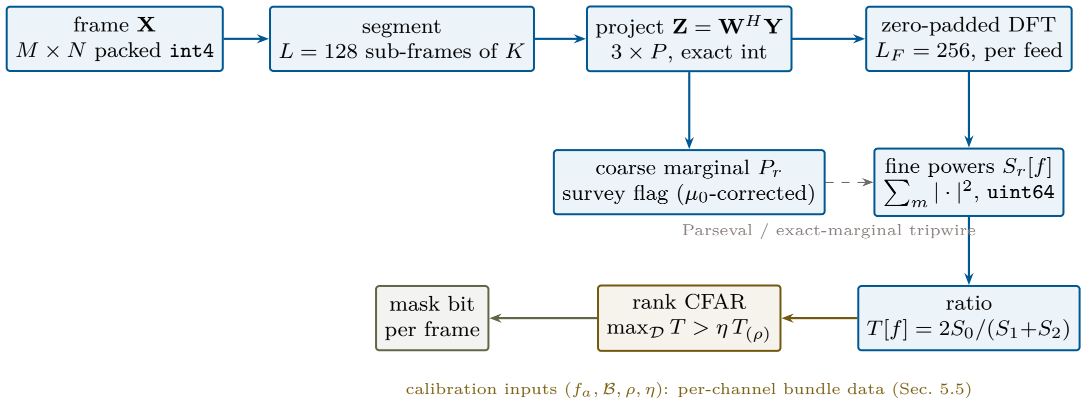

Figure 5.2: The pilot-proxy processing chain of this section. All quantities upstream of the recorded diagnostics are exact integers; the designated window, bulk mask, rank, and multiplier enter as per-channel calibration data. The coarse marginal P_r is retained as the survey flag and doubles, through the Parseval relation of Eq. (5.12), as the pipeline's internal verification tripwire (Section 6.6).

the ℓ-th sub-frame column vector of the m-th feed, where $\ell \in \{0, 1, \ldots, L-1\}$:

$$\mathbf{s}_{m,\ell} = \begin{bmatrix} x_{m,\,\ell K} & x_{m,\,\ell K+1} & \cdots & x_{m,\,\ell K+(K-1)} \end{bmatrix}^T. \tag{5.5}$$

Concatenating all sub-frames across all feeds yields the reshaped data matrix $\mathbf{Y} \in \mathbb{C}^{K \times P}$, where $P = ML = 262144$ represents the total number of sub-frames across the array:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{s}_{0,0} & \cdots & \mathbf{s}_{0,L-1} & \big| & \mathbf{s}_{1,0} & \cdots & \mathbf{s}_{1,L-1} & \big| & \cdots & \big| & \mathbf{s}_{M-1,0} & \cdots & \mathbf{s}_{M-1,L-1} \end{bmatrix}. \tag{5.6}$$

The segmentation is purely an indexing view — the sub-frames tile the frame exactly ($KL = N$, Section 4.1), so no samples are discarded, duplicated, or windowed, and the reshape involves no data movement in implementation. The temporal projection of the data frame onto the R weight profiles is evaluated via the matrix inner product:

$$\mathbf{Z} = \mathbf{W}^H \mathbf{Y} \in \mathbb{C}^{R \times P}, \tag{5.7}$$

where $(\cdot)^H$ denotes the Hermitian (conjugate transpose) operator. Each column $p \in \{0, 1, \ldots, P-1\}$ of $\mathbf{Z}$ contains the complex response of the target and reference profiles for a specific feed $m = \lfloor p/L \rfloor$ and temporal sub-frame index $\ell = p \bmod L$. This single product — $P = 262144$ length-K complex projections per 41.94 ms frame, three profiles each — is the pipeline's dominant arithmetic cost, and the stage the packed-integer kernel geometry of Section 4.1 is designed around. Everything downstream operates on the 3×262144 projected values rather than the 2048×16384 raw samples: the projection is also a $\sim$43$\times$ reduction in value count ($\sim$11$\times$ in bytes at the widened integer precision) that renders the remaining stages computationally minor.

5.3 Coherent Spectral Integration

The projection of Section 5.2 is coherent only *within* each K-sample sub-frame. A first-generation reduction may stop there: square each projection and sum the powers over all sub-frames and feeds (the coarse statistic of Section 5.4). Such a reduction, however, discards the phase relationship *between* successive sub-frames, which for the signal of interest is deterministic rather than random.

A frequency-stable pilot line at residual offset $\Delta\nu$ from the synthesized tone advances its phase by exactly $2\pi\Delta\nu K$ between consecutive sub-frames of the same feed: the L successive projections of a feed rotate deterministically, and integrating them coherently before squaring recovers deflection that incoherent summation forfeits. The available gain is substantial: coherent integration of L sub-frames improves the deflection signal-to-noise ratio by up to $10\log_{10}\sqrt{L} \approx 10.5$ dB over the incoherent baseline. Realizing it requires matching the unknown residual rotation — which is precisely a spectral analysis across the sub-frame axis.

To exploit this temporal coherence, the projected matrix $\mathbf{Z}^T \in \mathbb{C}^{P\times R}$ is partitioned vertically into M distinct feed blocks $\{\mathbf{B}_0, \mathbf{B}_1, \ldots, \mathbf{B}_{M-1}\}$, where each block $\mathbf{B}_m \in \mathbb{C}^{L\times R}$ collects the L sequential temporal projections for feed m, defining the triple-indexed entries $z_{m,\ell,r} \equiv z_{r,\,mL+\ell}$:

$$\mathbf{B}_m = \begin{bmatrix} z_{m,0,0} & z_{m,0,1} & z_{m,0,2} \\ z_{m,1,0} & z_{m,1,1} & z_{m,1,2} \\ \vdots & \vdots & \vdots \\ z_{m,L-1,0} & z_{m,L-1,1} & z_{m,L-1,2} \end{bmatrix}. \tag{5.8}$$

Coherent temporal integration is performed by applying a zero-padded Discrete Fourier Transform (DFT) of length $L_F = 2L = 256$ independently across the sub-frame axis ℓ for each feed m and weight profile r:

$$\widetilde{B}_m[f,r] = \sum_{\ell=0}^{L-1} B_m[\ell,r]\, e^{-j\frac{2\pi}{L_F} f\ell}, \qquad f \in \{0,1,\ldots,L_F-1\}. \tag{5.9}$$

Each spectral bin f coherently integrates the L sub-frame projections under a frequency offset hypothesis of $f/(L_F K)$ cycles per sample (interpreted modulo the sub-frame rate: bins above $L_F/2$ correspond to negative offsets): the DFT evaluates all L_F rotation hypotheses simultaneously, so no prior knowledge of the residual offset is needed — the matched hypothesis corresponds to the bin of maximum response, whose index directly measures the pilot's residual frequency (the survey's per-channel *anchor*, Section 5.5).

Two distinct scalloping mechanisms bound the recovered gain, and they are treated separately here. The first is the *sub-frame projection rolloff*: the K-tap projection has the Dirichlet response of Eq. (4.2), so a pilot at the edge of the fine span ($\pm f_s/2K$, i.e. half a coarse grid bin from the synthesized tone) is attenuated by $|D_K(1/2)| = 2/\pi$, a worst-case loss of 3.92 dB. This loss is a property of the segmentation geometry, not of the transform; it is minimized in practice because the synthesized tone absorbs each channel's nominal offset exactly (Section 4.2), leaving typical residuals far inside the span. The second is *fine-grid scalloping*: a residual falling between adjacent fine bins splits its energy. The zero-padding factor of 2 — fine-bin spacing $(f_s/K)/L_F \approx 11.9$ Hz — halves the worst-case bin-center mismatch; a further doubling would halve it again, but at double the transform cost and shared-memory footprint for a residual the decision stage already absorbs, so the factor-2 padding represents the practical optimum. What padding leaves is captured by the ± 2-bin designated window of Section 5.5, and this width is not an additional free parameter: on the window axis, a residual tone's response is the L-point analogue of Eq. (4.2), whose main lobe spans ± 1 window-rate bin — exactly ± 2 bins of the padded grid. The designated window is therefore the kernel's *main-lobe extent*: with the anchor measured to the nearest fine bin, the two dominant bins straddling any sub-bin residual always lie inside it, evaluating the window as a maximum bounds the worst-case capture loss (measured below 1 dB), and slow anchor drift *within* a calibration epoch rides within the same margin. Era-scale changes do not: a station change moves the anchor by tens of fine bins — measured across the archive's calendar halves at 11 bins on channel 32 and 29 on channel 35 — and is handled as a calibration-epoch boundary with a re-derived anchor, not as drift (Section 9.10). The two ± 2 displacements in this architecture are thus dual applications of the same response function (Figure 4.1): the coarse-grid ± 2 of Section 4.1 places the references at the kernel's decayed skirts — an *exclusion* guard keeping pilot energy out of the background estimate — while the fine-grid ± 2 spans the kernel's main lobe — a *capture* window keeping pilot energy inside the detection set. Monte-Carlo evaluation of the complete evaluated geometry (reported in the results chapter) recovers 9.4–10.0 dB of the 10.5 dB coherent bound at the survey's operating false-alarm rates, confirming that the two scalloping budgets, the window maximum, and the finite-P_{fa} threshold account for essentially all of the gap.

One statistical consequence of padding must be carried forward: the $L_F = 2L$ output bins oversample an L-dimensional spectrum, so adjacent padded bins are statistically correlated, and only alternate spectral bins constitute independent noise samples. The null-estimation rule of Section 5.5 is built on the independent subset.

5.4 Non-Coherent Spatial Integration & Test Statistics

The remaining axis is space. Here the pipeline makes the opposite choice, *non-coherent* (power) integration across the $M = 2048$ feeds; this asymmetry is deliberate. Coherent combination across feeds would require the array to be phase-calibrated toward the interference source: a terrestrial transmitter in the array's near sidelobes, at unknown azimuth, reached by multipath that decorrelates across the aperture and varies with conditions. No stable steering vector exists to calibrate toward, and any attempt would couple the detector's sensitivity to the array's beamforming state — exactly the dependence the flagging system must not have (Section 4.1). Summing per-feed *powers* instead accepts a $\sqrt{M}$ (rather than M) deflection scaling in exchange for complete immunity to inter-feed phase: the detector works identically whether the array is calibrated or not, and multipath phase structure across the aperture — which would randomize a coherent sum — contributes all its power to a non-coherent one. With $M = 2048$, the non-coherent deflection is ample, and the primary contribution of the feed sum is statistical depth: it supplies the M in the degrees of freedom that concentrate the null distribution (Section 5.5).

The system computes two complementary integration products: a coarse time-domain statistic and a fine frequency-domain statistic. The *coarse marginal statistic* P_r accumulates total projected energy directly across all $P = 262144$ sub-frames:

$$P_r = \sum_{p=0}^{P-1} |z_{r,p}|^2 = \sum_{m=0}^{M-1} \sum_{\ell=0}^{L-1} |B_m[\ell, r]|^2 . \tag{5.10}$$

The *fine spectral statistic* $S_r[f]$ integrates spatial power per frequency bin f across all M feeds:

$$S_r[f] = \sum_{m=0}^{M-1} \left| \widetilde{B}_m[f, r] \right|^2 , \qquad f \in \{0, 1, \ldots, L_F - 1\}. \tag{5.11}$$

The two are not independent products: for the zero-padded DFT, Parseval's theorem equates each feed's total spectral energy to L_F times its sub-frame energy (the padded transform is L_F times an isometry on the L nonzero inputs), and summing over feeds yields the frame-level identity

$$\sum_{f=0}^{L_F-1} S_r[f] = L_F \, P_r. \tag{5.12}$$

The coarse statistic is thus the marginal of the fine spectrum: the fine analysis *refines* the coarse energy across frequency rather than measuring something different. This structural redundancy is exploited deliberately for verification. The exact-integer marginal identity — P_r recomputed from the projected row sums must equal the independently accumulated kernel power sums bit-for-bit — serves as the pipeline's internal verification tripwire, exercised continuously in verification runs and available as a runtime assertion: any fault in memory layout, accumulation, or kernel scheduling breaks an exact integer equality and is caught on the frame where it occurs, rather than surfacing as a subtle statistical bias discovered downstream. The Parseval relation of Eq. (5.12) additionally gates the floating-point FFT path at ULP level, and the fixed-point transform satisfies it to within its documented rounding contract. The coarse statistic P_r is retained as a low-overhead survey flag, while the fine spectrum $S_r[f]$ provides the candidate runtime detection refinement.

For each bin f, the fine test statistic $T[f]$ is defined by normalizing the target profile power by the summed background reference profile power:

$$T[f] = \frac{2\,S_0[f]}{S_1[f] + S_2[f]}. \tag{5.13}$$

This is the scale-invariant form anticipated in Section 5.1: numerator and denominator are the same functional of the same frame through instrument responses differing only by the reference displacement, so the unknown background scale — and, per bin, any spectrally smooth gain structure the slope cancellation of Section 4.1 removes — divides out. *Remark on profile norms:* Quantized weight profiles have slightly unequal vector norms, biasing the null center of the ratio away from unity by a factor common to all bins — measured across the configured channels at up to 1.5% (Section 4.3), which would exceed the coarse axis's entire decision margin if left uncorrected. The recorded coarse flag applies an exact rational norm correction (μ_0); the fine CFAR rule of Section 5.5 requires none, since the common factor cancels identically between the designated bins and the null-bulk baseline.

5.5 Statistical Distribution & CFAR Decision Rule

Idealized null distribution

Under the idealized null hypothesis (H_0: independent and identically distributed circularly symmetric complex normal noise across feeds and samples), the null distribution of $T[f]$ follows from a standard argument. The complex-normal convention used here is

$$z \sim \mathcal{CN}(0, \sigma_z^2), \qquad \mathbb{E}|z|^2 = \sigma_z^2, \qquad \frac{2|z|^2}{\sigma_z^2} \sim \chi_2^2. \tag{5.14}$$

Each projection is a fixed linear functional of Gaussian noise, so $z_{m,\ell,r} \sim \mathcal{CN}(0, \sigma_z^2)$ with $\sigma_z^2 = \sigma^2 \|\mathbf{w}_r\|^2$. Independence across feeds and sub-frames is an analytical approximation; the measured width of the null is used to reveal and absorb departures from it. The DFT is a fixed linear combination of L independent such variables, so each bin is again complex normal, $\widetilde{B}_m[f, r] \sim \mathcal{CN}(0, L\sigma_z^2)$, with variance independent of f. Each squared magnitude contributes β real degrees of freedom, where $\beta = 2$ for complex data ($\beta = 1$ for a real-valued field), giving $|\widetilde{B}_m[f, r]|^2 \sim \frac{L\sigma_z^2}{2}\chi_\beta^2$; summing over the M independent feeds,

$$S_r[f] \sim \frac{L\sigma_z^2}{2}\,\chi_{\beta M}^2. \tag{5.15}$$

The numerator of $T[f]$ therefore carries $\beta N_{\text{target}} M$ degrees of freedom ($N_{\text{target}} = 1$) and the denominator pools $\beta N_{\text{ref}} M$ across the two reference branches; the scale $L\sigma_z^2$ cancels in the ratio (equal weight norms in the continuous construction, Section 4.3), leaving the central F-distribution

$$T[f]\big|_{H_0} \sim F(\beta N_{\text{target}} M,\, \beta N_{\text{ref}} M) = F(2M,\, 4M) = F(4096,\, 8192). \tag{5.16}$$

Similarly, the coarse scalar ratio follows $F(\beta N_{\text{target}} P,\, \beta N_{\text{ref}} P) = F(524288,\, 1048576)$. These large degrees of freedom concentrate the idealized null distributions tightly about unity: mean 1.00024 with standard deviation 2.71% for the fine per-bin statistic, and mean 1.000002 with standard deviation 0.239% for the coarse scalar. Under the idealized model, sub-percent pilot excesses would therefore be significant at many standard deviations on the coarse axis.

Detection-side distribution and the coherent-gain identity

The distribution under the alternative hypothesis completes the statistical description. Under H_1, a pilot at the matched bin adds a deterministic mean to each feed's bin value: if the per-sub-frame projection acquires coherent amplitude a_m at feed m, the matched DFT bin accumulates La_m, and $S_0[f]$ becomes noncentral,

$$T[f]\big|_{H_1} \sim F'(2M,\, 4M;\ \lambda)\,, \qquad \lambda = \frac{2L\sum_m |a_m|^2}{\sigma_z^2}, \tag{5.17}$$

a noncentral F with the same degrees of freedom (the references remain central under the guard-band model of Section 4.1). The factor of two follows from Eq. (5.14): one complex scalar contributes two real degrees of freedom when σ_z^2 denotes total complex variance, $\mathbb{E}|z|^2$. The sum over m then accumulates the feed-dependent noncentralities. The mean excess is $\mathbb{E}[T] - 1 \approx \lambda/(2M)$. The coarse statistic absorbs the *same* total signal energy — by the Parseval identity of Eq. (5.12), its noncentrality is also λ — but spreads it against a null of $2P$ rather than $2M$ numerator degrees of freedom. The deflection ratio between the two axes is therefore

$$\frac{d_{\text{fine}}}{d_{\text{coarse}}} = \frac{\lambda/(2M)}{\lambda/(2P)} \cdot \sqrt{\frac{\mathrm{Var}[T_{\text{coarse}}|H_0]}{\mathrm{Var}[T[f]\,|H_0]}} = \sqrt{\frac{P}{M}} = \sqrt{L}, \tag{5.18}$$

recovering the $10\log_{10}\sqrt{L} \approx 10.5$ dB coherent-integration bound of Section 5.3 as a degrees-of-freedom concentration identity: the fine axis concentrates the same signal energy against L times fewer null degrees of freedom. The Monte-Carlo measurement of 9.4–10.0 dB recovered gain (results chapter) corresponds to this identity evaluated through the complete evaluated geometry, including scalloping, the window maximum, and finite-P_{fa} thresholds.

Measured departure from the i.i.d. model

Empirical system measurements reveal that the idealized model's tails cannot be trusted for threshold setting: the observed coarse null is ~19× wider than the i.i.d. F-model, while the fine per-bin null exhibits a near-ideal core contaminated by a heavy ~0.3% tail that inflates its width factor to ~13.8×. Both factors are raw standard-deviation ratios against the analytic i.i.d. widths, measured during calibration-era processing on the survey's one pre-verified transmitter-off cohort (channel 521's off-air epoch, 2020 captures); they are quoted to motivate the design, not as portable constants. Width factors of this kind are sensitive to the estimator and the cohort — a robust core-width estimate of the same nulls sits an order of magnitude lower, precisely because the tail carries the width — so the results chapter re-derives them from the final survey under a single pinned definition set (estimator, transient handling, cohort), reporting the robust-core equivalents alongside. The physical origins are receiver systematics — gain variations across the frame, spectral leakage from features elsewhere in the channel, and non-Gaussian interference — none of which the i.i.d. derivation models. Two design consequences follow. First, thresholds must be *calibrated from measured nulls* rather than computed from the F survival function; the analytic distribution is retained as the idealized baseline and as a lower bound on achievable width, not as the operating calibration. Second, the fine null's specific structure — a clean core with a thin contaminated tail — discriminates between estimator families: any threshold built from moments (mean and variance) of the observed bulk is dragged by the tail, while an estimator that simply *sorts* the bulk and reads off a value at a fixed position from the bottom is unmoved by any contamination that lands above that position, however extreme. This measured structure motivates the rank-based estimator formalized next.

Remark: why there is no taper. Both stages of the reduction are deliberately untapered, and the choice follows the same logic as the estimator above. Within the K-tap profile, uniform magnitude *is* the matched filter for a tone of known frequency: any taper trades deflection for sidelobe suppression — a Hann profile forfeits 1.8 dB — and sensitivity is the scarce resource for a beacon 11.3 dB below its own shelf. Along the window axis the classic rectangular penalty is scalloping, and the padded transform is the free correction: doubling the hypothesis grid bounds the worst-case straddle loss near 0.9 dB at no sensitivity cost, where a taper would pay decibels for the same coverage and widen the designated window besides. The failure modes tapering exists to prevent — leakage from strong structure elsewhere in the coarse channel — are instead absorbed downstream at zero sensitivity cost: the ratio statistic cancels spectrally smooth structure, the reference placement cancels the residual slope, guard and census-excluded bins remove the known offenders, and the rank estimator is immune by construction to the contaminated tail that survives them all. The design spends robustness where robustness is free, and sensitivity nowhere. The choice is also contractual: the profile norms are the exact integers behind μ_0, so any taper is a new weight bank, a new null calibration, and a new survey epoch.

Which axis decides, and which one only checks

The two axes are not two independent detectors. Under the declared known-tone, common-calibration model, the deflection identity above shows that the fine axis concentrates the same signal energy against L times fewer null degrees of freedom and improves the idealized deflection by $\sqrt{L}$. This is a model result, not a universal dominance theorem: mismatch, nonstationary backgrounds, and different calibration errors can change empirical ordering. Detection is therefore assigned to the fine axis only after out-of-sample calibration, while the coherent summation of Section 5.3 is the sensitivity mechanism the runtime design intends to use.

That second property sets the order in which the system could be brought up, and it is why the survey products analyzed in this work carry a coarse-rule mask. The fine decision needs a per-channel pilot anchor and a per-channel null, and measuring those is precisely what the survey exists to do; it could not be gated on quantities it was being run to produce. So the survey configuration falls back to the calibration-free rule, which rejects a frame exactly when the target profile carries more power than the mean of its two references — a raw positive excess:

$$\text{reject } H_0 \iff F > \mu_0, \qquad F = \frac{2P_0}{P_1 + P_2}, \qquad \mu_0 = \frac{2\|\mathbf{w}_0\|^2}{\|\mathbf{w}_1\|^2 + \|\mathbf{w}_2\|^2}, \tag{5.19}$$

where μ_0 is the exact rational norm correction of Section 4.3 and not a tuned quantity: it is the value F takes under pure noise given the quantized profiles' unequal norms, so Eq. (5.19) asks only whether the target exceeds its own null center. Evaluated as the integer cross-multiplication $P_0\|\mathbf{w}_1\|^2+P_0\|\mathbf{w}_2\|^2 > 2P_1\|\mathbf{w}_0\|^2+\dots$, it forms no quotient and involves no calibrated constant, which is why the survey-era contract records its threshold mode as *none* and its decision as exactly reproducible.

Two limitations follow, and both should be read as properties of the bootstrap configuration rather than of the method. It fires on *any* detectable pilot, with no way to ask for less — and its false-alarm price is measured: because the threshold sits at the null center, the rule holds $P_d \approx 1$ on every channel but spends 48.5% of verified-quiet time in the candidate-selection record (44.2% recomputed on the full 11,199-frame transmitter-off null of the completed archive). The measured coarse-versus-fine ROC and Youden-J analysis behind that record motivated the fine designated-set CFAR as the runtime decision, and it reproduces from the released products (`analysis/youden_j.py`): at a fixed null P_{fa} of 0.05, the fine designated-set rule holds $P_d \geq 0.997$ on channels 34 and 36 where the calibrated coarse points collapse to 0.06 and 0.70. The design intent was a threshold η placed above this floor, so that frames whose inferred shelf sits far below what the science can register would be retained. Realizing that needs each channel's null characterized well enough to place a threshold against, and at the time the survey ran that characterization did not exist — the measured nulls are wide, tail-dominated, and sensitive to the estimator and cohort used to summarize them (previous subsection), so no defensible per-channel η could then be derived. Section 9.8 now derives one per channel from the forecast instead of from the null. And it decides on the less sensitive axis, forgoing the ≈ 10.5 dB the coherent stage offers. Every masked fraction quoted from the survey in this work, including those the noise-tolerance analysis of Chapter 9 prices, is therefore a measurement of this calibration-free rule and not of the designed detector.

Chapter 9 changes how the first of those should be read. On the one channel where the full residual chain is measured, sweeping η upward on this coarse rule and propagating each setting through to the science shows that the floor is the *only* admissible operating point: the first relaxation that recovers a meaningful amount of data multiplies the surviving contamination by three orders of magnitude, because the coherence gain in the chain exceeds the proxy margin the relaxation was supposed to spend (Section 9.7). On that channel the calibration-free rule is also the correct one, and what looked like a limitation is a result. Whether it holds where the residual correlation time is shorter is exactly what the sweep answers per channel, and is why η remains an interface value rather than a frozen constant.

The fine decision, and what it is waiting on

The fine axis runs on every frame and its per-frame CFAR products are recorded, but in the survey configuration it does not gate: the emitted rejection is bit-identical to Eq. (5.19) on every valid frame and differs from the fine stage's own detections. What follows is therefore the decision the detector is designed around, held back only by the calibration the survey is running to supply. In the proposed runtime design the roles complete their exchange: the fine designated-set rule is the sole decision path, and the coarse marginal retains no decision role at all — it persists because it is nearly free (an exact integer sum the kernel already accumulates) and because it buys two things worth having: the Parseval tripwire of Section 6.6, and a development-mode sanity check that any build can evaluate without calibration data. A statistic that costs nothing, verifies the datapath, and once bootstrapped an entire survey earns its place in the products; it just never again decides anything.

Its form is a Constant False Alarm Rate test, and the two words that matter are *constant* and *false alarm rate*. A fixed numerical threshold cannot hold a fixed false-alarm rate when the background moves, and the background here does move — with receiver gain, with sky, with whatever else occupies the channel. A CFAR rule fixes the rate instead of the number: it estimates the background from the frame it is testing, and declares a detection only when a candidate bin exceeds that frame's own background by a set factor. The threshold floats; the false-alarm rate does not.

How the background is estimated is then the whole design, and the measured null structure of the previous subsection dictates it. Averaging the bulk is the textbook choice and is exactly wrong here: a 0.3% contaminated tail drags a mean or a variance on every frame it appears in, inflating the threshold and suppressing real detections. The alternative is to *sort* the bulk and read the background off at a fixed position from the bottom. Sorting is indifferent to how large the contaminated values are — only to how many there are — so a baseline taken from below them is untouched however extreme they get. This is the ordered-statistic CFAR family, standard in radar for precisely this situation [7].

Concretely: sort the $|\mathcal{B}|$ background bins of a frame into ascending order and take the ρ-th value, written $T_{(\rho)}$. The rank ρ is a position in that sorted list, and it is the estimator's single tuning knob. It has two readings, and holding both is what makes the choice intelligible. As a *quantile*, $T_{(\rho)}$ estimates the $\rho/|\mathcal{B}|$ point of the null distribution — $\rho = |\mathcal{B}|/2$ is the median, $\rho = |\mathcal{B}|$ is the maximum, and the baseline rises as ρ does. As a *breakdown point*, the estimator is unmoved by upward contamination of as many as $|\mathcal{B}| - \rho$ bins, so lowering ρ buys robustness. A calibration would trade these against each other: with $|\mathcal{B}| = 125$ and a tail occupying $\sim$0.3% of bins — under half a bin per frame on average — even a rank one or two below the maximum absorbs the typical frame, while a rank far below it estimates a quantile so low that the baseline stops representing the null's working range. Through the survey era, both ρ and the fine multiplier were blocked on the same missing null characterization as η. They no longer are: with the survey's per-bin `fstat_fine` in hand, the calibration has been prototyped end to end on the first-measured ten channels, and three of its results belong here. First, the anchors are *measured*: every channel's pilot sits hundreds of hertz to 1.4 kHz off nominal (-1371 Hz on channel 32, $+1299$ Hz on channel 35), comfortably inside the ± 1.5 kHz fine span; the estimator is the difference of per-bin medians between coarse-detected and coarse-quiet frames, so static structure common to both cohorts cancels, with a plain-median fallback (marked as such) on channels that never fall quiet. Second, the exchangeability guarantee holds on real data, not just on paper: on channel 29's quiet frames, a bulk bin tested against the rank of the remaining bulk exceeds it at the combinatorial rate $(|\mathcal{B}| + 1 - \rho)/(|\mathcal{B}| + 1)$ to three significant figures at every rank tested. Third, the pair (ρ, η_{q16}) is selected by the same objective that selects the coarse operating point (Section 9.8), under a discipline in which every stage is either data or a named ordering: feasibility must hold on the full archive *and* in each calendar half of it — the mask-side twin of the forecast's stability gate — cost ties within the 2% plateau are broken by the worst era-half cost, and only exact ties fall through to fixed orderings (η nearest one, then largest breakdown headroom $|\mathcal{B}| - \rho$). What that discipline exposed is that the operating point is per-*epoch* data rather than per-channel data: measured anchors move by tens of bins when stations change (Section 9.10), so the calibration block carries an epoch boundary exactly where the runtime-contract rule already requires one.

The remaining definitions specify which bins are candidates and which are background. Let f_a denote the measured per-channel pilot anchor bin — the fine bin at which the survey localizes the channel's residual pilot offset. The designated detection window $\mathcal{D}$ encompasses ± 2 fine bins around the anchor:

$$\mathcal{D} = \{(f_a + k) \bmod L_F : |k| \le 2\}. \tag{5.20}$$

The background is estimated from the *usable null bulk*: the set $\mathcal{B}$ of bins that are (i) members of the independent (alternate-bin) subset of Section 5.3, (ii) outside the designated window $\mathcal{D}$, with no dilation beyond it — the runtime exclusion is exactly $[f_a - 2,\ f_a + 2]$, since the window half-width *is* the anchor's guard zone (one unpadded fine bin, equal to the pad factor in padded bins, per side): designated window and CFAR guard coincide by construction, (iii) not census-excluded as persistent RFI features, and (iv) statistically live in the current frame (nonzero reference denominator). For an even-parity anchor this exclusion removes three members of the $L_F/2 = 128$-bin independent subset, leaving $|\mathcal{B}| = 125$; for an odd-parity anchor it removes two, leaving 126. In the odd case the pilot's ± 3-padded-bin Dirichlet shoulders (the ≈ -13.5 dB first sidelobe of Eq. (4.2) on the window axis) land on the independent grid and remain bulk members in the on state — upward contamination of exactly the kind the rank baseline ignores by construction, with clause (iii) available to a calibration that prefers to census-exclude them explicitly.[1] With those two sets fixed, the baseline is the order statistic defined above, $\hat{T}_{\text{null}} = T_{(\rho)}$ over $\{T[f] : f \in \mathcal{B}\}$, and a frame is flagged as a detection if and only if any designated bin exceeds it by the calibrated factor:

$$\text{reject } H_0 \iff \max_{f \in \mathcal{D}} T[f] > \eta \cdot \hat{T}_{\text{null}}, \tag{5.21}$$

where the multiplier and the rank would be quantile-calibrated jointly against that channel's measured off-epoch null. Two thresholds are in play and they are not the same. Calibrating so that the empirical exceedance rate over verified transmitter-off epochs equals a target false-alarm probability fixes the *detection floor* — the smallest multiplier that holds P_{fa} — and that is a statistics question. What the survey should actually use is the *science* threshold, set by the contamination the measurement can absorb, and that is a cosmology question answered in Chapter 9. The proxy margin below shows the two can differ by tens of decibels in principle; Section 9.7 finds that on the one fully measured channel the residual chain consumes the entire margin, so they coincide there. Both constants are per-channel *data*, carried in the runtime calibration bundle with their calibration provenance; the decision arithmetic itself is frozen. The path by which the science threshold arrives is worth making concrete, because it closes a loop this dissertation opened in Section 1.7: the survey era ran the bootstrap placeholder — the positive-excess rule, the multiplier's $T = 1$ — and the mature forecast now replaces that placeholder with the tolerance-derived operating point, computed per channel by constrained optimization over the recorded statistics (Section 9.8). The multiplier enters the bundle as a Q16 fixed-point rational, $\eta_{q16} = \lfloor \eta \cdot 2^{16} \rceil$, so a forecast-selected threshold costs the exactness contract nothing: the comparison remains an integer cross product, and the threshold remains data rather than code.

Three properties justify this specific form. *Robustness:* the breakdown property above is what removes an entire class of operational complexity. No contamination-triggered fallback mode is needed, no outlier-rejection pass runs before the estimate, and no frame takes a different code path from any other; the estimator is unconditionally the same arithmetic on every frame, and the tail is handled by the shape of the statistic rather than by logic that has to detect it first. *Self-normalization:* the quantized-norm factor μ_0 of Section 4.3 multiplies every bin's statistic equally, hence cancels between $\max_{\mathcal{D}} T$ and $\hat{T}_{\text{null}}$; the fine decision needs no zero-point correction, and more broadly any per-frame, bin-common gain structure divides out — the rule is CFAR against the frame's own background level by construction. *Exactness:* because $T[f]$ is a ratio of integer power sums, rank comparisons and the threshold comparison can be evaluated in exact integer cross-multiplication without ever forming the quotients; the exact-integer kernel does precisely this,

[1] The survey products additionally carry a per-frame *diagnostic* CFAR (location, scale, threshold), estimated on the independent bulk with only the envelope-DC guard zone excluded (designated bin 0), because per-channel anchors were unmeasured before the survey ran; its median-location, left-side-scale policy is robust to the pilot then sitting *inside* its bulk. That survey scaffolding is distinct from the decision rule defined here, whose bulk mask is per-channel calibration data validated to exclude the measured $\mathcal{D}$.

making the emitted decision bit-reproducible and verifiable against an arbitrary-precision reference; Section 6.4 details this arithmetic.

Two degenerate-frame rules complete the definition, both fail-safe toward non-rejection. A designated bin whose reference denominator is zero cannot fire (the per-bin analogue of the coarse path's zero-reference rule); and a frame whose usable bulk $\mathcal{B}$ is not deeper than the calibration rank ρ is declared invalid and emits no rejection — the detector refuses to flag data against a background estimate it cannot form. Both conditions are vanishingly rare in nominal operation but are defined, deterministic outcomes rather than undefined behavior: in an unattended flagging system, an estimator that cannot form a valid background must default to retaining the data rather than rejecting it.

Detectable versus harmful: the proxy margin

One quantity closes the argument of this section, and it is the reason the method is viable rather than merely exact. The decision rule above is calibrated against the *pilot*; what threatens the measurement is the *shelf*. Equation (3.8) relates them at the transmitter, and the two occupy very different bandwidths: the pilot's power falls in one analysis bin of width f_s/K, while the shelf spreads over B_{DTV}. Comparing the two at the level relative to system noise that each produces,

$$\mathrm{SNR}_{\mathrm{shelf}} = \mathrm{PNR}_{\mathrm{bin}} + 10\log_{10}\left(\frac{P_{\mathrm{shelf}}}{P_{\mathrm{pilot}}}\right) - 10\log_{10}\left(\frac{B_{\mathrm{DTV}}}{f_s/K}\right), \tag{5.22}$$

which at the evaluated values (11.3 dB pilot-below-payload, 6 MHz allocation, 3.052 kHz analysis bin) evaluates to $\mathrm{SNR}_{\mathrm{shelf}} = \mathrm{PNR}_{\mathrm{bin}} - 21.6$ dB. The pilot therefore becomes detectable 21.6 dB before the interference it implies reaches the same level relative to noise — and that margin, not the detector's exactness, is what makes a per-frame decision useful: it is the gap inside which the operating threshold is placed. A rule set at the detection floor rejects every frame carrying a detectable pilot, including those whose implied shelf is tens of decibels below anything the science can register; a rule set at the science tolerance rejects only what matters and retains the rest. Both are the same detector with a different η. Whether the two differ in practice depends on how much of the margin the foreground-removal and integration stages consume before the residual reaches the power spectrum, which is measured per channel in Section 9.7. Quantifying the tolerance, and hence the operating point, requires propagating the shelf through the foreground-removal stage of the cosmological pipeline; Chapter 9 constructs that propagation and the tolerance it yields, and the conversion to per-channel thresholds, with the measured distribution of inferred shelf levels across the monitored channels, is developed in the calibration and results chapters. What this section establishes is a *transmitter-model screening margin*: it is large and computable from the nominal pilot-to-payload relation and the receiver analysis bandwidth. It does not yet establish that the received shelf follows the pilot with a fixed ratio across an allocation, nor that the scalar pilot-power proxy maps directly onto a post-filter complex-visibility residual. Those two transfer measurements are explicit closing gates in Sections 9.2 and 9.4.

5.6 A Worked Example

The pipeline is illustrated with one frame of survey data from channel 506 (physical channel 36), captured 2025 July 31 at 15:52:22 UT, drawn from the channel's calibration-era processing cohort of 7010 valid frames; the frame sits at the median of that cohort's masked population, and the channel's survey-measured anchor is $f_a = 62$ (an always-on channel, so its anchor comes from the fallback median estimator rather than the on-minus-off difference; Section 5.5). The frame's fine spectrum and coarse statistic are quoted from the recorded survey product, which evaluates the statistics of Section 5.4 in the analyzer's floating-point path; the bulk quantities below are then evaluated from that recorded spectrum under the bulk definition of Section 5.5, and the exact-integer kernel evaluates the same quantities as exact integers (Chapter 6).

Across the designated window $\mathcal{D} = \{60, \ldots, 64\}$, the fine test statistic is

$$T[60..64] = (1.066,\ 8.730,\ 18.615,\ 7.598,\ 1.083),$$

displaying the main-lobe structure of Section 5.3: a peak at the anchor, correlated shoulders on the adjacent padded bins, and near-unity values by the alternate (independent) bins. The usable null bulk for this frame contains $|\mathcal{B}| = 125$ bins with median 1.009 and 90th percentile 1.165. The designated-window maximum is therefore 18.6, a factor of 18.4 above the bulk median: the frame is rejected for any plausible calibration pair (ρ, η), and the recorded coarse flag concurs ($F/\mu_0 = 1.258 > 1$).

Figure 5.3 displays both frames' complete fine spectra with the designated window, usable bulk, and rank baseline drawn; a secondary leakage feature near bin 168 is visible in the exemplar frame, and the rank baseline is insensitive to it — the robustness property of Section 5.5 exhibited incidentally on real data.

A more instructive companion is the cohort's weakest valid frame (2025 May 16). Its coarse statistic is $F/\mu_0 = 0.897$ — below the positive-excess boundary, so the coarse flag retains the frame — yet its designated-window maximum is 2.59 against a bulk median of 0.833, a ratio of 3.1: the fine designated-set rule rejects it at any multiplier below $\sim$3. Of the cohort's 7010 valid frames, only three fall below the coarse boundary, and this is one of them; on the fine axis it remains an unambiguous detection. (The full survey trawl re-processes a superset of these captures — 38 160 frames for this channel, of which 34 fall below the boundary, a comparable rate — and the example's quoted values reproduce digit-for-digit in the superset product, each fine spectrum deriving deterministically from the exact integer row sums of Chapter 6.) This is the coherent-integration sensitivity of Section 5.3 exhibited on survey data rather than in simulation, and it is the concrete evidence motivating the fine designated-set CFAR, rather than the coarse flag, as the candidate runtime decision. The calibration pair (ρ, η) required before deployment is set by the campaign of the calibration chapter; the values here are illustrative and do not themselves define an operating point.

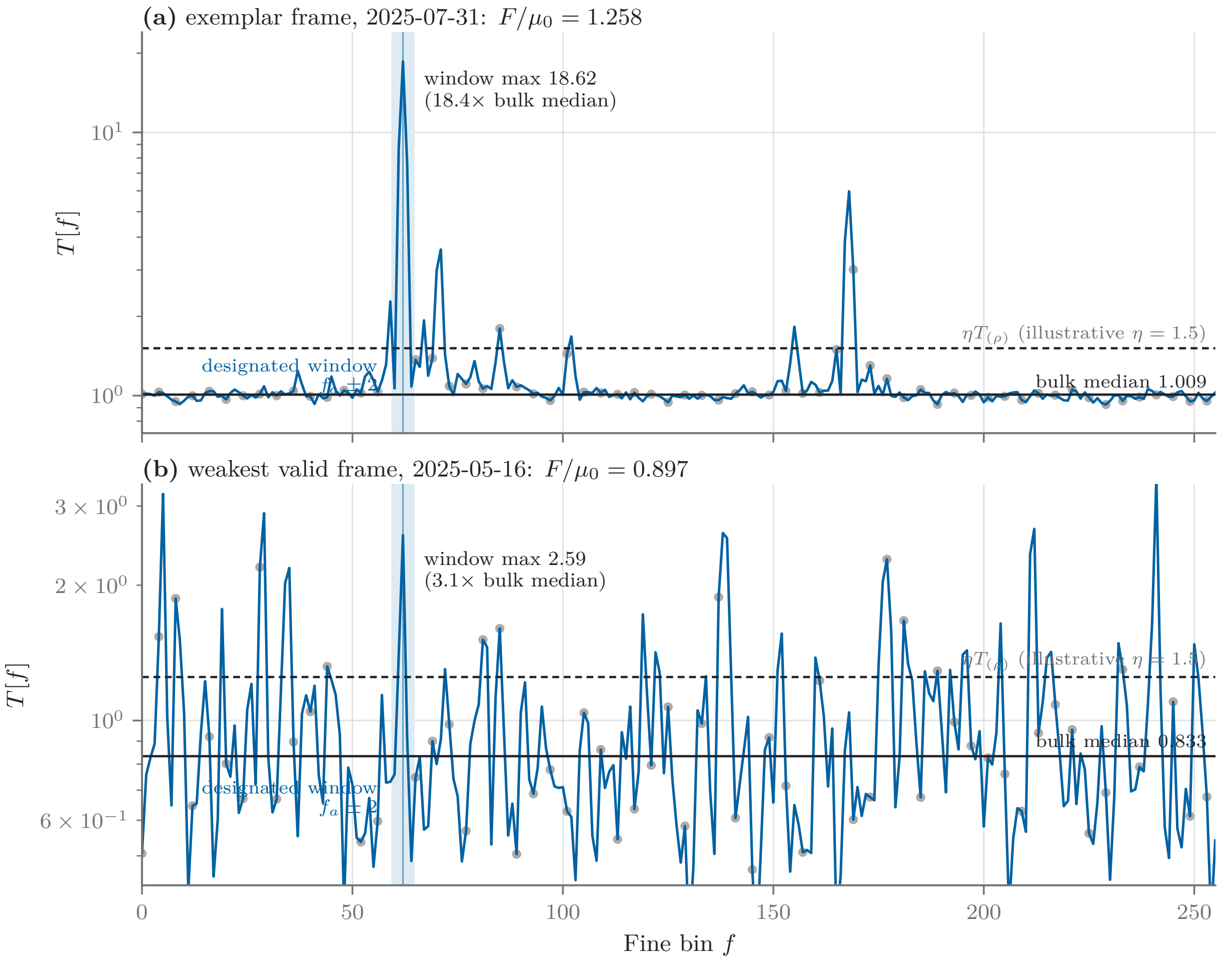

Figure 5.3: The worked example of this subsection: fine spectra $T[f]$ for the two channel-506 frames, with the designated window $f_a \pm 2$ shaded, usable-bulk bins marked, and the bulk median (solid) and illustrative threshold $\eta\, T_{(\rho)}$ (dashed) drawn. (a) The exemplar masked frame: window maximum 18.6, 18.4× the bulk median. (b) The cohort's weakest valid frame: the coarse statistic sits below the positive-excess boundary ($F/\mu_0 = 0.897$), yet the pilot stands at 3.1× the bulk median on the fine axis — the coherent-integration sensitivity of Section 5.3 on survey data.

Chapter 6

Exact-Arithmetic Implementation & Verification

6.1 The Exactness Contract

The implementation is governed by a single engineering contract: *the production-candidate GPU datapath and an independent reference implementation must produce bit-identical outputs for identical inputs* — not statistically consistent outputs, not outputs agreeing to a tolerance, but the same integers. This standard is unusual for scientific signal processing and is justified here in detail, since the remainder of this section depends on it.

Floating-point pipelines cannot meet it across implementations. Library FFTs on different platforms use different factorizations and operation orders; compilers contract multiply–add sequences into fused instructions at their discretion; parallel reductions reassociate sums nondeterministically. Two correct floating-point implementations of the same mathematics therefore differ in their final bits, and any comparison between them must adopt a tolerance. Tolerances carry three compounding costs in an unattended system: every verification gate becomes a statistical judgment; small systematic defects (an off-by-one window, a swapped conjugate, a biased rounding) can persist undetected within the tolerance; and when two eras of the system disagree, benign arithmetic drift cannot be distinguished from genuine regression. Exact integer arithmetic removes all three costs: equality is decidable per frame, and any defect that changes an *exercised* output becomes exactly detectable at the first comparison it reaches. This is an implementation guarantee, not a claim of complete test coverage or scientific correctness: an unexercised control path can still contain a defect, and bit equality cannot validate the signal, calibration, or cosmological transfer models. Reproductions across supported hosts, compiler versions, and hardware generations are expected to be exact once the same frozen specification and artifact manifest are used, so a mismatched bit is unambiguous evidence of disagreement even though a match is only evidence for the cases exercised.

The scope of the contract is as follows. Every quantity of Chapter 5 — the projections $\mathbf{Z}$, the spectra $\widetilde{B}_m$, the power sums $S_r[f]$ and P_r, and the mask decision itself — is computed in integer arithmetic with accumulator widths proven sufficient in Section 4.1, so no intermediate value rounds, saturates, or overflows. The single point at which the computation departs from real-valued mathematics is the fixed-point FFT's butterfly rounding, and that departure is itself part of the frozen specification (Section 6.3): deterministic, portable, measured, and identical in every implementation. Floating point appears only in derived diagnostics (the recorded F-statistics as `float32` quotients of exact integers), never upstream of a decision.

6.2 The Packed Fixed-Point Datapath

The detector consumes packed 4+4-bit complex samples, one byte per sample, real component in the high nibble (Section 2.2). The receiver's native encoding is excess-8 (offset binary); the ingest adapter repacks each byte losslessly to two's complement (per byte, XOR 0x88), and the kernel unpacks by sign-extending each two's-complement nibble; weights are stored in the same packed format on the same $[-7, 7]$ grid the receiver's 15-level quantizer emits. The coordinate convention noted in Section 4.2 is fixed here once for the whole system: the receiver's spectrally inverted channel ordering is absorbed by synthesizing weight tones with the conjugate (negative-exponent) rotation and time-reversing each detector window before projection; both choices are recorded as explicit fields of the runtime calibration bundle's detector contract, so the convention is auditable configuration data.

The projection stage — the dominant cost identified in Section 5.2 — runs on packed 8-bit dot-product instructions (DP4A), which accumulate four 8×8-bit products per instruction into a 32-bit register. The mapping from complex arithmetic onto these lanes requires care. Two consecutive packed samples unpack in registers into two lane words,

$$a_{\mathrm{re}} = (x_{0,\mathrm{re}},\ x_{0,\mathrm{im}},\ x_{1,\mathrm{re}},\ x_{1,\mathrm{im}}), \qquad a_{\mathrm{im}} = (x_{0,\mathrm{im}},\ -x_{0,\mathrm{re}},\ x_{1,\mathrm{im}},\ -x_{1,\mathrm{re}}),$$

while the corresponding two weight taps are pre-packed once per weight upload into a single shared lane $w = (w_{0,\mathrm{re}},\ w_{0,\mathrm{im}},\ w_{1,\mathrm{re}},\ w_{1,\mathrm{im}})$. One DP4A of a_{re} against w then accumulates $x_{\mathrm{re}}w_{\mathrm{re}} + x_{\mathrm{im}}w_{\mathrm{im}}$ for both taps — the real part of $x\,\overline{w}$ — and one DP4A of a_{im} against w accumulates $x_{\mathrm{im}}w_{\mathrm{re}} - x_{\mathrm{re}}w_{\mathrm{im}}$, the imaginary part. The full $K = 128$-tap complex projection for one window and one profile is thus 64 tap-pairs × 2 instructions, mapping exactly onto a 64-thread block with no idle lanes (the warp-exact geometry promised in Section 4.1); the 3×64 pre-packed weight lanes occupy 768 bytes, preloaded into shared memory per block. A scalar unpack-multiply-accumulate path implementing the identical arithmetic is retained behind a compile flag as the direct-validation reference for the packed path.

Completed dot-product components are bounded by $14336 < 2^{14}$ (Section 4.1) and held in 32-bit integers; squared magnitudes fit 32 bits; power accumulations use unsigned 64-bit throughout. The recorded coarse flag is likewise exact: the norm-corrected threshold comparison of Section 5.4 is evaluated as an integer cross-multiplication, $P_0 \cdot (\|\mathbf{w}_1\|^2 + \|\mathbf{w}_2\|^2) > \|\mathbf{w}_0\|^2 \cdot (P_1 + P_2)$, with the exact integer weight norms carried in the runtime calibration bundle — the rational form of μ_0, applied without ever forming a quotient, and forced to the non-rejection outcome when the reference power is degenerate.

6.3 The Frozen Fixed-Point Transform: the `fxfft` Family

The coherent integration of Section 5.3 requires an FFT that is bit-identical between the production-candidate kernel and the reference — which, per Section 6.1, no floating-point library can provide. The transform is therefore implemented within the project and frozen as specification `fxfft256` v1:

- *Structure:* a 256-point forward DFT in the negative-exponent convention of Section 5.3, radix-2 decimation-in-time, bit-reversed load of the 128 zero-padded window sums, natural-order output, in a fixed butterfly order.
- *Twiddles:* $W_n[k] \approx e^{-2\pi i k/n}$ in Q15 fixed point, $C[k] = \mathrm{nearest}(32768\cos(2\pi k/n))$, $S[k] = \mathrm{nearest}(-32768\sin(2\pi k/n))$ for $k < n/2$, frozen as *source literals* — no runtime trigonometry,

and the table is checkable by hash. No entry sits at a rounding tie, so "nearest" is unambiguous; the extremal values ± 32768 are held in 32-bit storage and are exact under the rounding rule.

- *Butterfly:* $t = \text{round15}(W[k] \cdot b)$, then $(a, b) \to (a + t,\, a - t)$ in 32-bit integers, with the complex product formed exactly in 64 bits before rounding.
- *Rounding:* $\text{round15}(v) = \lfloor (v + 2^{14})/2^{15} \rfloor$ per component. This rule breaks exact half-step ties toward $+\infty$; it is neither round-to-even nor symmetric round-away-from-zero. The verified build records the CUDA compiler version and explicit `-std=` language mode supported by that toolchain [34]. In C++20, right shift of a negative signed integer is specified as arithmetic division rounded toward negative infinity [33]; the host/device conformance gate additionally asserts `(-1 >> 1) == -1` and exercises negative half-step boundary vectors. Older language modes are not accepted solely on the strength of an implementation convention.
- *Scaling:* none. Correctness rests on a no-overflow lemma: one butterfly maps the working infinity norm M to at most $(1 + \sqrt{2} + 2^{-15})M + 1$, so eight stages from the input contract $|\cdot| \le 2^{20}$ remain below $2^{30.7} < 2^{31} - 1$. Evaluated row sums obey $|\cdot| \le 14336 < 2^{14}$, leaving six bits of spare contract headroom; the reference implementation enforces the contract and raises on any excursion rather than wrapping.

One master table, every supported length. The transform length is not a free parameter of the method but a consequence of the frame and window lengths, $L_F = 2N/K$; across the receivers and frame sizes contemplated here it takes values from 256 to 2048. Rather than freeze an independent table per length, the specification freezes a single master of $M/2$ entries and derives each member from it. The construction is an exact identity, not a re-derivation: a radix-2 decimation-in-time of length n requires $W_n[k]$ only for $k < n/2$, and $W_n[k] = W_M[kM/n]$ whenever n divides M; because the Q15 rounding is applied to the same real value either way, the length-n entries are *bit-identical* decimations of the master. Adding a supported length therefore introduces no new rounding decisions, and the absence of rounding ties — verified once on the master, whose closest approach to a tie is 5.8×10^{-4}, some eight orders of magnitude above the double-precision epsilon at that scale — is inherited by every member. The stride folds into the existing index arithmetic: stage s reads $\text{master}[\, t \ll (\log_2 M - s)\,]$, which at $L_F = 256$ selects exactly the entries the original 128-entry table held. The master is generated offline and emitted as source literals for the device, since runtime trigonometry would reintroduce precisely the cross-platform floating-point non-determinism the frozen specification exists to eliminate; a hash pins the table, and a build gate compares the emitted header against the generator.

The no-overflow lemma is likewise per-length rather than inherited. Applying the per-stage bound $\log_2 L_F$ times and requiring the result below 2^{31} gives the admissible input norm, which *tightens* as the transform grows: 2^{20} at $L_F = 256$, then 2^{18}, 2^{17} and 2^{16} at 512, 1024 and 2048. The eight-stage contract is not reusable at nine stages. All four admit the implemented row-sum bound of 2^{14}, and each is asserted at compile time against the accumulator width actually configured.

Downstream of the transform, exactness resumes unconditionally: spectral magnitudes squared and their sums over $M = 2048$ feeds are computed in unsigned 64-bit integers, with evaluated magnitudes small enough that the frame-level sums sit far below overflow. The specification is realized three times — a numpy integer reference, a portable C port, and the CUDA device implementation — and all three are required to reproduce a golden-vector corpus of 56 input/output pairs spanning synthetic families, survey-measured pilot offsets, and genuine `int4`-dot row-sum arithmetic, byte-for-byte. The corpus, the twiddle table hash, and the rounding rule together define the transform: any arithmetic change constitutes a new specification version.

The cost of the fixed-point rounding is measured directly. Against the analyzer's `complex128` pipeline, the frozen transform's spectrum differs by at most 6.3×10^{-4} of the spectrum peak across all golden families, and the end-to-end fine statistic moves by at most 5.9×10^{-4} relative (-0.0026 dB at a measured detection peak) — five orders of magnitude below the coherent-integration effect the transform exists to capture. A deterministic-float alternative (fixed operation order, fused multiply–add disabled) was considered and rejected: although it can be made deterministic on a single platform, its determinism depends on compiler flags rather than on the specification itself, and it departs from the integer arithmetic on which every other verification gate in this section relies.

6.4 Exact Rational Decision Arithmetic

The decision rule of Section 5.5 is a chain of comparisons among ratios of 64-bit integers. The implementation never forms the ratios. Each per-bin statistic is carried as an integer pair $(n_f, d_f) = (2S_0[f],\, S_1[f] + S_2[f])$, and every comparison is evaluated by cross-multiplication in fixed-width arithmetic: $n_i/d_i < n_j/d_j$ becomes $n_i d_j < n_j d_i$, computed exactly as 128-bit products (high and low 64-bit halves) — well within bounds, since the Parseval identity caps evaluated power sums near 2^{55}. The rank selection uses these comparisons in a counting form: bin i holds rank ρ iff the number of bulk bins strictly below it does not exceed ρ while the count including ties does. This formulation is order-free by construction — the selected *value* is unique even under exact ties, and because tied bins are equal as rationals, any representative yields the same downstream decision, so the result is independent of thread scheduling by construction rather than by locking. The final threshold comparison, $n_f \cdot 2^{16} \cdot d_\rho > \eta_{q16} \cdot n_\rho \cdot d_f$ with the multiplier in Q16 fixed point, is a comparison of triple products; it is evaluated in explicit 192-bit arithmetic (each side as three 64-bit limbs), exact for arbitrary 64-bit operands, independent of the operand bounds. The degenerate rules of Section 5.5 are implemented as stated: zero-denominator bins are excluded from the bulk and can never fire in the designated set, and a bulk not deeper than ρ forces the non-rejection outcome.

The reference for this arithmetic is an arbitrary-precision implementation (Python integers), itself gated against a rational-number oracle. The 192-bit comparison is written as explicit schoolbook limb products and carry propagation: each 64×64 product contributes a low and high limb, carries are added with overflow detection, and the terminal carry is required to be zero because each side is the exact product of three 64-bit operands represented in three 64-bit limbs. Boundary vectors cover 0, 1, $2^{64} - 1$, repeated maximal limbs, exact ties, and operands chosen to generate carries at every stage. A fuzz gate then drives 2×10^6 pseudo-random operand sets through both the fixed-width code and the arbitrary-precision reference. The randomized agreement is strong empirical evidence, but the completeness claim rests on the explicit limb construction and terminal-carry invariant rather than on fuzzing alone.

6.5 The Fused Kernel

The production-candidate form computes the entire pipeline — packed samples in, one mask bit out — in a single kernel launch. The motivation is the deployment footprint of Section 2.3: a staged implementation must materialize the 3×262144 complex row sums to device memory (≈ 6 MB per frame) and re-read them for the transform, while the fused form keeps each stream's row sums entirely in on-chip shared memory, touching global memory only for the 3×256 exact power sums and the final mask word.

The kernel assigns one 64-thread block per stream (2048 blocks per frame) and proceeds in four phases. *Phase 1:* the block walks the window axis with a thread stride, each thread taking whole windows and computing their projections for all three profiles via the DP4A path of Section 6.2 — no cross-thread reduction, and no requirement that the window count divide the block size — writing the $3 \times L$ complex row sums to shared memory and accumulating per-thread contributions to the coarse marginals in registers. *Phase 2:* the block reduces and atomically adds its coarse marginals to the frame totals — making the exact-integer counterpart of the Parseval identity (Eq. (5.12)) an *internal* property of the launch — and, when a debug tap is bound, exports the row sums; in production the tap is null and row sums never exist off-chip. *Phase 3:* the block executes the frozen transform cooperatively for each profile: bit-reversed load into a shared 256-point buffer, then eight butterfly stages of 128 butterflies spread across the 64 threads, barrier-synchronized between stages. Within a stage, butterflies write disjoint index pairs, so the produced bits are identical for any thread schedule — the specification's fixed butterfly order is preserved not by serialization but by the disjointness of the writes. Squared magnitudes are then accumulated atomically into the global fine power sums, exact because integer addition is associative and commutative — the accumulation *order* is nondeterministic, the *result* is not. *Phase 4:* a per-frame completion counter folded into the mask word is incremented by each finishing block behind a memory fence; the last-arriving block observes the full count, re-reads the finalized power sums, and executes the decision arithmetic of Section 6.4 cooperatively — bulk census, rank selection, designated-set comparison — overwriting the counter with the mask bit. The public launch wrapper owns a stream-ordered zero of the mask word immediately before the launch, and the raw kernel entry point is internal. A caller that bypasses this contract can begin from a false completion count, select the wrong finalizer, or prevent any block from observing the expected count; the outcome is therefore invalid rather than a stale-but-usable mask. The acceptance suite deliberately pre-fills the word with a nonzero pattern and verifies that the wrapper restores correct behavior. The per-channel calibration inputs (anchor, window width, bulk mask, rank, multiplier) enter as kernel arguments from the runtime calibration bundle: the library compiles in no channel policy, so calibration campaigns modify data rather than code.

The wrapper also enforces the remaining launch preconditions: output power arrays are zeroed or freshly allocated in the same stream; calibration dimensions match the compiled geometry; all buffer spans are large enough for the declared frame; and a CUDA error or failed post-launch tripwire marks the frame invalid rather than silently retaining or rejecting it. These are part of the API contract because the exact kernel cannot repair an incorrectly initialized host-side state.

Figure 6.1 sketches the launch structure and its global-memory contract.

The staged entry points (row sums, fine powers, marginals as separate launches) remain in the library as the validated intermediates; the fused kernel's acceptance criterion is bit-equality with their composition, and the debug taps let any deployment reproduce the staged outputs on demand.

6.6 Verification Methodology & Measured Performance

Verification is layered so that each property is established by the cheapest gate that can decide it, and every layer's criterion is exact equality.

1. *Frozen-specification gates.* The 56 golden vectors pin the transform across all three implementations; the twiddle table is hash-checked; a generator cross-check reproduces the table from first principles in the test suite.

2. *Reference-chain gates.* The decision reference is gated against a rational-number oracle across

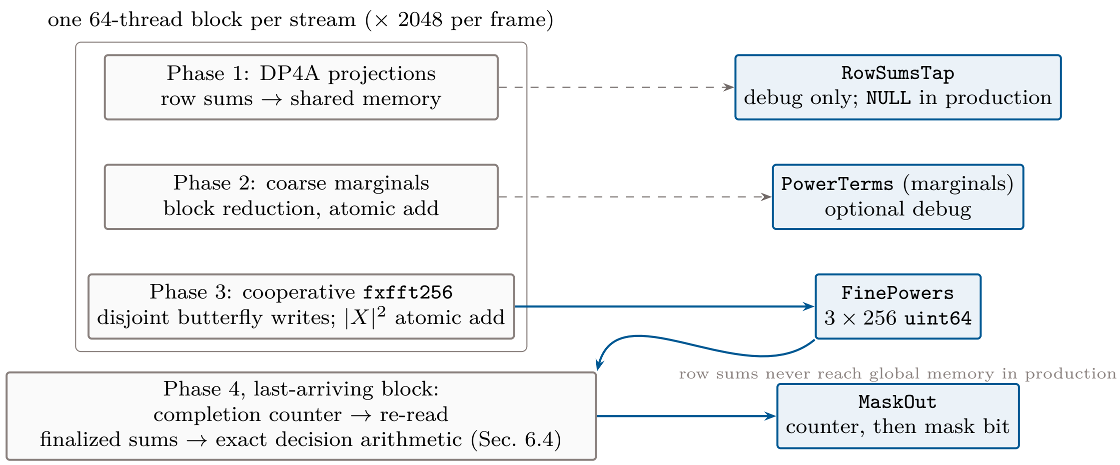

Figure 6.1: Launch structure of the fused kernel (Section 6.5). Each block computes one stream's row sums entirely in shared memory, contributes exact atomic sums to the global fine powers (and, when bound, the debug taps), and increments a per-frame completion counter folded into the mask word; the last-arriving block re-reads the finalized sums and executes the exact decision arithmetic, overwriting the counter with the mask bit. Blue boxes are the only global-memory traffic.

randomized, tied, degenerate, wrapped, and near-overflow cases; the wide-limb comparisons are fuzz-gated against arbitrary precision over 2×10^6 trials.

3. *Concurrency emulation.* Before any GPU execution, the device source is compiled unmodified into a threaded CPU harness: one POSIX thread per GPU thread, a barrier for each `__syncthreads`, mutexed atomics, the scalar dot path substituted for DP4A. This executes the kernel's true concurrent structure — including the cooperative transform and the last-block epilogue — under real thread interleaving, and its outputs are required to match the reference chain bit-for-bit across noise, injected-pilot, wrap-anchor, degenerate-bulk, and tap-unbound scenarios, stably across repeated concurrent runs.

4. *On-hardware gates.* On the benchmark GPU, the fused kernel must reproduce the composed staged path and the reference bit-for-bit; an injected pilot must fire the mask; the production (tap-unbound) form must equal the debug form; and twenty repeated launches must produce identical bits — the one property the CPU emulation cannot probe, since it exercises the completion-counter epilogue under genuine grid scheduling.

5. *Production tripwire.* The exact marginal identity is the acceptance tripwire for a live integration: the coarse power sums are computed twice, through independent code paths, and compared as integers every frame. Any fault in memory layout, accumulation, scheduling, or a future regression breaks an exact equality on the frame where it occurs.

6. *Cross-era reproduction.* Two processing cohorts, built independently with different library binaries and CUDA toolkits, processed overlapping archived events; joined on event identity, the compared statistics agreed with zero mismatches. For the final reproducibility record, "zero" is reported together with the joined event and frame counts, join completeness, invalid and missing records, exact fields compared, and immutable cohort manifests. Those counts live in the immutable processing ledger rather than in this document, so the mismatch count is not turned into an undocumented denominator here. Under a tolerance regime the comparison

Table 6.1: Legacy paired implementation-loss measurement. The entries are the horizontal difference between the full-precision and packed-`int4` crossings on the same trials; negative values indicate that the packed crossing was slightly more favorable. These are measurements of the earlier 512-row, coarse-statistic geometry, not of the present two-stage detector.

Operating crossing	Point estimate (dB)	Paired-bootstrap 95% interval (dB)
Fixed threshold, $P_\mathrm{d} = 0.5$	-0.047	$[-0.121, +0.022]$
Fixed threshold, $P_\mathrm{d} = 0.9$	-0.022	$[-0.100, +0.051]$
Positive excess, $P_\mathrm{d} = 0.9$	-0.063	$[-0.238, +0.121]$

would require a distributional judgment; under the exactness contract each joined field reduces to equality.

7. *The build is not the artifact.* One further finding completes the procedure: compiled binaries are *not* byte-deterministic — clean rebuilds of identical sources differ in embedded compiler artifacts. The verified binary itself, content-addressed by cryptographic hash and recorded in the processing ledger, therefore serves as the reference artifact; a rebuild is a new artifact that must re-pass the verification gates before use.

Measured performance on the benchmark-class GPU (A100) is summarized as follows. Against the 41.94 ms frame cadence, the staged three-launch composition costs 2.14 ms; the fused kernel completes the samples-to-powers datapath in 0.77 ms (a 55× real-time margin), with the debug tap adding only $\approx 5\ \mu$s when bound; and the full production-candidate form including the decision epilogue costs 0.90 ms — a 47× margin, with the entire decision stage contributing +0.13 ms because only one block per frame executes it. The complete evaluated detector therefore occupies approximately 2% of its real-time budget.

6.7 Controlled Injection and Measured Legacy Implementation Loss

Exact equality establishes what the implementation computed; it does not establish how much sensitivity the representation costs. An earlier detector generation therefore performed a paired offline injection experiment that remains useful as a validation pattern. A standards-chain 8-VSB waveform was produced with the GNU Radio ATSC randomizer, Reed–Solomon encoder, interleaver, trellis encoder, field-sync multiplexer, and RRC pulse shaping. Complex AWGN was added, the signal was passed through the four-tap sinc–Hamming reference PFB, and the samples were quantized to offset-binary `int4` at a declared 3σ clipping level. The sweep covered shelf SNR from -38 to -24 dB and pilot offsets of -1, 0, and $+1$ kHz, with 1000 trials at each of the 45 grid points. Each trial carried both the packed-integer statistic and a matched full-precision-weight control, so the implementation comparison was paired rather than inferred from independently generated curves.

Table 6.1 disfavors quantization as the dominant term in the observed +0.27–0.32 dB ideal-to-measured crossing envelope, although the widest interval still permits an implementation contribution near 0.24 dB. A same-seed CPU/GPU spot check at -31 dB supplied an orthogonal gate: all 200 packed trials agreed exactly in the statistic and in both sets of decisions (126/200 fixed-threshold and 197/200 positive-excess detections). The weight-bank hash also matched between the CPU sweep and GPU session. This is stronger than comparing summary plots, but it remains a test of the earlier kernel and decision geometry.

The experiment also exposed why an injected signal must be audited independently of the detector. The nominal golden capture contained a generator startup transient: its first 5.6 ms block carried 9.7 dB less pilot power than the stationary mean, the second was 0.4 dB low, and subsequent blocks were stable. After cropping to the stationary span, two independent estimators agreed within 0.007 dB; the steady pilot was 0.136 dB above the nominal ratio and required an amplitude retrim by 0.9845. A full-capture scalar had therefore mixed generator nonstationarity with detector loss. The durable protocol is to crop, measure, and trim the exact span consumed by the evaluator before generating the sweep, and to report waveform calibration, model discrepancy, and implementation loss as separate terms.

A still earlier CUDA prototype reported a shelf-equivalent mean bias of −0.309 dB and maximum absolute bias of 0.59 dB, together with 216.23 blocks/s (29.02 GB/s of packed input) against its stated 3051.76-blocks/s requirement. Those numbers characterize another kernel and another loss definition; they are not contradictory to the paired crossing shifts and are not portable correction constants. Appendix B records the provenance and reuse boundary for both experiments.

The present transform-only comparison of −0.0026 dB at a measured peak is narrower still: it isolates the frozen transform, not the complete signal chain. Before a sensitivity number becomes a headline result for this dissertation, the paired experiment must be regenerated with the current $K = 128$, $L = 128$, full-2048-input two-stage geometry, the designated-set statistic, empirical null calibration, and the forecast-selected rational threshold. The legacy study supplies the experiment design and an honest prior bound; it does not substitute for that current-geometry measurement.

Chapter 7

The Survey: Captures, Trawl, & Products

The survey is the bridge between a verified detector and a scientific claim. It applies the detector to archival telescope data, preserves enough intermediate information to revisit the decision later, and records where the archive cannot support an inference. This chapter therefore distinguishes three objects that are easy to conflate: the *available archive*, the *processing inventory*, and the *population of telescope time* about which one might want to make a statement. The first two are measurable from logs. The third requires a selection model, because the archive consists of triggered snapshots rather than a uniform observing stream.

7.1 Triggered Baseband Captures and Their Selection Function

The raw material is the archive of full-array baseband captures recorded by the telescope's transient backend [4]. A capture contains the packed 4+4-bit channelized stream for all $M = 2048$ inputs and spans seconds rather than a continuous day. The trawl completed in August 2026, and the archive now carries products for all 23 ATSC allocations (channels 14–36), each complete against its archive holdings: 9,214 distinct capture events probed, of which 8,983 contribute at least one valid frame; 170,374 event–channel processing units; 750,461 detector frames; a median of three $N = 16384$ detector frames per acquisition; calendar coverage from December 2018 through July 2026; three units quarantined at the archive (two truncated, one with an unreadable header); and a single detector-kernel cohort across every product. Per-channel event counts are far from uniform — 1,543 to 9,045 — and the deficit is concentrated on exactly the two most contaminated channels, for the selection reason discussed below. Those figures describe the working archive at snapshot time; the immutable release ledger, rather than this prose, is the authoritative denominator.

The captures were not selected by the DTV detector. They were initiated by the transient system and filtered by telescope operations, data availability, archive retention, and later inventory rules. One of those filters is now measured and deserves emphasis because it is *endogenous to the contamination itself*: the telescope curtails baseband collection on its most contaminated coarse channels, and captures on the channel-30 monitored bin cease entirely after September 2023 — an operations exclusion driven by the DTV occupancy this survey measures. Archive completeness is therefore correlated with the quantity under study; the thin channel-30 and channel-24 inventories are a selection effect, not a property of the transmitters, and a channel excluded from collection can never report a subsequent transmitter sign-off from this archive (five sign-offs or station departures were recorded on collected channels during the survey span, so the possibility is not hypothetical).

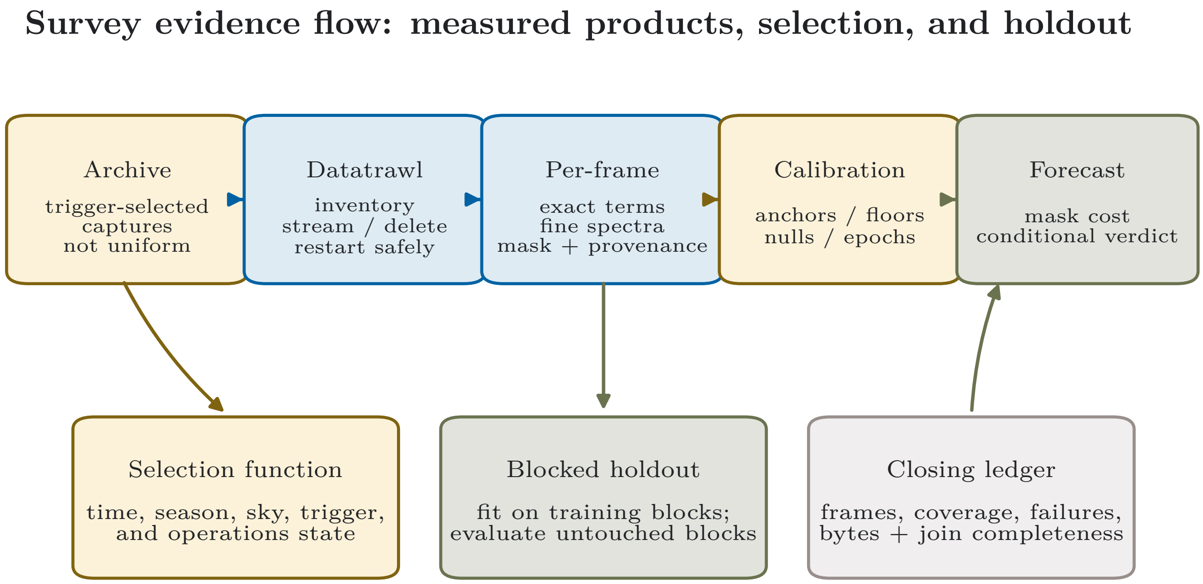

Figure 7.1: The survey evidence flow. Triggered baseband captures are inventoried before processing; the restartable trawl produces exact detector terms, fine spectra, decisions, and provenance; calibration and forecast products are derived from those records. The dashed boundary is load-bearing: a triggered archive is not automatically a random sample of transmitter occupancy. Selection diagnostics and blocked holdout validation are therefore part of the scientific product, not optional metadata.

Let $I(t, c)$ be the indicator that telescope time t on channel c enters the survey. The occupancy quantity measured directly by the survey is

$$\Pr(\text{DTV state} \mid I = 1, c), \tag{7.1}$$

not the unconditional time occupancy $\Pr(\text{DTV state} \mid c)$ unless the inclusion process is ignorable for the variables controlling propagation and transmitter state. Time of day, season, sidereal time, weather, trigger class, commissioning state, and station-era boundaries are all plausible dependencies. The archive can still support a strong detector and calibration study, but its masked fractions are explicitly *archive fractions* until the selection comparison is completed.

Two consequences recur later. First, the snapshot duration caps any contiguous block statistic that can be reproduced from this archive. Second, the sparse spacing leaves a correlation-time blind interval between the frame duration and typical acquisition separation. Chapter 8 therefore allows the correlation estimator to return a measurement, a bound, or a refusal rather than interpolating across that gap.

7.2 A Restartable, Bounded-Storage Trawl

The trawl processes every eligible capture through the same detector arithmetic described in Chapters 5 and 6. It is organized in three nested units: an inventory entry identifies an archived

Table 7.1: Survey inventory fields and their status at the current snapshot. "Not supplied" is an evidence state, not a numerical zero. The final analysis release should generate this table directly from the immutable processing ledger.

Inventory field	Draft value/status	Required release record
Calendar coverage	Dec 2018–Jul 2026	First/last valid UTC and gaps by year and channel
Capture events per analyzed channel	1,543–9,045 (9,214 distinct events; all 23 allocations)	Exact discovered, eligible, processed, skipped, failed, and quarantined counts
Frames per acquisition	median 3	Full distribution and valid/invalid frame counts
Longest contiguous statistic used here	1.38 s	Distribution by acquisition and reason for truncation
Raw and derived storage	not supplied	Bytes discovered, transferred, cached, deleted, and retained
Trawl throughput and wall time	not supplied	Frames/s, data rate, GPU time, retries, and end-to-end wall time
Selection diagnostics	incomplete	Coverage by UTC, local time, season, sidereal time, trigger class, and operations state

acquisition, a processing unit identifies a contiguous readable segment, and a detector frame is the N-sample decision quantum. This separation allows the operational machinery to restart without changing the scientific definition of a frame.

The processing environment cannot stage the entire archive locally. The trawl therefore follows a bounded-storage cycle: discover an eligible source, pull one unit, validate it, process all complete frames, write a content-addressed product, verify the product, and remove the staged source only after the write is durable. A unit that fails validation is quarantined with its exception and provenance; it is not silently omitted or counted as a quiet frame. Resume logic treats an already verified product as complete only when the source identity, detector geometry, weight and transform hashes, schema, and science-relevant configuration match. A source-code change that leaves those immutable tokens unchanged can be recorded without forcing a scientifically identical re-run; a kernel, geometry, or schema change creates a new cohort.

The trawl architecture is intentionally detector-agnostic at the operational layer. Inventory, transfer, retry, checkpoint, quarantine, and deletion belong to the data-management system; the pilot proxy enters through an analyzer interface that consumes a frame and emits a declared product schema. This boundary is why failures in archive access do not become hidden branches in the signal-processing code, and why the same operational core can support other large telescope reductions.

7.3 Products: Statistics, Not Only Verdicts

Each product using schema `pilotproxy_detector_datatrawl_v3` retains, per frame, the exact target and reference power terms; floating diagnostics such as the coarse ratio and inferred shelf level; validity and rejection fields; the complete 256-bin fine statistic; diagnostic CFAR location and scale fields; and any-bin detection locations. Product-wide metadata retain the source identity, weight-bank and transform hashes, resolved reference placement, exact integer norms, detector and schema versions, and the within-channel spectra used for the broadcast characterization.

This choice makes thresholding reversible. A product that stored only a mask bit could answer what one historical policy rejected. A product that stores the integer terms and fine spectrum can answer what a later per-epoch policy *would have* rejected, provided the later policy uses quantities already in the schema. The science threshold, rank, bulk mask, and designated window can therefore be recalibrated without fetching the raw capture again. It also keeps implementation verification separate from policy: exact power terms can agree bit-for-bit even when a new calibration bundle intentionally changes the mask.

7.4 Reproduction and the Required Denominator

Two independently built processing cohorts, using different library binaries and CUDA toolkits, processed overlapping archived acquisitions. On records joined by event identity, the compared detector statistics had zero mismatches. That is the expected payoff of the exactness contract, but "zero" is not a complete result without its denominator. The final release record must include

- the number of discovered, eligible, and successfully joined events and frames;
- join completeness and the treatment of duplicates, missing products, and invalid frames;
- the exact fields and byte representations compared; and
- the source, binary, toolkit, weight, transform, schema, and cohort hashes.

Those values live in the immutable processing ledger and are quoted only where that ledger supplies them. Appendix A records the comparison as measured but incompletely reported until the immutable ledger is attached.

7.5 Selection-Aware Calibration and Holdout Design

Randomly splitting individual frames would overstate independence: neighboring frames within a capture share receiver state, transmitter state, and propagation, and frames across a station era can share a persistent offset. Calibration and evaluation are therefore blocked at the acquisition or sidereal-day level and stratified by measured station epoch. Within each epoch, one block set localizes anchors and selects (ρ, η); disjoint blocks estimate false alarms, masked fraction, residual levels, and cost. A stricter leave-one-year-or-era-out check asks whether the selected operating point transports across calendar structure.

The holdout design cannot make the archive uniform, but it prevents a second, avoidable optimism: evaluating a threshold on the same correlated frames that selected it. The final campaign reports both the within-archive blocked result and the coverage diagnostics of Table 7.1. Only then can an archive fraction be promoted to an estimate of observing-time occupancy, and only over the population for which the coverage comparison supports that move.

Chapter 8

Calibration: Anchors, Nulls, & Operating Points

Calibration turns the detector's fixed arithmetic into per-channel, per-epoch policy data. It does not retune the pilot to whatever line happens to be strongest: the synthesized target remains the nominal ATSC pilot. Calibration localizes the residual offset around that target, characterizes the empirical null, identifies intervals for which a floor or correlation time is measurable, and selects an operating point under the cosmological cost constraint. Every estimator is allowed to refuse when the archive lacks the population it requires.

8.1 Anchor Localization

For fine bin f, let T_f^{on} and T_f^{quiet} denote the sets of recorded fine statistics in coarse-detected and coarse-quiet frames. The primary anchor estimator is

$$\widehat{f}_a = \arg\max_f \left[\mathrm{median}(T_f^{\mathrm{on}}) - \mathrm{median}(T_f^{\mathrm{quiet}})\right]. \tag{8.1}$$

Static structure common to both cohorts cancels, so the contrast isolates the feature associated with the transmitter state rather than simply returning the largest fixed instrumental bin. A channel with no defensible quiet cohort uses a plain-median fallback and is labelled as such; that fallback is adequate to place a designated window around an always-present line but does not create a transmitter-off null.

The current products place the dominant anchors from hundreds of hertz to approximately -1.37 kHz from the synthesized tone, within the ± 1.526 kHz fine span; the lower-band completion extends the same picture, with its persistent carriers anchored between -0.70 and $+1.26$ kHz. Calendar splits also reveal station changes rather than gradual drift: channel 32 moves by 11 fine bins and channel 35 by 29, and channel 17's recovered anchor shifts between its two on-epochs as the later transmitter configuration comes to dominate the archive. The correct calibration object is therefore (c, e), channel and epoch, not channel alone. When an epoch boundary is known, the bundle carries separate anchors; a wide union window is a temporary compatibility measure, not a substitute for an epoch-specific calibration.

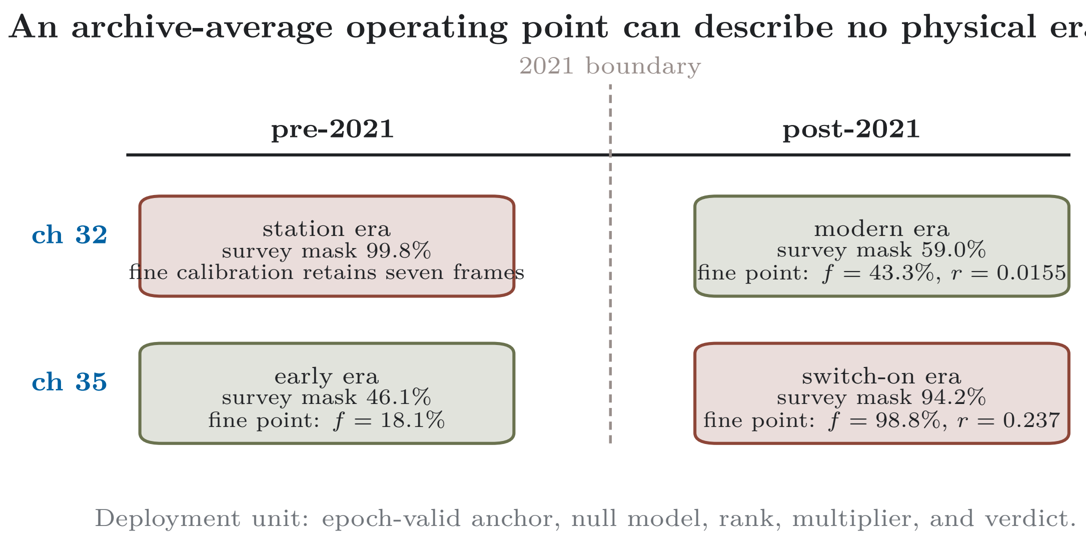

Figure 8.1: Why an archive-average operating point can describe no physical era. A station departure can move a channel from the occupancy wall into a usable region, while a switch-on can move another channel in the opposite direction. Anchor, bulk mask, threshold, and cost are therefore calibrated and versioned by epoch. The numbers shown are the prototype epoch-split values; the Q16 thresholds come from the epoch-restricted fine operating-point rerun (Table 8.1). The frozen export additionally carries the epoch pairs of the dated sign-off channels 19, 20, 26, and 27 (Section 9.12); the figure keeps the original two worked examples.

8.2 A Floor Is Not the Smallest Observed Value

A sensitivity floor is a property of a specified null population and estimator. It is not the minimum value observed in a mixture of quiet and contaminated frames. For a channel with verified quiet data, calibration reports the empirical null distribution of the selected statistic, its sample size, the block structure used for uncertainty, the tail quantiles relevant to the false alarm requirement, and the corresponding shelf-level mapping. A robust core width and an ordinary standard deviation are both useful because a thin contaminated tail can leave the former nearly ideal while inflating the latter.

Channels that are effectively always on are censored in the wrong direction for an empirical off-state floor. For them, the calibration can still report:

1. an analytical or injected-signal sensitivity under the declared noise model;
2. the distribution of retained-frame proxy levels under a chosen policy; and
3. an upper bound on a quiet-state floor if an independently verified off epoch exists elsewhere.

It cannot call the least contaminated retained frame an off-state measurement. This distinction is especially important for channels 35 and 36, for which the full-archive null population is absent or severely limited. Parenthesized or bounded values in Chapter 9 remain diagnostic rather than silently becoming measured floors.

The fine null is calibrated empirically because the independent-feed Gaussian model of Chapter 5 is only a baseline. Feed covariance, receiver gain structure, spectral leakage, and intermittent non-Gaussian features change the effective width and tail. The exchangeability check for an order statistic is performed on held-out quiet blocks: a bulk bin tested against the rank of the remaining bulk should exceed it at the combinatorial rate implied by that rank. Agreement is evidence for the rank calibration on that cohort, not a universal guarantee across station epochs.

8.3 Correlation Time and Refusal Semantics

The residual chain needs a coherence scale, but the archive contains dense samples only within a short capture and sparse samples between captures. The estimator therefore uses same-day or same-epoch structure functions with bootstrap resampling over sidereal-day blocks and reports one of three outcomes:

$$\widehat{\tau}_c \in \{\text{measured interval, one-sided bound, refused}\}. \tag{8.2}$$

A refusal is propagated as a conservative sidereal-day cap in the screening forecast and is visually and numerically distinguished from a measurement. Channels 31 and 35 currently carry measured values near 34 and 45 minutes, respectively; channel 33 carries an upper bound; and channels 29 and 34 remain sensitive to the unresolved cadence gap. Because the verdict for channel 29 can change by orders of magnitude across that gap, a contiguous-scan campaign is a required measurement rather than an optional precision improvement.

8.4 Per-Epoch Operating Points

For each channel and epoch, the calibration searches candidate decision policies and minimizes the forecast cost

$$\mathcal{C}(\pi) = \frac{1 + r_{\mathrm{var}}(\pi)}{1 - f(\pi)} \tag{8.3}$$

subject to the separate coherent-bias constraint $r_{\mathrm{sys}}(\pi) \le r_{\mathrm{sys,tol}}$ and the stability gates of Chapter 9. Until the visibility transfer is measured, the survey reports r_{proxy} and carries $r_{\mathrm{var}} = g_{\mathrm{var}} r_{\mathrm{proxy}}$ and $r_{\mathrm{sys}} = g_{\mathrm{sys}} r_{\mathrm{proxy}}$ as explicit calibration closures rather than silently setting them equal. On the coarse axis the policy is a threshold η; on the fine axis it is $(f_a, \mathcal{D}, \mathcal{B}, \rho, \eta_{q16})$. Feasibility must hold on the full calibration block and on its era-preserving halves. Among policies within a 2% cost plateau, the selection minimizes the worst holdout cost before applying fixed tie-breaks. This prevents a numerically sharp optimum from being chosen on a statistically flat surface.

The rerun also reports the calendar-late residual that certifies or refuses each pair: $r_{\mathrm{late}} = 0.008$ on the era-stable channel 31 (pair 1.50/98304 — selected, feasible, and still the vacuous occupancy-wall point of Table 9.4); 0.016 on channel 32 (more precisely 0.0155 against its 0.0156 tolerance — feasible on the late half by a hair, the knife edge the coarse margin already showed); and 0.237 on channel 35, the switch-on era's refusal of a pair its quiet era supports.

Table 8.1 makes the consequence concrete. Channel 32's current epoch is more favorable than its archive average, while channel 35's is less favorable. A single archive-wide threshold would blend physically different transmitters, anchors, and duty cycles. The runtime bundle therefore carries an epoch validity interval and refuses to load a calibration outside it.

Table 8.1: Prototype epoch split for channels whose station environment changes the operating regime. The $(\eta^*, \mathrm{Q16})$ pair is per channel, from the epoch-restricted rerun of the fine operating-point selection (`fine_operating_point.py`; the released `fine_operating_points.csv` is authoritative): one pair per channel, with feasibility required on the full archive and in each calendar half, so the same pair appears against both eras and the per-era residual is what certifies or refuses it.

Channel	Epoch	Coarse masked	Fine masked f	Fine residual r	η^* / Q16
32	pre-2021 station era	99.8%	7 frames retained	not certified	1.15 / 75366
32	post-2021 departure	59.0%	43.3%	0.0155	1.15 / 75366
35	pre-2021 quiet era	46.1%	18.1%	dilation tier	7.61 / 498656
35	post-2021 switch-on	94.2%	98.8%	0.237	7.61 / 498656

8.5 Legacy Sensitivity Scale and the Cost of Tail Cutting

The earlier positive-excess detector admits a useful dimensionless sensitivity coordinate. If a shelf SNR s moves the mean statistic by Cs and the idealized null width is

$$\sigma = \sqrt{\frac{1}{R} + \frac{1}{2R}}, \qquad k = \frac{Cs}{\sigma}, \tag{8.4}$$

then the detection curves at different row counts collapse approximately onto one curve in k. In the retained development study, the width crossing $k = 1$ gave $P_\mathrm{d} = 0.833$ at -34.30 dB for the measured 512-row trials; the analytically scaled full-frame value was $P_\mathrm{d} = 0.841$ at -47.85 dB. The latter was not a deployment-scale injection measurement and is retained only as the scaling prediction that the current-geometry sweep must test.

The same coordinate clarifies why moving a keep threshold deep into the left tail is not a substitute for sensitivity. Under the shifted-Gaussian model, a threshold two null widths below the mean retains 2.3% of clean frames rather than 50%: approximately $22\times$ less exposure and $4.7\times$ worse map noise at fixed survey time. The contamination fraction among retained frames improves by only $1.2\times$ for $k = 0.1$, $1.6\times$ for $k = 0.3$, and $5.4\times$ even at $k = 1$. Below the width crossing, both signal and null are retained at nearly the same rate, so purity changes slowly while exposure falls exponentially. Moreover, low statistic values need not be especially pure: power entering a reference cell depresses the ratio, so reference contamination can populate the far left tail.

This legacy calculation explains the role of the coarse mean crossing as a parameter-free health diagnostic, not as the final science policy. The present fine-stage rule instead calibrates its rank and multiplier on measured transmitter-off epochs and selects the operating point against the cosmological cost of Chapter 9. Injection retention, a pilot-free floor, reference-cell vetoes, and per-channel occupancy priors are the measurements that constrain the $k \ll 1$ regime. Appendix B preserves the older derivation and its source products so this design argument can be reused without importing its superseded detector geometry.

8.6 Closing Calibration Campaign

The final campaign produces one immutable record per channel–epoch pair: anchor and uncertainty; designated and bulk masks; null sample count and block definition; false-alarm curve; selected rank and Q16 multiplier; masked fraction and residual on disjoint holdout blocks; correlation-time outcome; proxy-transfer version; and all source/product hashes. It also includes simultaneous spectra across each 6-MHz allocation and contaminated complex visibilities before and after the sidereal filter. Those latter products close the two transfer gates that a pilot-only calibration cannot close by itself.

Chapter 9

Noise Tolerance: Pricing the Mask for BAO Cosmology

Chapters 3–6 built a detector and verified its exercised digital behavior at the bit level; nothing yet has asked what it is *worth*. This section constructs the machinery that answers that question in the survey's own currency — integration time to a fixed BAO measurement — and it is built under an explicit falsifiability requirement: the machinery must be able to return a verdict in either direction. Two negative verdicts are admissible in advance. The residual DTV contamination surviving the full pipeline may overwhelm the BAO signature at any realizable integration time, in which case the band should be excised and the detector's contribution is the *measurement* that justifies excision; or the masking required to suppress the residual may discard so much data that the integration time to a fixed target becomes unrealistic, in which case the operating point — not the method — is wrong. A framework that can only return "the detector helps" is advocacy, not analysis, and every construction below is arranged so that the data can contradict it.

The scope contract is equally explicit. The cosmological pipeline downstream of the mask — delay filtering of spectrally smooth foregrounds, per-sidereal-day common-mode (crosstalk) subtraction, and foreground-wedge excision — is *assumed to operate as published* [18, 19, 20]. These stages are other subsystems' contributions; this work neither re-derives nor stress-tests them. What this work owns is frame selection and the per-frame contamination measurement, and the question this section prices is conditional in exactly that sense: *given* that everyone else does their job, does the pilot-proxy mask make BAO recoverable in the DTV band, at what cost, and compared against what alternative? One further scope note belongs here rather than in a footnote, because it qualifies every masked fraction quoted below. The survey products this section consumes were produced with the calibration-free coarse rule of Section 5.5, not with the fine coherent decision the detector is designed around — the fine rule requires per-channel anchors and nulls that the survey itself is running to measure, so it could not gate the run that produces them. Every f in this section is therefore a property of that bootstrap configuration, and it decides on an axis roughly 10.5 dB less sensitive than the designed one. The direction of that gap is not converted into an automatic credit. A more sensitive axis can improve the receiver-operating characteristic, but at a fixed numerical threshold it can also change the masked fraction. Fine-stage results are therefore quoted only where the recorded fine spectra and a held-out calibration support them; the coarse fractions below describe the bootstrap products, not a proven pessimistic bound on the final detector.

9.1 The Band at Stake, and the Forecast Frame

The hyperfine line redshifts as

$$z(\nu) = \frac{\nu_{21}}{\nu} - 1, \qquad \nu_{21} = 1420.406 \text{ MHz}, \tag{9.1}$$

so the post-repack ATSC core allocation, channels 14–36 spanning 470–608 MHz, occupies $1.34 \leq z \leq 2.02$ — not an edge of CHIME's band but the middle of its high-redshift BAO range. Matter still dominates the expansion over this interval; the scientific value is the long cosmological lever arm and the complementary tracer, volume, and systematics of 21 cm intensity mapping. Modern spectroscopic surveys, including DESI galaxy, quasar, and Lyα samples, overlap substantial parts of the range rather than leaving it without competition [30, 31]. The trawl completed in August 2026: all 23 allocations carry archive products (channels 14–36, $z = 1.336$–2.022). The block with the complete residual-chain analysis (Section 9.11) remains the first-measured ten contiguous allocations, channels 27–36, covering $z = 1.336$–1.592; the thirteen lower-band channels are summarized at the survey and epoch level, and carried through the same residual chain, in Section 9.12. This is what is at stake: roughly a third of the survey's usable redshift depth lies under licensed megawatt transmitters, and the question "mask, excise, or integrate through" must be answered channel by channel with numbers rather than policy.

The forecast engine is the Fisher-matrix framework of Bull et al. [17] (*RadioFisher*), evaluated on CHIME's instrument specification, with the BAO significance defined through the amplitude of the extracted acoustic feature and distance errors (σ_{D_A}, σ_H) through the standard $(\alpha_\perp, \alpha_\parallel)$ dilation parameters. It is configured to the convention of the Overview paper's own BAO forecast [1]: sensitivity set by the system temperature and the as-built array, with no foreground residual, no wedge excision, and BAO-shift-only parameter flags. That convention is adopted deliberately and it fixes the scope of everything below. *The only two things this section models are the system temperature and the DTV contamination.* Every other stage is assumed to do its job perfectly — which means, precisely, that its cost is not charged and *its benefit is not claimed.* The consequence that matters is the second half: the delay filter suppresses the DTV shelf as well as the foregrounds, and Section 9.4 declines to take that suppression, because a budget cannot bank a discount from a stage it does not carry. Absolute forecast times inherit the customary optimism of Fisher forecasting — smooth foreground residuals, no calibration systematics — and are quoted in on-sky hours. The robust currency, used for every verdict in this section, is the *time penalty relative to the clean survey*: the ratio of integration times to reach the same target with and without contamination. Ratios cancel the shared optimism; absolute times do not.

That convention is not an invention of this work. The Overview paper's own forecast is described there as an idealized one, for the precision with which CHIME could measure the expansion history *in the absence of foregrounds or systematics* [1], and this section deliberately inherits it so that the DTV term is the only thing added to a baseline the collaboration has already published. The quantity that baseline reports is the fractional error on the volume distance,

$$\ln D_V = \tfrac{2}{3} \ln \alpha_\perp + \tfrac{1}{3} \ln \alpha_\parallel, \tag{9.2}$$

and the table below evaluates it from the same instrument specification, so the starting point of everything below is auditable rather than asserted:

z	0.85	1.05	1.25	1.45	1.65	1.85	2.43
$\sigma(D_V)/D_V$ [%], clean, 1 on-sky yr	0.470	0.502	0.549	0.620	0.712	0.859	1.039

How closely this tracks the published curve is worth stating precisely, because three of the four ingredients are provably shared and the fourth is not. The forecast code is the same: the branch used here differs from the one that produced the published forecast by twenty-eight lines in a single file, every one of them inside a conditional on `expt['P_res']`, so with no residual template requested the two execute identical instructions. The experiment specification is the same object, and the as-built $n(u)$ file is byte-identical between the branches. The fiducial $P(k)$ differs in provenance only — regenerated from a Python Boltzmann solver against the same cosmology and renormalized to the same σ_8, rather than transferred as Fortran output.

The fourth ingredient is the estimator, and it genuinely differs. The published errorbars are formed by transforming each bin to (D_V, F), assembling *all* redshift bins into one Fisher matrix with the per-bin quantities expanded and the shared nuisance and equation-of-state parameters marginalized globally, and reading $\sigma(D_V)$ off the diagonal of a single inverse. The table above instead inverts each bin independently over $\{A, \sigma_{\rm NL}, \alpha_\perp, \alpha_\parallel, b\sigma_8, f\sigma_8\}$ and forms $\sigma(\ln D_V)$ analytically from Eq. (9.2). The two agree on which parameters are marginalized and on the distance combination; they differ in that the published estimator couples the bins through parameters they share, and the per-bin one does not. So this row is the right *baseline for a per-bin perturbation study* — which is what the rest of this section is — but it is not the published estimator, and no claim of numerical agreement with the published curve is made. Reproducing that curve exactly requires adopting the combined multi-bin construction. This is a closing validation gate rather than a cosmetic cross-check: the clean error bars, residual tolerances, policy ranking, and selected η^* are all recomputed with both estimators, and any channel whose class changes is reported as estimator-sensitive. The present numbers remain a transparent per-bin perturbation study, but every later verdict is conditional on that comparison until it is attached to the release bundle.

9.2 The Footprint: One Bin Measured, Fourteen Protected

Everything priced in this section is a decision taken in one CHIME coarse channel and applied across roughly fifteen. That asymmetry is not incidental plumbing — it is where the method's leverage comes from, and it is also where its one untested premise sits — so it is worth stating in the geometry the forecast actually uses before any cost is computed.

An ATSC allocation is 6 MHz wide against CHIME's coarse raster of $f_s = 390.625$ kHz, so one allocation covers $6/f_s = 15.36$ coarse channels, of which the RRC payload occupies $B_N/f_s \approx 13.8$ (Section 3.3) and the remainder is allocation guard. The pilot is nominally described as 309 kHz above the lower allocation edge in ATSC A/53 and is generated here from the standard's exact frequency relation, Eq. (3.6), giving 309.440559 kHz. It falls inside one coarse channel rather than straddling a boundary for every configured allocation. At band scale the same picture repeats 23 times: 470–608 MHz is 23 contiguous allocations spanning 353 coarse channels, and the detector reads 23 of them. *The monitored bins are* 6.5% *of the contaminated band, and the decisions they carry cover the other* 93.5%. Figure 9.1 draws the geometry at both scales.

Three properties make the transfer from the monitored bin to the rest legitimate rather than merely convenient, and the third is an assumption rather than a result.

The screening template is flat in spectral density. The pilot measurement is converted to a nominal payload spectral density using Eq. (3.8); the forecast then assigns that density to the payload-bearing coarse channels. This makes r a per-bin quantity and gives a clear, reproducible masking footprint. It does *not* prove that the received shelf is flat: transmitter filtering, channel response, and frequency-selective multipath can change both level and shape across 6 MHz. Uniform application can therefore over-mask a cleaner bin or under-protect a locally enhanced one. The

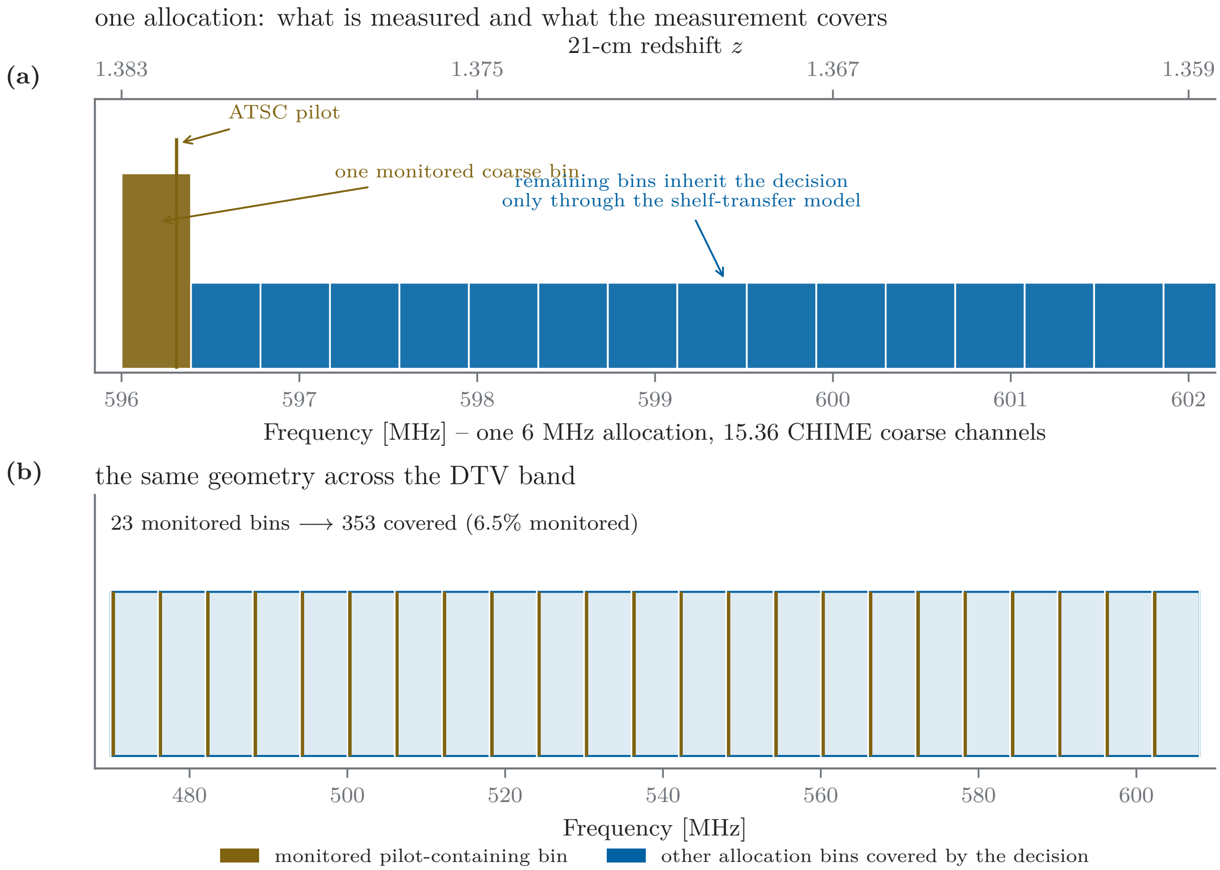

Figure 9.1: The footprint geometry. (a) One ATSC allocation at CHIME's coarse raster: the pilot-containing bin read by the detector against the remaining bins covered by the same decision. The different heights distinguish the monitored bin from the shelf-covered bins and are schematic, not measured power. Flatness across the received allocation is a transfer assumption to be tested, not a consequence of monitoring one pilot bin. (b) The same geometry across the DTV band: 23 monitored bins carrying decisions for 353.

simultaneous across-allocation measurement described below is the gate that decides which occurs.

The monitored bin is the one bin worth spending. Two properties make the pilot-containing coarse channel a natural standing excision candidate in the present model. First, it is the most contaminated: the pilot carries $P_{\mathrm{shelf}}/13.4$ while one coarse channel receives $P_{\mathrm{shelf}}/13.8$ of the payload, so the pilot line is within 3% of exactly as strong as the shelf in one bin — a coincidence of the broadcast standard against this receiver's raster — and the bin containing it holds 3.1 dB more DTV power than its neighbors. Second, and decisively, the delay filter cannot help it. The shelf's suppression under a delay cut (the 3.6–11.4 dB of Section 9.4) exists *because* the shelf is band-limited to $B_N = 5.38$ MHz, whose delay-space envelope has its first null at $1/B_N = 186$ ns and therefore concentrates below any cut near that scale; a pilot confined to a single coarse channel is a near-delta in frequency, hence uniform in delay, and leaks into every delay the analysis retains — including the acoustic ones. Because a narrow line is not preferentially confined to the low-delay region removed from a smooth shelf, the current analysis treats the pilot-containing coarse channel as a standing frequency excision; that treatment should be verified against the actual channelized response used by the BAO pipeline. That excision is the method's standing cost — one bin per allocation, 6.5% of the DTV band — and it is levied on the bin the analysis would have had to discard anyway.

The transfer itself is an assumption, and the products in hand cannot close it. Equation (3.8) is a nominal transmitter model derived from the standard's pilot bias and equiprobable payload symbols; the received ratio scatters about it through frequency-selective propagation across 6 MHz, since nothing requires a multipath null to fall on the pilot and on the far edge of the shelf alike. The survey products record only the pilot's own coarse channel — their integrated spectra are the 16384-point fine structure *within* that one channel, not a view across the allocation — so nothing in the present data verifies that the shelf is where the proxy says it is. Closing this requires the deployment to record, for a subset of frames, integrated spectra in coarse channels distributed across the allocation, masked and unmasked, and to confirm that the shelf removal in those channels tracks the level inferred from the pilot. That measurement is the direct test of the premise the method is named for. It is specified here as a requirement on the implementation, and it is not attempted from the trawl archive, whose per-channel products were never built to carry it.

The leverage and the assumption are the same statement seen from two sides. A method that measured contamination in the bins it protects would need no proxy and would offer no leverage; the pilot buys a factor of 15 in coverage at the price of one inference, and the reach of every verdict below is ultimately bounded by how well that inference is checked. What follows prices the decision under that transfer model. Results are labelled conditional until its measured distribution is propagated through the forecast.

9.3 The Cost Side: Masking Alone

A per-frame mask enters radiometric cost through the masked fraction f and a *stochastic equivalent-variance ratio* r_{var} for contamination surviving in the kept frames. Discarding frames scales the effective integration as $t \to t(1-f)$; a noise-like residual scales the variance as $\sigma^2 \to \sigma^2(1+r_{\mathrm{var}})$. The integration time to reach a fixed statistical-error target therefore obeys

$$\frac{T(f, r_{\mathrm{var}})}{T_{\mathrm{clean}}} = \frac{1+r_{\mathrm{var}}}{1-f}. \tag{9.3}$$

This is the statistical-cost functional. A coherent residual systematic is not an added variance and is carried separately as r_{sys} through the Fisher-bias calculation. The present screening analysis derives one scalar proxy r_{proxy} and, pending direct visibility calibration, applies it to both roles: $r_{\mathrm{var}} = r_{\mathrm{sys}} = r_{\mathrm{proxy}}$. That equality is a declared modelling closure, not an identity. The two failure directions remain visible: $f \to 1$ consumes the data, while either residual component can make the retained data unusable.

Within a forecast redshift bin the per-channel weights $w_c = (1-f_c)/(1+r_{\mathrm{var},c})$ must be reduced to a band average, and the reduction is not unique — it encodes a physical assumption about how surviving frames populate the modes the analysis measures:

$$\bar{w}_{\mathrm{inv}} = \langle w \rangle_B, \qquad \bar{w}_{\mathrm{four}} = \left\langle w^{-1} \right\rangle_B^{-1}, \tag{9.4}$$

where $\langle \cdot \rangle_B$ is the bandwidth-weighted mean over the bin. The inverse-variance form $\bar{w}_{\mathrm{inv}}$ holds when kept frames are distributed so that every mode sees the average depth — the correct convention for masking that is uncorrelated with sidereal time. The harmonic form $\bar{w}_{\mathrm{four}}$ holds when radial Fourier modes require simultaneous coverage across the bin, so the thinnest slice throttles every mode that crosses it; it is the pessimistic convention, and a single near-fully-masked channel drags the entire bin

($w \to 0$ forces $\bar{w}_{\text{four}} \to 0$ while $\bar{w}_{\text{inv}}$ barely moves). Both are implemented; the measured-masking forecasts below move by under 0.3% between them, because the measured masking is either mild (most channels) or so severe that the channel is excised outright and leaves the average entirely. Excision is priced separately and differently: an excised channel is removed from the bin's *volume* (Fisher information scales linearly with surviving bandwidth), never from its noise, because no integration time recovers a band that is not observed.

Exposure-weighted across the survey's quarterly rate record, the measured masking costs the survey 3.2% additional integration time to any fixed BAO-amplitude target — the full-survey figure is benign because 400–470 and 608–800 MHz carry no DTV allocation and dominate the mapping speed. The binding numbers are per-bin. In $z = 1.40$–1.50, the bin most affected, the measured masking multiplies the time to a bin-level 5σ BAO extraction by 1.35 ($0.177 \to 0.238$ on-sky years); Fig. 9.2 traces the same cost as a function of a uniform masked fraction across the DTV band. One channel makes the excision logic concrete: channel 30 (566–572 MHz, $z = 1.48$–1.51) is masked $\approx$97% of the time in the quarterly rate record — 98.4% in its own survey product, the one channel where the two records of Section 9.3 agree — so integrating through it to clean-equivalent depth costs $\times 33$ in time on that slice — ten years of integration buys 3.6 clean-equivalent months — while excising it costs 22% of one bin's volume and 0.8% at survey level. Excision wins by an enormous margin, and the general lesson generalizes: *a channel is either cheap to keep or cheaper to cut; it is never worth outwaiting.*

Which masking rate, and the gap between two of them

Those figures require a provenance statement, because two masked-fraction records exist for these channels and they do not agree. The quarterly rate table from which the forecast above is built reports exposure-weighted fractions of 0.023, 0.138, and 0.012 on channels 34, 35, and 36; the survey products, evaluated under the rule carried in their own detector contract, give 0.991, 0.837, and 0.999 on the same channels and very nearly the same frames — disagreements of 43×, 6.1×, and 87×. The gap is not coverage: joined quarter by quarter, the two records agree on frame counts to within a few percent and place channel 35's transmitter switch-on in the same quarter (2021.75, at 0.145 against 0.151). The gap is the detector epoch, and it is now fully identified: the quarterly table was produced by the survey-composition reduction of 2026-07-18 under the coarse rule $F > \hat{\mu} + 0.012\,\mu_0$ — about five null widths above each channel's empirical zero point — from the *first* production trawl, whose weight bank (SHA-256 `b0dce17a`, retained as the `legacy_halfband` bank) assumed the coarse-channel center at Nyquist and was therefore mistuned by $f_s/2$ relative to the verified center-at-DC frame convention. At the line, that mistuning suppresses every pilot by 39–47 dB — except channel 30's, whose pilot sits 847 Hz from $f_s/4$, where the error self-cancels, which is why channel 30 is the one channel on which the two records agree. The quarterly table is therefore a record of the legacy detector epoch, not a DTV occupancy measurement; the ingestion path now carries that rule and epoch explicitly, so the limitation is visible rather than implied, and it is kept only because the published forecast figures were built from it.

Table 9.1 brackets what this costs. Substituting the three measured products into the published record moves the affected bin from 1.35× to 1.48× — tolerable, and the honest current statement. Extending the bootstrap rule across the band is not tolerable: at $f = 0.942$ every DTV channel crosses the excision threshold, and since $z = 1.40$–1.50 maps to 568–592 MHz, entirely inside the allocation, that bin is not made expensive but *lost.* The three measured channels are the only evidence available on which of these the bootstrap rule actually produces band-wide, and all three sit above 0.83. This is the sharpest available argument that the operating point cannot be left at the detection floor: at the floor, on the evidence in hand, the band does not survive. It is the same

Table 9.1: The forecast under each available masking record. The published figures use the quarterly rate table, generated in the legacy $f_s/2$-mistuned detector epoch (its rule and weight bank are identified in the text); the bootstrap rule is the positive-excess rule of Section 5.5, evaluated on the survey products. The last row assumes all 23 channels behave like the three measured, which is a bound rather than a measurement — but it is the bound that matters, because at that rate every DTV channel crosses the excision threshold and the $z = 1.40$–1.50 bin lies entirely inside the allocation.

masking record	survey 5σ		$z = 1.40$–1.50 bin	
	on-sky yr	penalty	on-sky yr	penalty
clean (no DTV)	0.0241	1.000	0.177	1.000
quarterly rate table, 23 ch (published)	0.0249	1.032	0.238	1.347
with products substituted on ch 34–36	0.0257	1.063	0.261	1.479
bootstrap rule band-wide ($f = 0.942$)	0.0299	1.238	never	∞

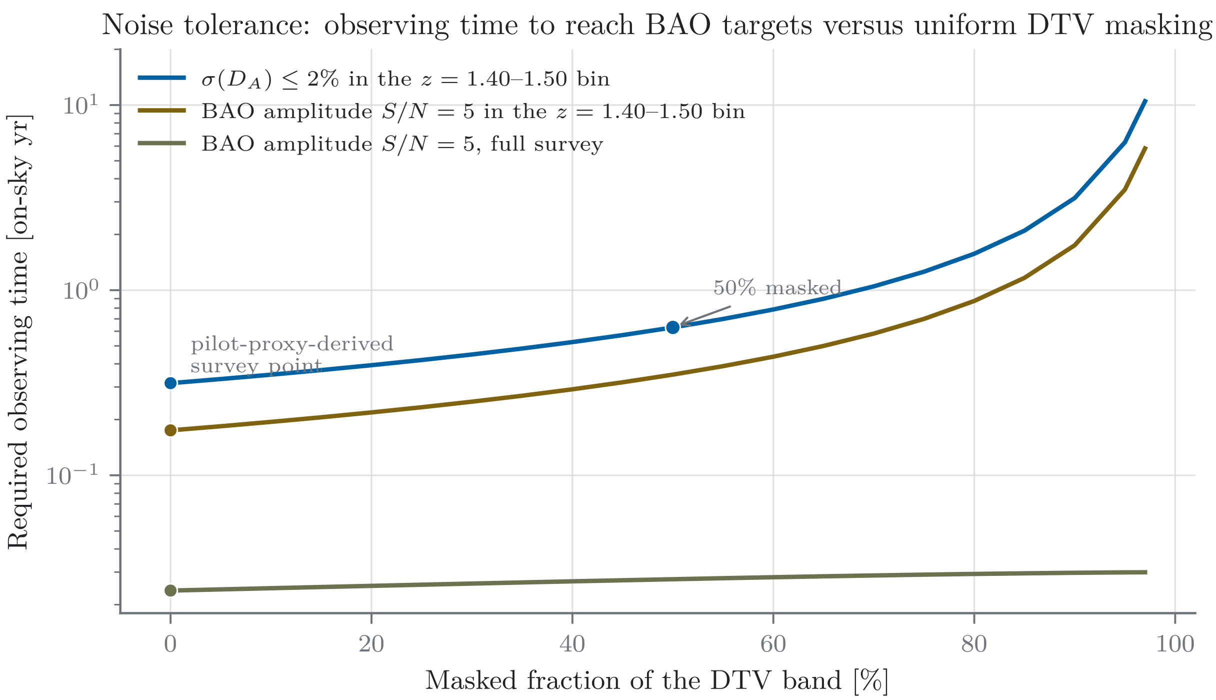

Figure 9.2: Integration time to fixed BAO targets versus masked fraction in the DTV band, at $r = 0$ (masking cost only, Eq. (9.3) with the band averaging of Eq. (9.4)). The measured pilot-proxy operating point sits at the left of the curve at survey level (3.2% penalty) but at 1.35× in the worst-affected redshift bin. This figure prices what the mask *costs*; what it *buys* requires the residual chain of Section 9.4.

conclusion Section 9.5 reaches for channel 34 from the residual side, arrived at independently from the rate side, and it is why Eq. (9.15) is a prerequisite for selecting a runtime operating point rather than a refinement of one.

Equation (9.3) with $r_{\text{var}} = 0$ prices only the cost of masking, and on that accounting the mask looks nearly free — which is precisely why cost-only accounting is insufficient. The benefit

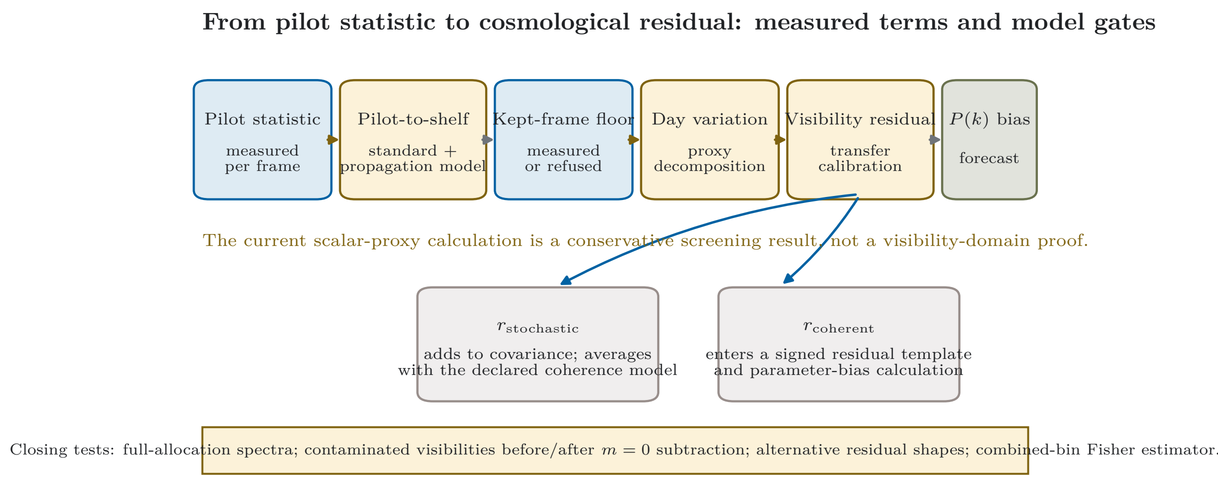

Figure 9.3: The residual chain and its evidence boundary. Pilot power, archive timescale structure, and correlation-time outcomes are scalar measurements. Across-allocation transfer and the mapping to complex visibility covariance and bias are separate, load-bearing calibrations. The current numerical forecast closes those arrows with a stated unity transfer; it does not turn them into measured equalities.

side requires knowing what contamination survives in the frames each policy keeps, and that is a measurement problem the next subsection turns into a four-term chain.

9.4 The Benefit Side: the Residual Chain

The survey does not directly observe the final visibility-domain contaminant. It observes a scalar pilot-power proxy, and the screening chain propagates that proxy through a timescale model. Define

$$p_{\mathrm{kept}} \equiv 10^{(S_{\mathrm{kept}} - S_{\mathrm{delay}})/10}, \tag{9.5}$$

$$r_{\mathrm{proxy}} \equiv p_{\mathrm{kept}} \left(\rho_{\mathrm{intra}} n_{\mathrm{coh,intra}} + \rho_{\mathrm{fast}} n_{\mathrm{coh,fast}} \right), \tag{9.6}$$

$$n_{\mathrm{coh,intra}} \equiv \frac{\min(\tau_c, T_{\mathrm{sid}})}{T_{\mathrm{frame}}}, \qquad n_{\mathrm{coh,fast}} \equiv 1. \tag{9.7}$$

Here S_{kept} is the kept-frame shelf level in dB relative to per-frame system noise under the pilot-to-shelf model; S_{delay} is any delay-filter credit booked on both sides of the calculation; ρ_{intra} and ρ_{fast} are normalized second-moment shares of the scalar proxy defined below; and τ_c is the intra-day correlation-time outcome. The numerical tables use r_{proxy} as a screening envelope. The physically distinct forecast inputs are

$$r_{\mathrm{var}} = g_{\mathrm{var}} r_{\mathrm{proxy}}, \qquad r_{\mathrm{sys}} = g_{\mathrm{sys}} r_{\mathrm{proxy}}, \tag{9.8}$$

where g_{var} and g_{sys} map the scalar proxy to stochastic covariance and coherent visibility bias. Those transfer coefficients are not measured by the present pilot-only products; the screening results below set both to unity and are conditional on that normalization. Direct complex-visibility measurements are required to replace it.

The delay filter, and the one assumption this section makes

Foreground removal in 21 cm analysis exploits spectral smoothness: astrophysical foregrounds are confined to low delay τ (the Fourier dual of frequency), while cosmological structure extends to higher delay through the mapping

$$k_{\parallel} = \frac{2\pi\,\nu_{21}\,H_0 E(z)}{c\,(1+z)^2}\,\tau \tag{9.9}$$

[18]. A delay cut $\tau_{\rm cut}$ therefore removes DTV power — which, arriving through a ~few-μs multipath spread but dominated by the direct path, is itself spectrally smooth over the 390 kHz coarse channel — along with the foregrounds. The published analyses cut at $\tau_{\rm cut} = 200$ ns, which suppresses the coarse-channel-confined shelf by 11.4 dB; but the same cut removes the low-$k_{\parallel}$ modes carrying the acoustic feature, and the hybrid foreground-removal demonstration states directly that the standard filter excludes any sensitivity to BAO [19]. A BAO analysis in this band must therefore *relax* the filter: preserving the second acoustic peak ($\tau \approx 104$ ns at these redshifts, via Eq. (9.9)) admits a cut buying 8.2 dB of shelf suppression; preserving the first peak ($\tau \approx 56$ ns) buys only 3.6 dB. This is the structural irony on which the whole problem rests: *the analysis choice that makes BAO measurable is the same choice that re-admits DTV*, so the two problems — foreground filtering and RFI masking — cannot be solved independently.

Under the Figure-31 convention of Section 9.1, however, the correct value to use for $S_{\rm delay}$ is *zero*, and that is what every number below adopts. The forecast carries no foreground removal, so the chain may not bank suppression from a filter the forecast does not model; the range 3.6–11.4 dB is retained not as an assumption but as a bound on the help that would become available if the filter were carried on both sides of the calculation. Setting it to zero is conservative with respect to this one suppression term — it makes every scalar-chain residual 2.3× larger than the first-peak value would. It does not make the chain assumption-free: the pilot-to-allocation transfer, scalar-to-visibility transfer, and residual template remain modelling choices. Any filter carried self-consistently in both the residual and Fisher models can reduce the smooth component, but the effect on the full visibility residual must be measured rather than inferred from the scalar shelf alone.

A note on the deployed cut, and the era it does not describe. One consequence of the trade above deserves to be stated on its own, carefully, because it is prospective rather than retrospective. Nothing in it identifies an error in any published CHIME result: no released analysis relies on the low-$k_{\parallel}$ region, and for the analyses actually published the 200 ns cut is the appropriate choice. The observation is about what that choice has quietly done to everyone's *experience* of the interference environment. Every characterization of RFI cleanliness earned to date — inspected maps, flagger performance, residual levels judged acceptable — has been earned in data filtered at 200 ns, which is to say with 11.4 dB of incidental DTV suppression silently included. A BAO analysis must hand back 3.2–7.8 dB of that discount, so the DTV residual a BAO pipeline will face is four to six times the one present experience exhibits — and it arrives at exactly the moment the tightest tolerances of Section 9.7 come into force. An assessment of "whether RFI is under control," calibrated on the current pipeline, therefore does not transfer to the BAO configuration: it understates the problem by precisely the suppression that must be traded away to make BAO measurable at all. In the most cautious reading this is bookkeeping — a reminder that cleanliness observed in today's maps is a property of today's filter, not of the band. In the least cautious reading it flags a risk worth naming: that the band could be judged manageable from experience gathered in the one configuration that will not be used to measure BAO. The present work takes no position between those readings — there is not enough information to take one — but the masking infrastructure it builds is aimed, deliberately, at the harsher configuration: the one CHIME has not yet operated in, and the only one in which the ruler can be read.

The sidereal filter: a scalar-proxy decomposition

The analysis subtracts a per-sidereal-day mean from each complex visibility [18]. The present archive does not contain the corresponding baseline-resolved complex visibilities; it contains a dimensionless scalar shelf power ratio. The decomposition below is therefore a model of the proxy's timescale structure, not a direct measurement of the visibility attenuation. Write

$$p_{dui} = \mu + a_d + b_{du} + \varepsilon_{dui}, \tag{9.10}$$

for frame i, acquisition u, and sidereal day d, with centered random effects a_d, b_{du}, and ε_{dui}. Define the proxy second-moment normalization

$$Q_p = \mu^2 + \mathrm{Var}(a) + \mathrm{Var}(b) + \mathrm{Var}(\varepsilon), \tag{9.11}$$

and

$$\rho_{\mathrm{intra}} = \frac{\mathrm{Var}(b)}{Q_p}, \qquad \rho_{\mathrm{fast}} = \frac{\mathrm{Var}(\varepsilon)}{Q_p}. \tag{9.12}$$

This construction is dimensionally consistent: every term in Q_p is a squared power-ratio fluctuation. It locates the scalar proxy's slow and fast variability relative to the sidereal-day boundary. It does not establish that the same fractions survive a linear operation on complex baseline visibilities, because the pilot-power product has discarded cross-feed phase and baseline structure.

The current screening calculation treats $\mu^2 + \mathrm{Var}(a)$ as the proxy share removed by the daily-mean operation and the remaining two shares as survivors. On channel 35 this assigns approximately 99.0% of the proxy's second moment to the removed side, corresponding to the quoted 19.97-dB scalar-chain suppression. That number is useful for prioritizing measurements, but it is not a measured visibility-domain suppression. The closing test records contaminated complex visibilities before and after the actual per-day operator and estimates g_{var} and g_{sys} by baseline, frequency, and epoch. The operator is an explicit analysis choice applied to data; the transit geometry also makes a truly static telescope-frame mode difficult to distinguish, but that fact does not by itself prove the scalar attenuation used here.

Coherence: how the survivors average

Thermal-noise variance in an integrated estimate falls approximately as $1/N_{\mathrm{frames}}$. In the scalar screening model, a component correlated for τ_c supplies fewer independent realizations and is amplified relative to the fast component by $n_{\mathrm{coh,intra}}$ of Eq. (9.5); the explicitly defined fast factor is $n_{\mathrm{coh,fast}} = 1$. The sidereal-day cap prevents the scalar model from assigning unlimited coherence to structure that the per-day operator is intended to remove. It is a bookkeeping convention, not proof that every longer-lived visibility contaminant is exactly removed. Keeping the slow and fast terms separate avoids applying the slow component's amplification to the noise-like sub-acquisition fluctuations.

τ_c is measured, not assumed, from the same-day structure function of the shelf: variance between same-day acquisition pairs as a function of lag, with the correlation time read from the lag at which the structure function reaches its plateau, and uncertainties from a bootstrap over whole sidereal-day blocks (resampling frames would fabricate independence the data does not have). The estimator refuses rather than extrapolates: a channel whose apparent τ_c moves by more than a factor-of-stability threshold across trim percentiles is declared *episodic* — the shelf is not a stationary process with one correlation time — and the chain falls back to the sidereal-day cap, reported as a bound rather than a measurement. The screening model treats surviving intra-day proxy fluctuations as independent across days. Civil-time transmitter schedules precess through sidereal time by about 3.93 min per solar day, and meteorological propagation is not naturally locked to the sky; these facts motivate the assumption but do not guarantee it. A sidereal-coherent component is therefore one of the alternative residual templates and direct visibility tests required below.

9.5 The Decision: Four Policies, One Ordering

A survey holding a DTV-contaminated channel has exactly four options, and the detector must be priced against all of them, not against a straw man: *keep* everything ($f = 0$); *excise* the channel ($f = 1$; infinite time-to-measurement in band, volume loss at survey level); flag with the *incumbent* methods already in the pipeline (Section 9.6); or flag on the *pilot*. Each policy supplies a masked fraction f and a scalar residual proxy r_{proxy}. Equation (9.8) then maps that proxy to the two quantities the forecast actually needs: r_{var} for statistical cost and r_{sys} for coherent bias. The present tables use the unity-transfer screening closure, so their single reported residual should be read as r_{proxy} rather than as a direct visibility measurement.

At the *frame* stage — kept-frame mean shelf power, no downstream chain — keeping everything wins on all three initially analysed channels, and it is worth stating this plainly rather than hiding it: the raw scalar shelf ratios are small ($r_{\text{proxy}} = 0.085$, 3.9×10^{-4}, and 1.35×10^{-3} on channels 35, 34, and 36, the Table 9.6 on-air shelves). Were those ratios already calibrated as purely stochastic visibility variance, integrating through them would be cheaper than cutting around them, with nominal $(1 + r_{\text{var}})$ penalties of 1.09, 1.00, and 1.00. The qualification matters: frame-level scalar power is not yet the final visibility residual.

The scalar coherence model is what overturns the frame-level ordering, and channel 35 shows why the timescale measurement matters. Its structure function yields $\tau_c = 45$ min (68% interval 32–53 min), hence $n_{\text{coh,intra}} = 6.4 \times 10^4$. Under the proxy decomposition and unity-transfer closure, the modest -10.7 dB on-air shelf becomes $r_{\text{proxy}} = 54.8$ under the keep policy, whereas masking at the archive rule spends $1/(1 - 0.837) = 6.1\times$ before a retained-frame residual is charged. Channel 35 has no empirical off-state floor, so its exact masked-policy residual remains unmeasured. Substituting a neighboring floor illustrates a roughly $9\times$ mask-over-keep advantage in the scalar screening model, but it is not a channel-35 visibility result. The defensible conclusion is narrower and still useful: a sub-noise shelf can become survey-limiting when its measured timescale is long, and τ_c is a load-bearing discriminator that must be carried into the direct visibility calibration.

Channel 34 shows the opposite face, and the framework must report it with equal prominence. Its bootstrap-rule masked fraction is 0.991, so masking costs at least $110\times$ regardless of how clean the kept frames are. The positive-excess rule sits at the detector's empirical decision floor and rejects essentially every frame carrying a detectable pilot; on this channel that is almost every valid frame. The keep-policy scalar chain is itself only a bound because τ_c is refused and replaced by the sidereal cap. Thus the archive does not identify a favorable policy; it identifies an operating-point problem and a missing cadence measurement. The role of Eq. (9.5) is to convert each candidate threshold η into (f, r_{proxy}), after which the calibrated transfers, statistical cost, and bias constraint can be evaluated separately. A runtime threshold is selected only after that full, held-out calculation.

Table 9.2: Every flagger on the same frames: shelf power removed from the kept population (dB), with the masked fraction each spends to do it. Measured within acquisitions on the survey products; the per-frame shelf is the pilot matched filter's own measurement, common to all rows (frames with no detection are assigned the sensitivity floor, a conservative choice that penalizes all flaggers equally, including the pilot proxy). Duty cycle is the fraction of valid frames carrying a positive pilot excess.

	shelf removed (dB)			masked fraction f		
	ch 34	ch 35	ch 36	ch 34	ch 35	ch 36
duty cycle	0.991	0.837	0.999			
MAD 1.8× (per acquisition)	0.09	0.01	0.02	0.067	0.056	0.063
spectral kurtosis, 3σ	6.40	0.22	8.73	0.331	0.203	0.395
pilot proxy (bootstrap rule)	18.98	63.54	35.96	0.985	0.826	0.999

9.6 The Incumbent Comparison: Why a Reference Beats a Baseline

The four-policy ordering requires measuring the incumbents, not asserting them. Two families cover what 21 cm pipelines commonly use: outlier rejection on integrated power at a few-second cadence with a MAD-scaled threshold (median $+ 1.8 \times$ MAD in the published CHIME configuration [18]), and spectral kurtosis [21], which tests departures from a calibrated Gaussian-like power distribution. A high-duty-cycle transmitter can contaminate the reference population of a block statistic: the median then follows the occupied state and a steady excess becomes difficult for a MAD cut to distinguish. Spectral kurtosis has a different limitation. Its expectation is power-invariant for an ideal stationary Gaussian process [21], so a stationary power excess whose filtered, coded 8-VSB samples remain close to the calibrated null shape can evade it. A real 8-VSB waveform is synchronized, filtered, coded, and quantized, and is therefore not mathematically identical to a Gaussian process; the limitation is empirical and model-dependent, not a theorem that every DTV shelf is exactly the SK null. The pilot statistic supplies an external frequency reference and therefore does not estimate its line location from the occupied power baseline. Synthetic tests and the common-frame comparison of Table 9.2 quantify the advantage for the present cohort without claiming universal dominance over every cyclostationary or decoding-based detector.

Three caveats keep this table honest. First, the incumbents' blocks are confined within single acquisitions — the survey archive is 8,983 usable snapshots of median 3 frames spaced hours apart over 7.6 years, and any block statistic spanning those gaps measures the archive's sampling pattern, not the sky — which caps them at 1.38 s and means *the published few-second flagging cadence cannot be reproduced on this dataset at all*; the contiguous-scan campaign exists in large part to close exactly this gap. Second, the SK null is calibrated per channel so its median block sits at unity, handing the incumbent the best normalization available, tuned on the very data it is tested against — a courtesy the pilot proxy does not receive. Third, where the shelf *is* intermittent (channels 34 and 36 carry episodic structure), SK does real work — 6.4 and 8.7 dB — and the comparison narrows honestly: the incumbent regime is bursty contamination, the pilot regime is steady contamination, and the DTV band is predominantly the latter. Real work, however, is not the same as sufficient work. Carried through the residual chain and compared against the tolerance evaluated in Section 9.7, SK's several decibels leave a residual four orders of magnitude above what the growth-rate measurement can absorb; the dB column understates the gap because it is measured against the raw shelf rather than against what the science requires.

The deeper asymmetry, however, is not the dB column. Every quantity in Eq. (9.5) — the kept-frame anchor S_{kept}, the timescale split, τ_c itself — exists because the matched filter measures a per-frame shelf level through the proxy relation. An energy-threshold flagger emits a bit; the pilot proxy emits a *measurement*. Even on a channel where the optimal policy turns out to be "do not mask" or "excise," that verdict is only computable because the contamination was measured, which is why the detector's contribution survives its own negative verdicts: the alternative to the pilot proxy is not a cheaper flagger, it is not knowing r.

9.7 Propagation to Cosmological Parameters

The statistical and systematic residuals enter different parts of the forecast. Masking scales the thermal covariance through $\bar{w}^{-1}$ (Eq. (9.4)), and a calibrated stochastic residual adds $r_{\mathrm{var}}C_N$ to that covariance. A coherent residual instead enters through a signed template

$$C_{\mathrm{res}}(k,\mu;z) = A\,r_{\mathrm{sys}}\,C_N(k,\mu;z;T^*), \tag{9.13}$$

where $C_N(T^*)$ is frozen at the target integration used to define the tolerance. This is a *noise-shaped screening template*, not a claim that DTV residuals physically share the full k, μ, redshift, baseline, or sidereal structure of thermal noise. The amplitude A is appended to the Fisher matrix through $\partial \ln C_{\mathrm{tot}}/\partial A = C_{\mathrm{res}}/C_{\mathrm{tot}}$. Marginalizing over A propagates uncertainty in that declared template; fixing it and evaluating

$$\Delta\theta_i = \Delta A \sum_j \left(F^{-1}\right)_{ij} F_{jA} \tag{9.14}$$

propagates an unmodelled coherent component into a first-order parameter shift. The statistical cost and coherent bias therefore remain distinct even when the present screening closure sets their scalar normalizations equal.

Residual-shape and time-scaling gates

The noise-shaped template is one member of a required sensitivity bank, not the final physical model. The closing forecast is repeated with, at minimum, frequency-localized, low-$k_\parallel$, wedge-like, sidereal-coherent, and empirically estimated visibility templates. A channel retains its class only if the class is stable across the templates justified by the calibration data; otherwise it is labelled template-sensitive. The direct visibility campaign supplies both the template shape and the transfer coefficients of Eq. (9.8).

Time scaling must be tied to the definition of the residual. A fixed physical contaminant does not automatically fall with $C_N(t)$, and its significance can grow as statistical errors shrink. By contrast, the existing convergence figure evaluates a *noise-normalized family* in which the residual template is rescaled with $C_N(t)$ at each time. In that family the induced bias and the statistical error can fall at similar rates, leaving a nearly flat bias significance. Those are different hypotheses, not contradictory conclusions. Equation (9.13) therefore defines the tolerance at the fixed target T^*; any statement about convergence away from T^* explicitly names which residual amplitude is held fixed.

The bias tolerance

$$r_{\mathrm{tol}}(T^*) = \max\left\{r_{\mathrm{sys}} : |\Delta\alpha(r_{\mathrm{sys}})| \le \zeta\,\sigma_\alpha(T^*)\right\} \tag{9.15}$$

— the largest residual whose induced bias stays below a stated fraction ζ of the statistical error at the survey's target integration T^* — is then the number that closes the loop of this entire dissertation:

r_{tol} fixes the science contamination tolerance invoked in Section 5.5, the proxy margin converts it to a per-channel threshold η, the threshold determines (f, r_{proxy}), the calibrated transfer determines $(r_{\text{var}}, r_{\text{sys}})$, and Eq. (9.3) prices the result against keep, excise, and incumbent.

The tolerance, evaluated

Equation (9.15) is not blocked on channel coverage: r_{tol} is a property of the survey and the forecast, and only its conversion into per-channel thresholds needs the channels. Evaluating it on a Fisher bank carrying the `_Pres` row — built with $P_{\text{res}} = 1$, so the template amplitude is in units of the reference-time noise power and ΔA is r_{sys} — gives the result in Table 9.3. Two features of it are worth more than the number.

First, *the growth rate binds, not the acoustic dilations.* At $\zeta = 1$, $f\sigma_8$ tolerates $r \approx 1.5 \times 10^{-3}$ while $\alpha_\perp$ and $\alpha_\parallel$ tolerate roughly ten times more, and the ordering holds in every DTV redshift bin at every integration time. This is what should be expected on reflection: the acoustic scale is a feature *position*, protected by the oscillatory signature that defines it, and a smooth additive residual displaces it only weakly; the growth rate is read off an *amplitude*, which is precisely what an additive residual counterfeits. A framework that had priced only the BAO dilations would have concluded the band was ten times safer than it is.

Second, *the dilations cannot be read off the grid unguarded*, and the reason is physical rather than numerical. The response $D_p(T) \equiv \partial p / \partial r_{\text{sys}}$ can pass smoothly through zero as sample variance overtakes thermal noise. Near such a crossing Eq. (9.15) can report a spuriously large tolerance. For each parameter p and central time T, define $D_- = D_p(0.9T)$, $D_0 = D_p(T)$, and $D_+ = D_p(1.1T)$. With a declared numerical scale ϵ_D, an entry is accepted only if

$$|D_0| > \epsilon_D, \qquad \operatorname{sgn}(D_-) = \operatorname{sgn}(D_0) = \operatorname{sgn}(D_+), \qquad \max_{s \in \{-,+\}} \left| \frac{D_s}{D_0} - 1 \right| \leq 0.20. \tag{9.16}$$

The release records ϵ_D, the three derivatives, and the rejection reason. On the present 19-point grid the gate refused several dilation entries and no $f\sigma_8$ entries; a 32-point grid refined through the crossings reproduced the shared Fisher matrices to all printed digits and did not change the qualitative result. The crossings are structure, not a coarse-grid artifact.

What survives the gate is usable, and it is the second tier of the tolerance story. The stable dilation entries that remain in every DTV bin have per-bin minima of $r_{\text{tol}}(\alpha_\perp) = 0.014$–$0.035$ at $\zeta = 1$ — an order of magnitude looser than $f\sigma_8$, exactly the asymmetry Figure 1.4 promised. The two tiers do different work. The $f\sigma_8$ tolerance of Table 9.3 is the binding requirement for the full measurement; the stable dilation tolerance is the requirement for the *ruler alone*, and it is the constraint the per-channel threshold optimization of Section 9.8 runs on, because it is the one a no-delay-filter analysis can actually meet. The screening structure that results — dilation-tier candidates on some channels, and growth-rate candidates only once the delay filter is booked on both sides of the calculation (Section 9.9) — is the two-tier headline of this section.

Applying $r_{\text{tol}} = 1.5 \times 10^{-3}$ to the policies of Section 9.5, through the scalar chain and unity-transfer screening closure, produces a stringent conditional result. Figure 9.4 shows the policy ordering at the target time, while Fig. 9.5 shows the behavior of the particular noise-normalized time family used by the current forecast. Neither figure is a substitute for the visibility-transfer and template tests stated above.

Within the scalar screening model, no evaluated policy clears the binding growth-rate tolerance on a channel with a defensible kept-frame floor. Channel 33 has the strongest scalar-chain evidence because its floor and correlation-time bound are both data-derived: keeping everything gives $r_{\text{proxy}} = 2.35$, the MAD cut 2.38, spectral kurtosis 0.573, and the pilot proxy 0.0359. Against

Table 9.3: The residual tolerance r_{tol} of Eq. (9.15) at the published criterion $\zeta = 1$ [6], per DTV redshift bin, taken over the entries that survive the stability refusal described in the text. $\alpha_\perp$ and $\alpha_\parallel$ are omitted: both are refused at several integration times in every bin, and where they survive they tolerate about ten times more than $f\sigma_8$, so they never bind. Within the current reference-time, noise-shaped template family, the tightest accepted value occurs near one on-sky year; the table reports that minimum without promoting its time dependence to a universal physical scaling law. The $f\sigma_8$ entry survives the gate at every grid point in every bin; the dilations' surviving entries are the second tier taken up in the text and in Section 9.8.

z bin	1.30–1.40	1.40–1.50	1.50–1.60	1.60–1.70	1.70–1.80	1.80–1.90	1.90–2.04
r_{tol} ($\times 10^{-3}$)	1.50	1.53	1.60	1.71	1.81	1.94	1.76

1.5×10^{-3} these are 1,566×, 1,587×, 382×, and 24× over. Channels 32 and 34 retain the same ordering, with proxy bounds of 44× and 1,603× over. These are not caveat-free visibility verdicts: they inherit the proxy transfer, unity coefficients, noise-shaped template, and per-bin estimator. They do show that the bootstrap coarse floor is not, by itself, a comfortable growth-rate operating point under the declared screening assumptions.

Two things survive that verdict and both matter. The first is the ordering. The pilot proxy leaves a residual 16 to 66 times smaller than the best alternative on the same frames, and 44 to 65 times smaller than keeping the data; within this common scalar screen, no other evaluated policy is within one and a half orders of magnitude of it. The second is that the verdict is a property of the *bootstrap coarse rule* (Section 5.5), which is what produced these products and which decides on an axis roughly 10 dB less sensitive than the fine coherent stage the detector is designed around. Section 9.11 walks the remaining gap.

A sensitivity floor is not the minimum observed value

One methodological point belongs here because the corrected result depends on it entirely. The masked-policy residual is anchored by the kept-frame *sensitivity floor*: the shelf level the estimator reports on frames where no pilot is detectable. Measuring it requires a null population — valid frames with no detection that nonetheless carry a finite shelf estimate — and four of the first-measured ten channels have none at all, while two more have only six and two candidate null frames.

The tempting substitute is the smallest shelf among frames that *were* detected, and it is wrong in a way that is easy to miss and expensive to keep. The inferred shelf tends to zero as the detection statistic approaches its null center from above, so the minimum detected shelf is not a physical floor but the weakest positive excess that happened to occur in the sample; it falls without bound as the detection count grows. On channel 35 it evaluates to −75.99 dB against measured floors of −44.95 to −48.54 dB on the channels that have them — thirty decibels of sampling artefact, which propagated into a masked residual four orders of magnitude too small and inverted that channel's verdict from failing to passing. The analysis framework now refuses to compute a masked residual without a measured floor rather than substituting one, and an explicit override is the only way to stand a value in from elsewhere. Channels 35 and 36 accordingly carry *no* masked-policy verdict in Table 9.6, which is the honest state of the evidence rather than a gap to be filled.

Three screening outcomes remain possible per channel: the residual lies inside tolerance under all required validation gates; the crossover exists but depends on an unresolved measurement; or no admissible operating point is found and the channel becomes an excision candidate. The present archive populates the latter two classes and identifies conditional candidates in the first.

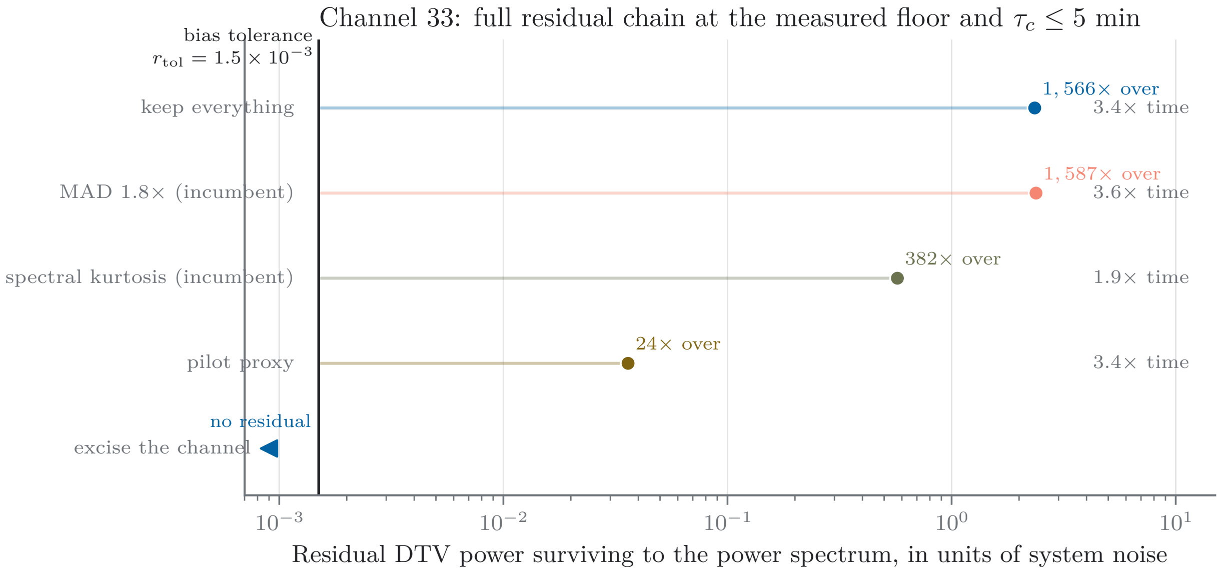

Figure 9.4: Every policy on the axis that decides between them, for channel 33 — the one channel whose kept-frame floor and correlation time are both constrained by data. The rule is the bias tolerance of Eq. (9.15) at the published criterion. Integration time rides as an annotation rather than a second axis, because every failing policy fails on residual whatever its time cost. Excision sits at the left edge with no residual and no measurement: it wins the axis shown and loses the one that cannot be drawn. The pilot proxy is the best of the four by a factor of sixteen and still lands 24× outside.

Spending the tolerance: limited freedom in the scalar screening model

The loop is only closed if the tolerance can be converted into an operating point, so the last step is to spend it. Channel 33 is the one to spend it on: it is the only channel with both a measured floor and a bounded correlation time, so every term in its chain is data. Sweeping the coarse threshold and carrying each setting through Eq. (9.5) gives:

η	1 (floor)	1.2	1.5	2	5
masked fraction f	0.805	0.158	0.092	0.048	0.007
kept-frame shelf (dB)	−45.3	−34.8	−33.3	−31.6	−29.6
residual $r_{\rm proxy}$	0.0334	0.369	0.524	0.772	1.24
$r_{\rm proxy}/r_{\rm tol}$	22	246	349	514	828

One bookkeeping note reconciles this table with the verdicts: the sweep books the kept set at its measured mean (−45.3 dB at the floor setting), while the policy comparison and Table 9.6 book every kept frame at the measured floor (−44.95 dB), the more conservative convention; the 0.3 dB difference is why the floor row here reads 22× where the verdict reads ≤ 24×. The bootstrap detection-floor rule is the best available setting by a wide margin, and nothing above it competes: relaxing η by a fifth recovers 65% of the data and multiplies the residual by eleven. For this channel and this scalar-chain sweep there is no competitive intermediate operating point: either the detectable-pilot population is removed and the retained set sits near the measured floor, or the readmitted frames dominate the mean. A different visibility transfer or residual template could

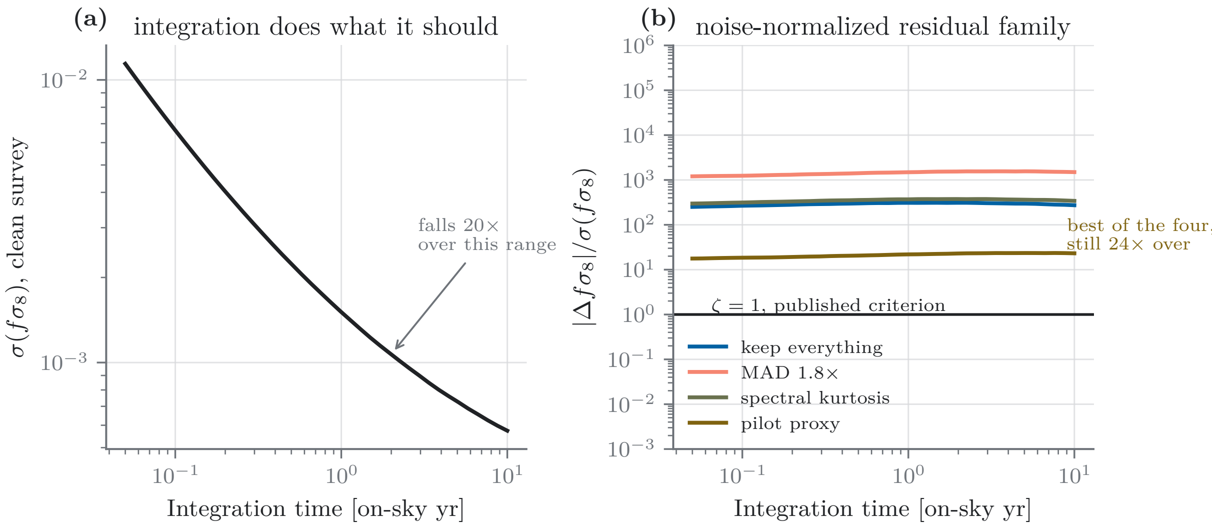

Figure 9.5: Time scaling under the current noise-normalized residual family. At each plotted integration time the template amplitude is defined relative to that time's thermal-noise covariance, so the induced bias and $\sigma(f\sigma_8)$ fall at similar rates and their ratio is nearly flat. This figure does not describe a fixed physical contaminant, whose significance can grow as the statistical error shrinks. The tolerance used for the channel decisions is therefore defined at the fixed target time T^* rather than inferred from this curve as a universal convergence law.

move the numerical constraint, which is why the conclusion is attached to the screening model rather than promoted to a detector-independent theorem.

This sharpens — and partly corrects — the operating-point argument of Section 5.5. The 21.6 dB proxy margin of Eq. (5.22) is real and still computable in advance from the broadcast standard. What that section could not know is that the margin is not free to spend, because the residual chain consumes it: on channel 33 the chain multiplies the kept-frame floor by +30.5 dB *net* — the coherence amplification already discounted by the ground filter's −7.6 dB, with no delay credit claimed — against a 21.6 dB margin. The general claim survives — a threshold should be set by the science tolerance rather than by P_{fa} — but on a channel whose residual outlives the common-mode filter the tolerance-set threshold and the detection floor coincide, and the bootstrap rule is selected by this arithmetic rather than by conservatism. It is also, on this evidence, not sufficient: the floor rule is the correct operating point and still leaves 22× too much residual. What closes that gap is not a different threshold, and Section 9.11 takes up what it is.

What ζ means, and why the verdict does not depend on it

The form of Eq. (9.15) is standard. Requiring that a systematic shift stay below the statistical error it competes with is the standard criterion for propagating an unmodelled effect through a Fisher forecast; Amara & Réfrégier state it as

$$b[\hat{p}_i] \;\leq\; \sigma[\hat{p}_i] \tag{9.17}$$

for the parameters of interest, and use the ratio b/σ as the metric throughout [6]. Equation (9.15) is that criterion with an explicit slack factor, and *this work adopts* $\zeta = 1$, which recovers it exactly. The threshold is therefore the community's rather than this work's, and the only question a reader has to weigh is whether the measurement clears it.

It is still worth saying what ζ means, because a survey may want to set its own. A bias Δ against a statistical error σ inflates the effective error to $\sqrt{\sigma^2 + \Delta^2} = \sigma\sqrt{1+\zeta^2}$, so ζ states how much precision the systematic may cost: 41% at the published $\zeta = 1$, 11.8% at 0.5, 4.4% at 0.3, 0.5% at 0.1. A second reading bounds it from the other side, since DTV is one line in a budget: a survey holding its total systematic below 0.5σ across N independent terms of comparable size allots each $0.5/\sqrt{N}$ — 0.289 at $N = 3$, 0.158 at $N = 10$. A collaboration with a long systematics list should therefore run tighter than the published criterion, and the framework computes $r_{\text{tol}}(\zeta)$ for whatever value is set; it does not presume to set it. Tightening to $\zeta = 0.3$, three times stricter, moves the binding tolerance from 1.5×10^{-3} to 4.5×10^{-4}.

Neither choice is load-bearing, which is the more useful point. r_{tol} scales linearly in ζ, and the ordering of policies does not depend on it at all: on channel 33 the pilot proxy leaves 0.0359 and the best alternative leaves 0.573, a factor of 16 apart, so no value of ζ reorders them. The absolute verdict is likewise robust over a wide range — the proxy would need $\zeta \geq 24$ to pass and the best alternative $\zeta \geq 382$, so declaring the mask adequate on present evidence requires accepting a bias twenty-four times the statistical error, which no reading of Eq. (9.17) supports. What the choice of ζ cannot do is change the sign of the answer, in either direction.

A note on two thresholds that are easy to confuse. This section's ζ and the detector's μ_0 are unrelated, and only μ_0 is derived. μ_0 is the null center of the coarse statistic, fixed exactly by the quantized weight-bank norms (Section 4.3); it is arithmetic on compile-time constants, carries no free parameter, and answers a question about the instrument — *is a pilot present in this frame?* ζ is a requirement a collaboration sets, lives in the Fisher forecast, and answers a question about the science — *how much bias can the measurement absorb?* The two meet only through the loop this section constructs, in which ζ fixes r_{tol}, which would fix a per-channel η, which scales μ_0. Of every threshold in this dissertation, μ_0 is the only one that is not a judgement call.

9.8 The Forecast-Selected Operating Point

Section 5.5 promised that the bootstrap placeholder — the positive-excess rule — would be replaced by a tolerance-derived operating point once the forecast matured. This subsection redeems that promise. For each channel, the threshold family $F > \eta\,\mu_0$ is swept over the recorded statistics and the survey-time cost of Eq. (9.3) is minimized subject to the bias constraint,

$$\min_{\eta} \ \frac{1 + r_{\text{var}}(\eta)}{1 - f(\eta)} \quad \text{s.t.} \quad r_{\text{sys}}(\eta) \leq r_{\text{tol}}(\alpha_\perp),\ \zeta = 1,$$

with the constraint on the *operable* tier — the stable dilation tolerance of the channel's own redshift bin — since that is the requirement a no-delay-filter analysis can meet. The conventions are fixed and few: the fine-stage credit is booked at its measured value (9.4–10.0 dB across channels, booked as 10; independently confirmed by the archived row-sum-level Monte Carlo at the deployed geometry, 9.32–9.77 dB at the two fixed false-alarm points — `tools/measure_fine_gain.py`, evidence under `docs/evidence/fine_gain_mc_2026-08-19`); the kept-frame bound is computed on both floor bases (the product's own null floor, and the stricter σ-null level a threshold at the null center can actually resolve) and both are reported, a basis disagreement being a finding rather than a nuisance; among near-optimal thresholds within 2% of the minimum cost the *smallest* η is selected (equal cost, more margin); and on channels whose τ_c is refused the result inherits the sidereal-cap bound, which is conservative in one direction only — a feasible η is truly feasible, but the reported optimum may be pessimistic, since a measured τ_c can only enlarge the feasible set.

Table 9.4 is the result of the current screening sweep, and it divides the first-measured ten channels into classes that the single number f or r_{proxy} alone would not reveal. Three channels are

Table 9.4: Coarse operating points selected under the current scalar screening closure, on the product-floor basis, with the measured fine-stage credit booked and the dilation-tier constraint applied to $r_{\rm sys}$. The table reports $r_{\rm proxy}$; the numerical optimization sets $r_{\rm var} = r_{\rm sys} = r_{\rm proxy}$ pending visibility calibration. *Margin* is $r_{\rm tol}/r_{\rm sys}$ and *cost* is Eq. (9.3). Values are candidate bundle data, not final runtime settings, until the combined-estimator, template, transfer, and blocked-holdout gates pass.

	η^*	$F >$	f	$r_{\rm proxy}$	margin	cost	note
ch 32	1.01	1.012312	51.8%	0.0111	1.4×	2.10	1.07× on the σ-null basis
ch 33	1.05	1.055216	47.5%	0.0133	1.2×	1.93	basis-sensitive: infeasible on σ-null
ch 35	1.30	1.296735	71.7%	0.0053	6.6×	3.55	archive-only: recent $f = 99.8\%$ (Sec. 9.10)
ch 31	1.15	1.152260	99.4%	0.0069	2.3×	177	formally feasible, vacuous: the occupancy wall
ch 27, 28, 29, 30, 34, 36: no feasible η at any setting — excision candidates under this screen							

threshold-feasible on the ruler. On channel 32 the optimum sits one percent above the detection floor — the floor is nearly the optimum, as the channel-33 sweep of Section 9.7 anticipated — with a 1.4× margin that thins to 1.07× under the stricter floor basis: real, but knife-edged. Channel 33 supports $\eta^* = 1.05$ on its product floor and *no* feasible threshold under the σ-null basis; that disagreement is precisely its unresolved wide-null structure, and the flanking-bin diagnosis specified for the closing campaign is what settles it. Channel 35 carries the widest margin (6.6×) at the highest optimized threshold ($\eta^* = 1.30$, the one channel where the proxy margin is genuinely spent rather than pinned at the floor) — but its chosen threshold masks 99.8% of the recent era, which is not a property of the threshold but of the channel's history (Section 9.10). Channel 31 is the cautionary row: the optimizer reports a feasible point, and it is vacuous — 99.4% masked at a 177× time cost is the occupancy wall wearing the costume of a solution, and the framework's cost axis is what unmasks it. Six channels admit no feasible threshold in the stated sweep: for them the scalar screening calculation returns an excision candidate under this constraint. Channel 31 is formally feasible but scientifically vacuous at a 177× cost, so the practical split is three threshold-feasible channels, one occupancy-wall solution, and six infeasible channels.

Figure 9.6 draws the available threshold curves in one plane and makes two limiting regimes visible. The bias wall is where this detector and calibration cannot demonstrate a sufficiently clean retained population; the occupancy wall is where the transmitter is present in nearly every archived frame. A channel is a screening-model recovery candidate only when its curve enters the box between them. The three feasible channels enter it; the occupancy-pinned channels run down the right-hand wall; and the two walls are nearly mutually exclusive in any one era because both are driven by the same variable — duty cycle — pulled in opposite directions. The same objective, re-run on the fine statistic `fstat_fine`, selects the fine stage's own operating pair (ρ, η_{q16}) under the discipline stated in Section 5.5; the coarse and fine selections are one framework applied to two axes; they enter a runtime bundle only after the validation gates above pass.

9.9 The Delay Filter Booked on Both Sides

Every verdict so far declines the delay filter's help, because the forecast convention carries no filter (Section 9.4). That is the conservative bookkeeping, but it leaves the growth rate unpriced under the analysis configuration that would actually measure BAO, and the framework can do better than a one-sided bound: the filter can be booked *self-consistently*, on both sides of the calculation at once. A delay cut $\tau_{\rm cut}$ maps to a parallel-wavenumber cut through Eq. (9.9) — in the forecast's own foreground switch, a mode exclusion at $k_{\rm FG} = \tau_{\rm cut} \times 400$ MHz in grid units — so each candidate

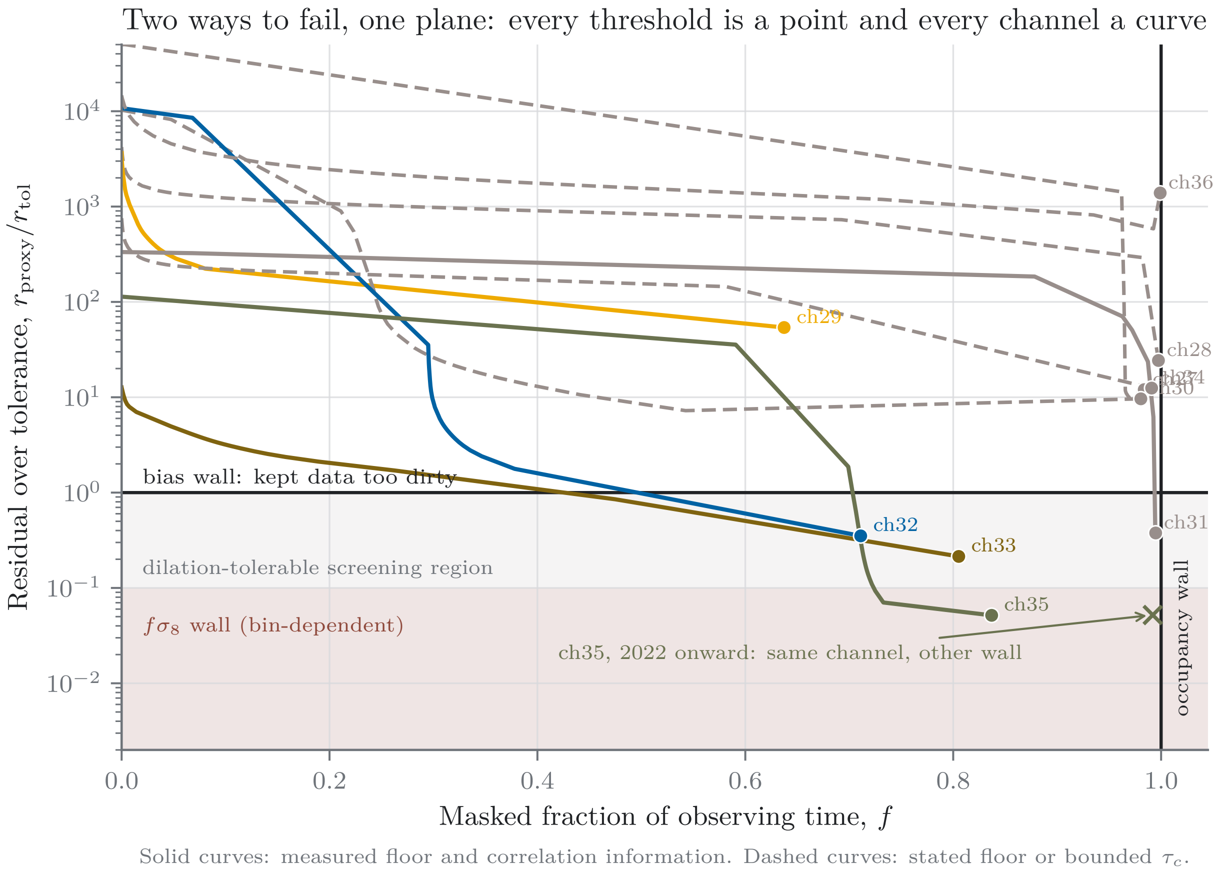

Figure 9.6: Every available threshold-sweep channel is a curve and every threshold is a point in the $(f,\ r_{\text{proxy}}/r_{\text{tol}})$ plane, with the residual on the fine-stage screening axis and the tolerance the stable dilation tier. The two tolerance boundaries are the two failure modes of this section: the *bias wall* (top edge: kept data too dirty) and the *occupancy wall* (right edge: nothing left to keep). Each dot is the positive-excess floor $\eta = 1$; raising η walks a curve right-to-left. Dashed curves carry a stated (rather than measured) floor or a τ_c bound; the shaded band is where the $f\sigma_8$ wall falls, 10–25× lower depending on bin. The × marks channel 35 restricted to 2022 onward: the same channel, moved to the other wall by a station switch-on (Section 9.10).

cut defines a *world*: the Fisher bank loses the modes below the cut (tolerances re-derived, stability gate re-applied), and the residual chain gains the matching shelf suppression from the same $\tau \leftrightarrow k_\parallel$ mapping. Nothing is invented: the cut values are the published 200 ns and the two BAO-preserving design points any analysis in this band faces (55 ns preserving the first acoustic peak, 110 ns the second), and the suppressions are the same 3.6/8.2/11.4 dB the chain already carries as unbanked bounds. One guardrail keeps the exercise honest: the worlds model the *mode-cut geometry* only — what sensitivity the analysis gives up and what shelf power the cut removes — never the filter's foreground-removal performance, which remains another subsystem's contribution.

Table 9.5 gives the strongest positive result of the current screening model: at the 110 ns design cut, the unity-transfer, noise-shaped calculation places all three parameters inside tolerance on channels 32, 33, and 35, with $f\sigma_8$ margins of 2.5–8× and dilation margins of 19–296×. This is evidence that a joint mask–filter configuration is worth validating; it is not yet a visibility-domain recovery claim. The result remains conditional on the across-allocation proxy, g_{var} and g_{sys}, the

Table 9.5: Three screening worlds: the growth-rate margin $r_{\rm tol}(f\sigma_8)/r_{\rm sys}$ on the threshold-feasible channels, with the residual at the minimum-residual operating point ($\eta = 1$, fine stage) and each world's tolerances re-derived from its own mode-cut bank (stability gate re-applied; the dilations pass for channels 32/33/35 in every world and are not tabulated). Margins ≥ 1 (bold) pass. Channel 29 is shown for contrast: its τ_c-capped residual fails every parameter in every world — filtering is not its problem.

$f\sigma_8$ margin	no filter	55 ns (3.6 dB)	110 ns (8.2 dB)	200 ns (11.4 dB)
ch 32	0.28	0.72	**2.5**	**9.3**
ch 33	0.46	**1.2**	**4.1**	**15**
ch 35	0.82	**2.2**	**8.1**	**35**
ch 29	fails all	fails all	fails all	fails all ($\alpha_\perp$ 1.2× over)

residual-shape bank, the combined-bin estimator, and epoch-specific holdout tests. Two structural features of the table deserve note. The $f\sigma_8$ tolerance itself *loosens* with cut depth — from 1.5×10^{-3} with no filter to 3.7×10^{-3} at the deployed cut — because the cut removes exactly the smooth low-$k_\parallel$ modes through which an additive residual counterfeits amplitude: the filter protects the growth rate twice, once by suppressing the residual and once by excising the modes it damages most. And the deployed 200 ns column, though it posts the largest margins, is not an operating point for this science — it is the configuration the hybrid demonstration states excludes BAO sensitivity [19] — which is why the 110 ns world is the operative one, and why the note of Section 9.4 matters: the cleanliness experience earned at 200 ns overstates what the 110 ns world will exhibit by the 3.2 dB difference, roughly a factor of two in residual. Channel 29's row is the reminder that filtering solves only the filtering-shaped problem: its failure is the τ_c refusal, and its verdict belongs to the contiguous campaign, not to any choice of cut.

9.10 Eight Years Is Not One Era

Every fraction quoted so far averages over an archive within which the broadcast environment measurably changed — the FCC incentive-auction repack was still relocating stations through mid-2020, and individual transmitters came and went throughout. The archive records this directly, and two channels carry step changes large enough to overturn their own averages. Channel 32's masked fraction falls from 99.8% (before 2021) to 59.0% (after): a station present through the repack era departs, and the channel's modern state is the moderately occupied one every earlier table describes. Channel 35 is the mirror image: 46.1% masked before 2021, 94.2% after — the switch-on that both masking records independently place at quarter 2021.75 — so its archive-average 83.7% describes neither era, and its optimized threshold (Table 9.4) is an *archive* statement whose recent-era column already confesses as much. Channels 29 and 31 are stable across the same boundary (59.9 → 65.4% and ≈100% throughout), and channel 33 swings without a step (88.4 → 78.2%). The anchors move with the stations: the per-era anchor measurement of Section 5.5 shifts by 11 fine bins on channel 32 and 29 on channel 35 between the archive's calendar halves — different transmitters at different offsets, not drift.

The completed lower-band trawl (Section 9.12) adds three facts to this era record from the first-measured block itself. Channel 27's transmitter is gone: its monthly median pilot excess runs 18–21 dB in every collected month through November 2020 and 0.03–0.4 dB in every collected month from October 2022 onward, so the transition is bounded inside the archive's 2021–2022 collection gap, and the always-on excision reading of its archive average describes an era that has ended. The same monthly record places channel 32's departure in the same gap (strong through November 2020, trace from early 2023), sharpening the calendar-half split quoted above. And channel 30's archive simply ends: its captures cease after September 2023 because collection on that coarse channel was curtailed by operations on account of the contamination itself (Section 7.1), so its post-2023 state is unmonitored rather than measured — an era boundary imposed by the instrument, not the transmitter. The consequence is that several of this section's verdicts are *era* statements, and the honest calibration unit for any future runtime is the epoch. Channel 35's usable epoch is its past: before the 2021 switch-on it was among the cleanest channels measured (46% masked with the widest tolerance margin in Table 9.4), and that archive is science in the bank; its present is excision-shaped. Channel 32's usable epoch is its present: the post-departure era carries the feasible operating point, and the station era contributes almost nothing to keep. The fine-stage calibration prototype quantifies both under its own conventions (fine-credit residual against the dilation tier): channel 32's modern era certifies at $f = 43.3\%$ with $r = 0.0155$ inside tolerance, while its station era retains seven frames — nothing to certify; channel 35's early era certifies at $f = 18.1\%$, while its post-switch-on era fails both walls at once ($f = 98.8\%$ *and* $r = 0.237$). Per-epoch operating points — anchor, bulk mask, rank, multiplier, and threshold — are therefore not a refinement but the correct reading of the runtime contract. Table 8.1 and Fig. 8.1 display the split explicitly; values absent from the supplied analysis products remain marked pending rather than being inferred from archive averages. The forward-looking corollary: verdicts in this section are archive verdicts, and a future runtime inherits the *current* epoch's, which for channel 32 is better than its average and for channel 35 is worse.

9.11 Measured Verdicts, and What the Archive Cannot Settle

Table 9.6 carries the measured state of the analysis: ten contiguous ATSC allocations, 548–608 MHz, $z = 1.336$–1.592. They do not distribute evenly across the framework's possible outcomes, and the pattern is more informative than any single channel.

Channel 33 carries the most constrained scalar-chain result because both its kept-frame floor and correlation-time bound are data-derived — 1388 null frames give a floor of -44.95 dB, and the structure function bounds τ_c at five minutes. Its scalar-screening result is that the mask leaves $24\times$ too much residual under the unity-transfer closure. *Channel 32* is close behind, with the strongest ground filter measured (22.8 dB, only 0.23% of its shelf power surviving as intra-day variation) but a refused τ_c; its $44\times$ is an upper bound. *Channel 34* fails by $1{,}603\times$, and the reason is worth isolating: its floor is measured and unremarkable, but its τ_c is refused and evaluated at the sidereal cap, giving a coherence amplification of 8.7×10^4 against channel 33's 1.1×10^3. On that channel the binding term is not sensitivity at all. *Channels 35 and 36* have no null population in 7.6 years of archive, so their masked residual is unmeasured; channel 35 carries one of the set's two fully measured correlation times (45 min), and channel 36 combines one of the largest surviving intra-day shares (28.7%) with one of the weakest ground filters (5.3 dB), which makes it among the least promising of the ten in the scalar screen.

The remaining five measured channels divide along the same lines, with two instructive extremes. *Channel 29* is the archive's tantalizing case: the faintest shelf measured (-43.6 dB on air), the largest null population (8119 frames, so its floor is the best-measured in the set), and a moderate masked fraction (63.7%) — yet its structure function refuses a correlation time, it carries the largest intra-day share (31.9%) through the weakest ground filter (4.6 dB), and at the sidereal cap its bound lands $6620\times$ over. No channel's verdict hangs more completely on a single pending measurement:

Table 9.6: The scalar residual chain, term by term, on the first-measured ten channels — the contiguous ATSC allocations 27–36 spanning $z = 1.336$–1.592. The reported levels are pilot-derived per-coarse-channel screening values transferred across the 15.36-bin allocation by the model of Section 9.2; they are not direct measurements in every protected bin. No delay-filter credit is claimed (Section 9.4). "cap" marks channels where the correlation-time estimator refused and the chain is evaluated at the sidereal-day bound, so their residuals are upper bounds rather than measurements; $\leq$ marks every quantity inheriting such a bound. *Null frames* is the population from which the kept-frame floor is measured: channels 27, 30, 35, and 36 have none, and channels 28 and 31 have too few to stand on (6 and 2), so their floors are shown parenthesized as indicative and their masked-policy residuals are bounded or blank rather than substituted (channel 28's parenthesized entries inherit the 6-frame floor). $r_{\mathrm{tol}} = 1.5 \times 10^{-3}$ at the published criterion.

	ch 27	ch 28	ch 29	ch 30	ch 31	ch 32	ch 33	ch 34	ch 35	ch 36
allocation (MHz)	548–554	554–560	560–566	566–572	572–578	578–584	584–590	590–596	596–602	602–608
redshift span	1.56–1.59	1.54–1.56	1.51–1.54	1.48–1.51	1.46–1.48	1.43–1.46	1.41–1.43	1.38–1.41	1.36–1.38	1.34–1.36
monitored bin (MHz)	548.438	554.297	560.156	566.406	572.266	578.125	584.375	590.234	596.484	602.344
masked at survey rule	98.1%	99.7%	63.7%	98.4%	100.0%	71.1%	80.5%	99.1%	83.7%	99.9%
on-air shelf (dB)	−33.1	−30.8	−43.6	+0.2	−7.9	−35.3	−33.8	−34.1	−10.7	−28.7
null frames	0	6	8119	0	2	2818	1388	168	0	0
kept-frame floor (dB)	none	(−47.1)	−48.2	none	(−54.8)	−48.5	−45.0	−45.6	none	none
intra-day share	0.40%	8.76%	31.85%	0.36%	0.66%	0.23%	15.70%	4.24%	0.99%	28.66%
ground filter (dB)	22.4	10.4	4.6	22.2	21.7	22.8	7.6	13.3	20.0	5.3
τ_c	cap	cap	cap	cap	34 min	cap	$\leq$ 5 min	cap	45 min	cap
r_{keep}	≤ 4.05	≤ 151	≤ 28.4	≤ 7700	52.4	≤ 1.39	≤ 0.474	≤ 33.7	54.8	≤ 787
r_{proxy}	—	(≤ 3.5)	≤ 9.93	—	—	≤ 0.0653	≤ 0.0359	≤ 2.40	—	—
$r_{\mathrm{proxy}}/r_{\mathrm{tol}}$	—	($\leq$2300)	≤ 6620	—	—	≤ 44	≤ 24	≤ 1603	—	—

the contiguous scan of its allocation is specified first for exactly this reason (Appendix checklist, item C), and a measured τ_c of minutes would move it by three orders of magnitude; if the campaign is never executed, the sidereal cap stands here as a permanent conservative bound. *Channel 31* contributes the set's second measured correlation time (34 min) and is nonetheless unusable: masked at 100.0%, it is the occupancy wall in pure form, and its formally feasible optimizer point (Table 9.4) is the vacuous one. *Channels 27, 28, and 30* are excision candidates of the always-on kind — channel 27, the first arrival from the deeper half of the band, hosts the set's nearest full-service primary (60 km) and is masked 98.1% with no null frame in 7.6 years, its anchor the closest to nominal yet measured (−119 Hz) — channel 30's shelf sits at system noise (+0.2 dB, the loudest measured, forty-four decibels above channel 29's), its masking near-total in both records.

What would close the gap

A negative verdict earns its keep by saying which measurement to go and get, so it is worth walking the remaining distance in terms that can be checked. Three levers are available and they bind differently per channel.

The first is the *decision axis*. Every number in Table 9.6 comes from the bootstrap coarse rule; the fine coherent stage the detector is designed around has a measured advantage of 9.4–10.0 dB (Section 5.5), and a more sensitive decision lowers the floor by the same amount. That alone moves channel 33 from 24× over to 2× and channel 32 from 44× to 4× — with one channel-specific caveat the fine-stage calibration itself surfaced: channel 33's excess is *spread* across the fine axis (a designated window at any single anchor covers only 2–5% of its coarse-masked frames), so on that channel the full designated-bin credit should not be assumed until its wide-null structure is diagnosed; its path to tolerance runs more robustly through the filter lever, where it passes with margin. The second is the *delay filter*, whose credit this section declines to claim on scope grounds (Section 9.4) but which the eventual analysis will apply — and which Section 9.9 now books self-consistently rather than as a one-sided conditional: at the second-peak cut, the current screen places all three parameters inside tolerance on all three threshold-feasible channels. The third is τ_c itself, and on channels 29 and 34 it is the only lever that matters — substituting channel 33's measured five-minute bound for the sidereal cap moves channel 34 from 1,603× over to clearing the tolerance with the fine stage alone, and channel 29, the faintest and best-floored channel in the set, moves by the same mechanism if its pending contiguous scan returns a correlation time of minutes. These projections are stated as conditionals rather than results; what they establish is that the gap is one to two orders of magnitude on three of the ten channels (with a fourth hostage to a single scheduled measurement), and that the improvements which would close it are ones already designed rather than hoped for.

Channel 36 is the exception and should be said plainly: it would need a floor below −85.9 dB, which no combination of the three levers reaches. On present evidence it is an excision candidate, and channels 27, 28, 30, and 31 join it — always-on or occupancy-pinned, with channel 30's shelf sitting at system noise — so the framework's contribution on five of ten channels is a measured basis for treating them as excision candidates rather than assuming they can be integrated through, which was always one of the outcomes this section was built to be able to return.

What the archive cannot settle

Five limits of the trawl archive are structural, and stating them fixes the requirements of the closing campaign. First, *cadence*: its snapshots cap block statistics at 1.38 s, so the incumbent comparison at its native few-second cadence, and any τ_c between seconds and the acquisition spacing, are unmeasurable here. Second, *null populations*: four of ten channels have none and two more have too few frames to stand on, and without a transmitter-off epoch — or the fine stage's per-frame null, which the coarse rule does not provide — their masked residual cannot be measured at all. Third, *refusals are load-bearing*: seven of ten correlation times are refused, and on channels 29 and 34 those refusals are worth three orders of magnitude in the verdict; a framework that extrapolated τ_c would have produced confident answers the data cannot support. Fourth, *footprint*: the products record only the monitored bin, so the proxy's transfer across the allocation is assumed rather than verified (Section 9.2). Fifth, *coverage*: at the time this analysis was frozen, ten of 23 channels were measured, and the spread already visible — on-air shelves from −43.6 dB to +0.2 dB, ground filters from 4.6 to 22.8 dB, intra-day shares from 0.2% to 31.9%, null populations from zero to 8119 frames, correlation times from 34 minutes to unmeasurable — said the thirteen still unmeasured at the freeze would not be interpolations of these ten. The August 2026 completion of the thirteen lower-band allocations (Section 9.12) confirms that prediction directly, with two dated sign-offs, an epoch-split persistent carrier, a collection-curtailed archive, and a band-edge pair of episodic faint carriers among them.

The section closes where it began. Under the present scalar-transfer and per-bin forecast model, the archive supports a two-tier screening result. With the bootstrap decision rule and no delay-filter credit, no evaluated policy clears the binding growth-rate tolerance on a channel with a defensible floor, although the pilot proxy gives the smallest residual of the compared policies. Under the looser dilation tier, channels 32, 33, and the early epoch of 35 are forecast-feasible candidates; the 110 ns world also places the growth-rate screen inside tolerance on those candidates. Five channels are excision candidates, channels 29 and 34 remain controlled by cadence measurements, and thirteen allocations have no archive products. None of the positive classes becomes a final recovery verdict until the across-allocation, visibility-transfer, residual-template, combined-estimator,

Table 9.7: The thirteen lower-band allocations at the August 2026 snapshot: monitored coarse channel, events processed, archive-average masked fraction at the survey (positive-excess) rule, transmitter-era summary from the monthly occupancy record, and the correlation-time outcome of the structure-function estimator evaluated on the transmitter-on epoch where one exists ("cap" denotes a refusal, evaluated at the sidereal-day bound). Channel 24's thin inventory reflects the collection curtailment of Section 7.1.

	monitored (MHz)	events	masked	era summary	τ_c
ch 14	470.312	8,788	94.8%	trace; episodic faint carrier at synthesis	cap
ch 15	476.172	8,347	99.9%	steady faint excess (≈1.0 dB median), no transition	cap
ch 16	482.422	9,045	91.6%	trace; warm-season propagation swell	cap
ch 17	488.281	7,646	99.9%	persistent; two on-epochs (≈12.8 then ≈16.4 dB excess)	cap
ch 18	494.141	8,892	92.3%	trace, rising since 2024; seasonal swell	cap
ch 19	500.391	5,447	91.8%	sign-off December 2024	cap
ch 20	506.250	8,781	99.8%	step down September 2022	167 min
ch 21	512.500	8,959	59.7%	trace	cap
ch 22	518.359	8,711	100.0%	persistent, steady (≈5.4 dB excess)	cap
ch 23	524.219	8,585	88.7%	trace	5 min
ch 24	530.469	1,767	100.0%	persistent; collection-curtailed archive	62 min
ch 25	536.328	8,762	88.9%	trace	cap
ch 26	542.188	5,754	95.3%	sign-off April 2023	cap

and blocked-holdout gates pass. What is complete here is the auditable machinery: the tolerance is computable, policy costs and residual screens are reported separately, refusal states remain visible, and every remaining measurement is tied to a channel, epoch, and decision it can change.

9.12 The Lower Band, Measured: the August 2026 Completion

Between the freezing of the preceding sections' analysis and this bundle's snapshot, the trawl completed the thirteen lower-band allocations, channels 14–26 ($z = 1.592$–2.022), under the same immutable schema, the same bootstrap coarse rule, and a single detector-kernel cohort shared with the first-measured block (channels 15 and 14, the last in the scan order, completed on 2026-08-19 with 8,347 and 8,788 events). This section reports what those products establish, first at the survey and epoch level and then — in Table 9.8 — through the same residual chain as the first-measured block. The chain evaluation was rebuilt for this purpose as a released generator whose self-test reproduces Table 9.6's published constants from the raw products to the printed digit; the extension to the new block runs on that validated basis, and what is stated is stated exactly, from the products.

Table 9.7 confirms, in the direction it was made, the coverage warning of Section 9.11: the new block is not an interpolation of the old one. It contains two transmitter sign-offs resolved to the month (channel 19 in December 2024 and channel 26 in April 2023, with channel 20's two-stage step-down in September 2022), one persistent carrier whose two on-epochs carry measurably different anchors — channel 17's recovered fine-axis anchor moves as the second epoch comes to dominate the archive, the same station-change signature as the channel-32/35 anchor motion of Section 9.10 — and a collection-curtailed archive on channel 24. The band-edge pair is new in kind as well: channels 14 and 15 carry faint carriers at the synthesized pilot (the completed channel-14 carrier lands exactly at synthesis) that pin the survey rule (94.8% and 99.9% masked) while releasing three-quarters of current-epoch frames at the working threshold — an episodic, propagation-modulated occupancy that is neither the trace family nor the live-carrier family. Three structure functions return measured or bounded correlation times (62, 167, and 5 minutes on channels 24, 20, and 23); the rest refuse.

Figure 9.7 shows the block's spectral face in the same construction as Figure 3.3.

Two further results are new in kind rather than only in coverage.

Transmitter-off epochs supply the missing floors. The first-measured block's binding limitation was the absence of null populations on its always-on channels (Section 9.11). A measured sign-off converts that absence into a dated, transmitter-off epoch, and the analysis package now accepts such epochs directly (a sign-off epoch mirrors the sign-on epoch the channel-35 calibration already used). The resulting off-epoch shelf floors — the 90th percentile of the off-epoch shelf distribution, a different and more defensible population than the per-frame null floor because the off state is established by the era rather than by the absence of a detection — evaluate to −33.5, −26.1, and −32.3 dB on channels 19, 20, and 26, and the same construction applied to the first-measured block's era boundaries gives −24.4 dB on channel 27, −31.5 dB on channel 32, and −26.2 dB on channel 35's pre-switch-on epoch. Channels that previously carried *no* masked-policy anchor now carry a measured one.

The archive-average masked fraction is an era artifact on most of the band. Because the products store the decision statistic rather than a mask bit (Section 7.3), any threshold can be re-evaluated exactly, post hoc, on any epoch. Restricting to the current epoch — every month after the last detected transition, January 2025 onward — and rethresholding at $F > 1.4\,\mu_0$ collapses the masked fractions of every channel without a live carrier to between 1.4% and 14.8% (the propagation-fed exceptions — the burst-type channel 36 and the band-edge pair 14–15, whose faint carriers ride episodic enhancements — settle at 33.7%, 24.8%, and 26.9%), while the live carriers (17, 22, 24, and the first-measured block's 30, 31, 35) remain at 98–100%: across the seventeen no-live-carrier channels of the complete 23-channel set, the surviving fraction of current-epoch valid frames rises from 10.7% under the survey rule to 89.6% at $\eta = 1.4$, with the retained frames carrying at most +1.46 dB of per-frame pilot excess by construction. Priced through the released per-bin forecast bank under the cost convention of Section 9.3 — masking plus the stochastic folding only, no bias constraint, no delay credit, unity transfer — the current-epoch survey rule costs ×1.22 at survey level and ×5.4 in the $z = 1.40$–1.50 bin, while the $\eta = 1.4$ rethreshold costs ×1.10 and ×1.59; extended over the full archive the survey rule again prices the worst bin out of measurement, as Table 9.1 anticipated. These are screening costs on the cost axis alone: they are not tolerance verdicts, and for the channels with live carriers they change nothing — the bias wall, not the occupancy accounting, is what their verdicts turn on. What they establish is narrower and still consequential: on the majority of the DTV band the deployed-floor masked fractions measure dead transmitters and faint residual excess rather than live occupancy, and a per-epoch, tolerance-derived threshold recovers most of that band's observing time at screening level.

The lower band through the residual chain

Table 9.8 extends the per-epoch chain evaluation of Table 9.6 to the completed lower band, and its first result is a confirmation: *every one of the block's screening statuses survives the chain unchanged.* The live carriers behave as their occupancy said they would — channels 22 and 24 have no null population at all, and channel 17's twenty null frames are too few to stand on, the channel-28 situation again — so all three remain excision candidates on the occupancy wall. The trace channels are the channel-29 family made plural: intra-day-heavy shares (33–53%) through weak ground filters (2.7–4.6 dB), verdicts hostage to refused correlation times, and thermal-end residuals that pass tolerance trivially — the bracket, not the shelf, is their entire verdict. Channel 21 is the family's superlative case and the survey's new tantalizing channel: the *largest null population in the entire archive* (14,647 frames, surpassing channel 29's 8,119), a well-measured −43.2 dB floor, and $\leq 22{,}100\times$ at the cap that a minutes-scale τ_c would collapse by three orders of magnitude; it joins

Table 9.8: The thirteen lower-band allocations through the residual chain of Section 9.4, on the same basis as Table 9.6 (bootstrap rule, unity transfer, no delay credit; $r_{\text{tol}} = 1.5 \times 10^{-3}$, the binding growth-rate tolerance). Daggered channels (19, 20, 26) are evaluated on their transmitter-off era — the science-relevant state — whose null population is established by the era rather than by non-detection, and whose "shelf" row is the off-era detected-frame median; their floors are the 90th percentile of *all* off-era frames, a deliberately conservative basis, so their r values are bounds even where τ_c is measured (channel 20). Parenthesized entries rest on floors with too few null frames to stand on (channel 17: twenty), following the Table 9.6 convention.

	ch 14	ch 15	ch 16	ch 17	ch 18	ch 19†	ch 20†	ch 21	ch 22	ch 23	ch 24	ch 25	ch 26†
allocation (MHz)	470–476	476–482	482–488	488–494	494–500	500–506	506–512	512–518	518–524	524–530	530–536	536–542	542–548
redshift span	1.98–2.02	1.95–1.98	1.91–1.95	1.88–1.91	1.84–1.88	1.81–1.84	1.77–1.81	1.74–1.77	1.71–1.74	1.68–1.71	1.65–1.68	1.62–1.65	1.59–1.62
monitored bin (MHz)	470.312	476.172	482.422	488.281	494.141	500.391	506.250	512.500	518.359	524.219	530.469	536.328	542.188
masked at survey rule	94.8%	99.9%	91.6%	99.9%	92.3%	91.8%	99.8%	59.7%	100.0%	88.7%	100.0%	88.9%	95.3%
shelf (dB)	−38.4	−27.6	−41.2	−5.6	−35.6	−6.8	−18.6	−39.1	−17.7	−37.1	−1.7	−40.5	−8.0
null frames	1151	0	0	20	2789	12742	23673	14647	0	0	0	0	16674
kept-frame floor (dB)	−48.8	none	none	(−50.1)	−41.3	−33.5	−26.1	−43.2	none	none	none	none	−32.3
intra-day share	52.58%	9.52%	44.32%	0.99%	40.57%	0.63%	1.44%	33.53%	1.31%	43.25%	0.33%	42.63%	1.77%
ground filter (dB)	2.7	10.1	3.4	19.9	3.8	21.9	18.3	4.6	18.8	3.5	23.5	3.5	17.5
τ_c	cap	cap	cap	cap	cap	cap	167 min	cap	cap	$\leq$ 5 min	62 min	cap	cap
r_{keep}	≤ 157	≤ 336	≤ 69.2	≤ 5550	≤ 231	≤ 2680	47.3	≤ 84.6	≤ 456	≤ 0.608	197	≤ 78.7	≤ 5730
r_{proxy}	≤ 14.2	—	—	(≤ 0.20)	≤ 61.8	≤ 5.70	≤ 8.57	≤ 33.1	—	—	—	—	≤ 21.2
$r_{\text{proxy}}/r_{\text{tol}}$	$\leq$ 9,470	—	—	($\leq$132)	$\leq$ 41,200	$\leq$ 3,800	$\leq$ 5,710	$\leq$ 22,100	—	—	—	—	$\leq$ 14,100

channel 29 at the head of the cadence priority. The band-edge pair splits across the same taxonomy: channel 14 joins the trace family at its extremes — the survey's largest intra-day share (52.6%) through its weakest ground filter (2.7 dB), with a well-measured −48.8 dB floor from 1,151 null frames and $\leq$ 9,470× at the cap — while channel 15's steady faint excess leaves *no* null population in 7.6 years, so, like the other zero-null channels (16 and 22–25), its masked residual cannot be priced from this archive and its unlock is the fine-stage per-frame null re-decision or the control-frequency trawl. Channel 23 inverts the pattern — the band's one benign correlation-time bound ($\leq$ 5 min) attached to an unmeasured floor — so its unlock is the fine-stage per-frame null re-decision rather than a cadence campaign. The sign-off channels' off-era rows are bounds at conservative floors: channel 20's is the cheapest refinement in the survey, since its τ_c is already *measured* at 167 minutes and only the floor basis (the 90th percentile of all off-era frames, most of which the mask would remove) is conservative.

The full-archive views of the sign-off channels are reported here only as a demonstration, because they are the era-mixture trap of Section 9.10 realized on data: evaluated era-blind, channels 19 and 26 land near 4× the binding tolerance and formally *pass* the dilation tier, because the sign-off step itself is classified into the DC and inter-day shares the ground filter removes — the transmitter's death masquerades as filterable structure — and channel 19's era-straddling record even returns a spuriously "measured" τ_c of 19 minutes across the step. Those numbers are artifacts of averaging across a transition, not verdicts, and they are the clearest argument in this chapter for why every row above is per-epoch.

The non-claims of Table 9.6 carry over verbatim: unity pilot-to-shelf transfer, no delay credit, the noise-shaped template, and the per-bin estimator. Within them, the completed band's chain result is symmetric with the first block's: no lower-band channel reaches conditional recovery at the deployed basis under the coherence cap, every trace and sign-off channel passes at the thermal end, and the distance between those two statements is precisely the measurement program of Section 9.11 — now with channel 21 beside channel 29 at its head.

The working threshold: development and limits

The threshold this section evaluates is the third step of a four-step development, and stating the ladder fixes what each step required. Step one is the idealized rule $F > 1$, which takes the ratio statistic's null mean as unity. Step two is the deployed conservative rule $F > \mu_0$, where μ_0 is each channel's *exact* null mean under its calibrated configuration — exact in its integer arithmetic, but calibrated per channel from the survey's own results, which is why the measured values straddle unity by up to $\pm 1.1\%$ (0.98533–1.01111 across the 23 channels). That calibration is not cosmetic: at the current-epoch operating point the step-one and step-two masked fractions differ by tens of percentage points on quiet channels (channel 23: 68.7% versus 92.6%; channel 21: 97.8% versus 69.6%), because $\eta = 1$ thresholds sit in the bulk of the null distribution where a one-percent shift in the threshold moves the fraction materially — the same property that prices the step-two rule at nearly half of verified-quiet time (Section 5.5). Step three is the working threshold $F > \eta\,\mu_0$ with the band-wide $\eta = 1.4$, selected from the threshold sweeps priced through the forecast bank; it caps kept-frame pilot excess at $\hat{\rho} = F/\mu_0 - 1 < 0.4$ (+1.46 dB) by construction. Step four — a per-channel excess threshold $\hat{\rho}$ selected from each channel's own residual chain against the tolerance — is the per-epoch evaluation this chapter leaves pending.

The working threshold's limitations should be stated as plainly as its benefits. *First*, it is a cost-axis screening instrument: its selection folded in unity pilot-to-shelf transfer, no delay-filter credit, and no bias constraint, so it prices occupancy, not safety. *Second*, the single band-wide value is provably no channel's optimum; per-channel optima exist in the sweeps but differ by orders of magnitude between the two ends of the coherence bracket, because they are functions of the refused correlation times — $\eta = 1.4$ is the point where the *decision* (clean versus live) is bracket-stable, not where any channel's cost is minimized, and the per-channel step-four thresholds are unidentifiable until τ_c is measured or permanently bracket-bounded. *Third*, the clean/live classification it induces is an occupancy gate (masked fraction below 15% at $\eta = 1.4$), and the propagation-fed channels — the burst-type 36 at 33.7% and the episodic band-edge pair 14–15 at 24.8% and 26.9% — sit outside both classes as the gate's standing exceptions. *Fourth*, η was tuned on the same archive it is evaluated on; the blocked-holdout gate is what converts it from an in-sample choice into a defensible operating point. *Fifth*, it is a coarse-rule threshold; the fine designated-window stage's measured 9.4–10.0 dB advantage would move every operating point if adopted. *Sixth*, its validity is epoch-scoped: μ_0 and the anchors behind it are calibration products of a transmitter era, the window rule (every month after the last detected transition) is part of the threshold's definition, and a new transition anywhere in the band resets it — which is one more reason the monitoring taps below stay on.

The per-channel disposition

The survey's operational deliverable to the collaboration is a disposition for each of the 354 CHIME frequency channels spanning the ATSC 1.0 range (470–608 MHz). This is a *policy recommendation at screening level*, layered on the evidence of this chapter without altering it, under two rules adopted for the handover: allocations without a live carrier keep all of their channels at the working threshold; excised allocations discard their interiors but keep any boundary channel shared with a kept allocation and keep their pilot-bin channel as a monitoring tap. One further rule follows from the inclusive posture: an edge shared by *two* excised allocations is discarded. Applied to the six excised allocations, the result is 273 channels kept and 81 discarded:

DTV ch	CHIME channels	DTV ch	CHIME channels
17	784–797	30/31	584–597
22	708–720	31	569–582
24	677–689	35	508–520

with the six pilot taps (freq_ids 521, 583, 598, 690, 721, 798) excluded from the discard ranges. Channel 36 is kept-and-masked at the working threshold under an inclusive-for-now rule — its excision case rests on the unverified coherence cap rather than a live carrier, two-thirds of its current-epoch frames survive the threshold, and keeping them costs 0.55% of survey time priced at the cap — with its chain status (excision candidate) unchanged and the choice flagged pending the burst-coherence measurement. The completed band-edge pair, channels 14 and 15, is kept-and-masked under the same rule: their excision case is likewise the unverified coherence cap (the screening arithmetic prefers excision at the cap by 0.3% of survey time, the thermal end prefers keeping, and the spread between the bracket's two ends is 0.6% — immaterial), roughly three-quarters of their current-epoch frames survive the working threshold, and keeping both costs 0.47% of survey time priced at the cap; their chain statuses (measurement-bound) are unchanged and the choice is flagged pending the same coherence measurement. The inclusive posture generally is an asymmetry argument: the mask is software and the archive stores the statistic per frame, so every keep is revocable downstream and certified by the blocked holdout and per-channel jackknife of the final analysis, while excision is the one irreversible act and is applied only where the recoverable fraction rounds to zero (0–2% on the six discarded allocations).

The kept edge channels have a geometric defense beyond sharing. The 8-VSB signal occupies 5.38 of its 6 MHz, with ∼310 kHz transition skirts at each allocation edge; of the ten kept edges bordering excised allocations, nine overlap the excised side only within the skirt — they never touch the full-power shelf — and one (freq_id 707, the 22/23 boundary) reaches 57 kHz into channel 22's shelf and is flagged accordingly. The skirt argument assumes the nominal rolloff; measured emission skirts can be dirtier, and the edge freq_ids have full archive holdings, so their occupancy is directly measurable with the same trawl — the first rung of the across-allocation transfer measurement the checklist already requires.

Beyond this survey's observable

Nothing in the tolerance machinery of this chapter is specific to the BAO measurement. The residual chain (shelf, ground filter, delay credit, coherence), the threshold sweeps, the refusal semantics, and the two-wall analysis consume the contaminant side; the science enters in exactly one place — the Fisher forecast that converts a residual level into a bias against a target observable's tolerance. The framework is therefore a *contamination-tolerance* instrument with a pluggable forecast backend: any measurement whose sensitivity to a contaminating residual can be Fisher-forecast can be substituted for the BAO bank used here, and every per-channel mechanism — epochs, floors, correlation times, thresholds, refusals — carries over unchanged. Nor need the contaminant be man-made: smooth-spectrum foreground residuals after delay filtering present the same two-wall structure — modes cut to avoid the foreground wedge are that plane's occupancy wall, filter leakage its bias wall — and the chain's delay-credit term is already a foreground instrument. The RFI instantiation against the BAO bank is the demonstrated case, not the boundary of the method.

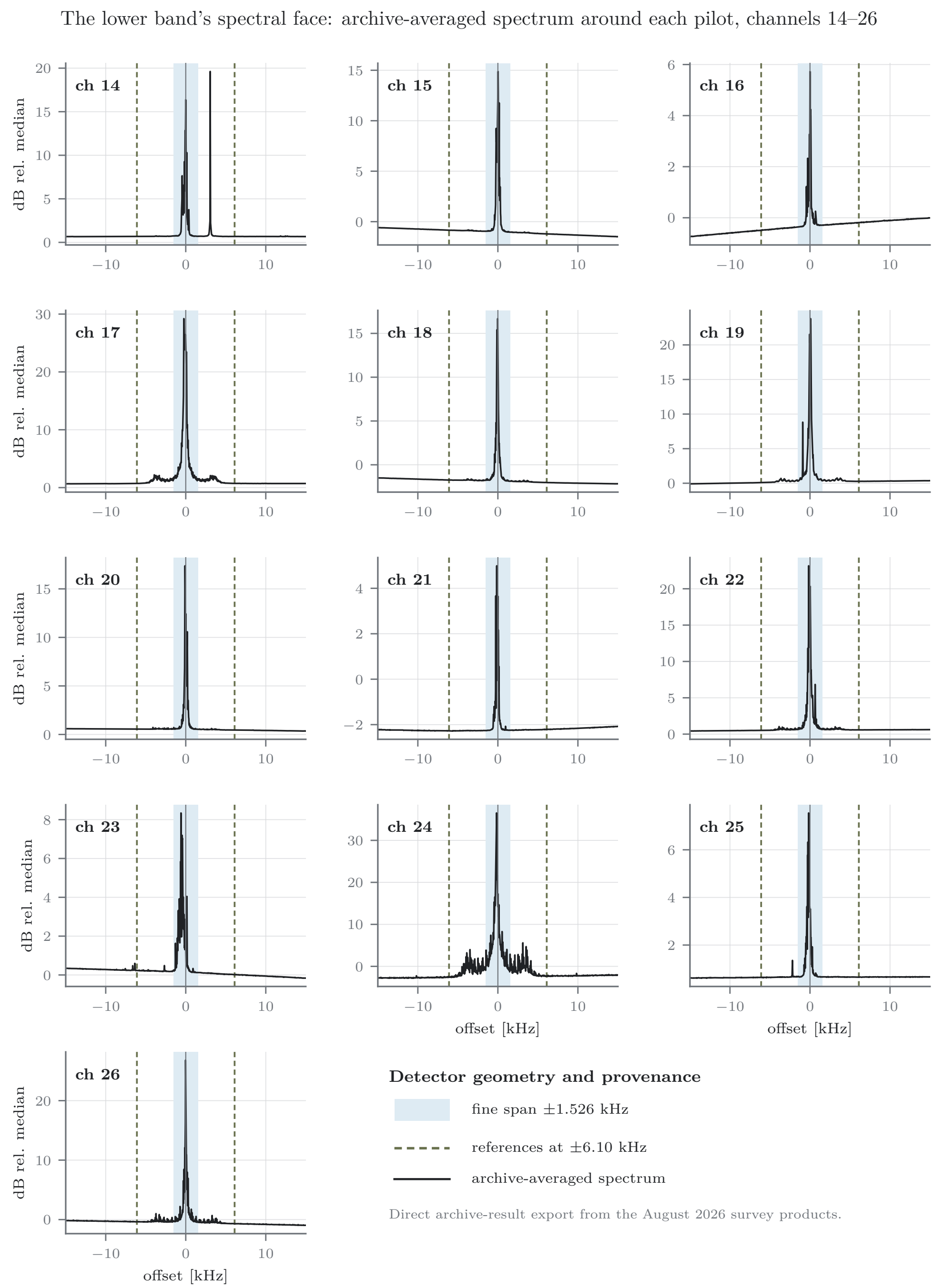

Figure 9.7: The lower band's spectral face at the August 2026 snapshot: archive-averaged spectrum within ±15 kHz of each synthesized pilot, from the products' stored integrated spectra, in the same construction as Figure 3.3 (dB relative to the channel median; shaded fine span ±1.526 kHz; dashed references ±6.1 kHz). Every dominant carrier in the block sits inside the fine span, within 0.6 kHz of synthesis — no lower-band channel repeats channel 33's off-nominal displacement — with archive-average lobes from 5 dB (channel 21) to 37 dB (channel 24) over the channel median. The sign-off channels (19, 20, 26) retain strong average lobes because the archive average is dominated by their transmitter-on epochs. The references sit in clean territory on every channel.

Chapter 10

Real-Time Integration Contract & Operations

This chapter specifies the real-time integration surface. It does not claim that the detector has already operated as a production `kotekan` stage. The fused kernel has been benchmarked and verified on deployment-class hardware; the remaining step is to bind it to the live buffer, synchronization, error, and operations contracts described here. Keeping that distinction explicit avoids using an offline archive result as evidence for a live system property.

10.1 Stage Boundary and Frame Alignment

The intended stage sits after channelization and before visibility accumulation. It consumes the same packed samples that contribute to a visibility integration and emits one validity/mask decision for the corresponding monitored DTV allocation. The acceptance condition is integer alignment: every detector frame must cover exactly one visibility integration or an explicitly declared integer number of them. A partial overlap is rejected at configuration time because it has no unambiguous mask semantics.

The benchmarked $N = 16384$ geometry has a 41.94-ms detector cadence at $f_s = 390.625$ kHz, whereas the published correlator uses a different integration length. The live stage therefore passes only after the engine's actual frame contract is pinned and one-to-one or integer-to-one alignment is demonstrated with injected frame identities. The detector can be regenerated for another power-of-two geometry, but the weights, transform bounds, golden vectors, and performance acceptance are then a new verified artifact rather than an informal runtime switch.

10.2 Initialization, Synchronization, and Failure Semantics

The host wrapper owns all state that the kernel assumes. Before each launch it zeros the fine and coarse accumulation buffers and the completion/mask word in the same CUDA stream, binds a calibration bundle whose epoch includes the frame time, and records the input-ready dependency. The downstream consumer waits on the detector completion event before reading either the mask or validity bit. The raw kernel entry point remains internal so a caller cannot omit the completion-word initialization described in Chapter 6.

A live system must distinguish a scientific non-detection from an execution failure. The following conditions set `valid=false`: geometry mismatch, invalid or expired calibration, zero-reference/bulk degeneracy beyond the specified detector rules, CUDA launch or asynchronous error, impossible

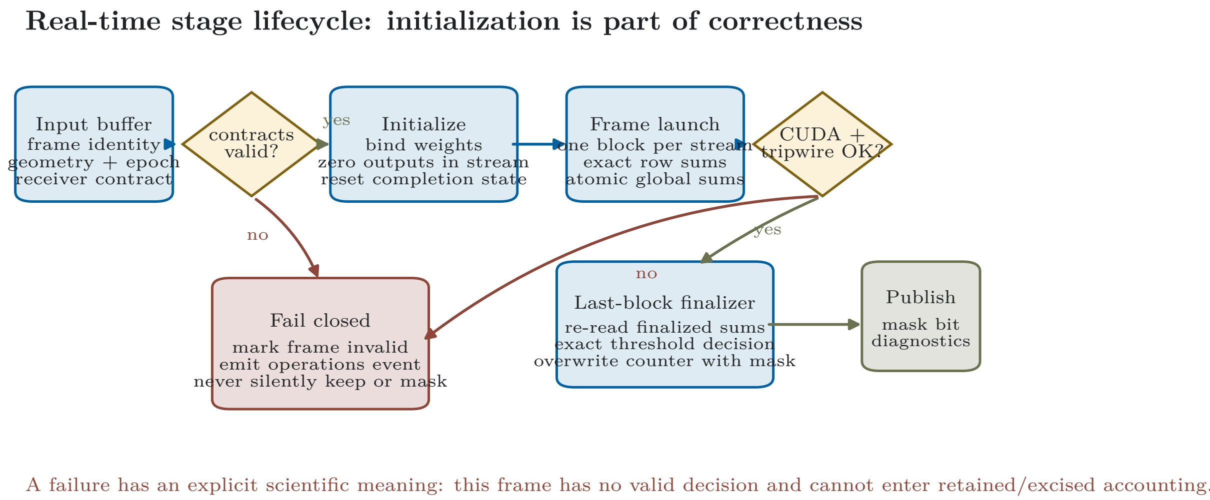

Figure 10.1: The proposed real-time stage lifecycle. Initialization is part of correctness: geometry and epoch validity are checked before launch, outputs are zeroed in the same CUDA stream, and a CUDA, completion-counter, or exact-tripwire failure produces an invalid frame and operations event. It never silently substitutes “keep” or “mask” for an execution whose decision contract was not satisfied.

completion count, exact-marginal tripwire failure, output checksum failure, or missing provenance. The stage records the reason and the affected frame range. It does not manufacture a clean decision. The collaboration’s downstream data policy may conservatively quarantine invalid integrations, but that policy is outside the detector bit and remains visible.

10.3 Operations Runbook

A concise run sequence is sufficient because the calibration and binary are content-addressed artifacts rather than editable files:

1. **Pre-run validation.** Verify receiver geometry, frame alignment, GPU capability, free device memory, bundle epoch, all hashes, and the signed pilot-injection convention. Reject mixed or expired artifacts.
2. **Acceptance burst.** Process a deterministic quiet vector, a signed-offset injected pilot, a zero-denominator case, and a nonzero-prefilled mask-word case. Require equality with the independent reference and the staged kernel path.
3. **Science mode.** Enable the stage only after the acceptance burst; record per-frame validity, mask, coarse/fine tripwire status, kernel duration, and calibration identifier.
4. **Respond to a tripwire.** Stop promoting affected frames to science products, preserve the input/product identifiers, capture a debug-tap reproduction, and compare against the reference before resuming.

Table 10.1: Real-time interface contract. Concrete C++ type names may change with the host framework; the semantic fields and failure outcomes may not.

Surface	Required contract	Failure outcome
Input samples	Contiguous packed complex `int4`; declared M, N, channel, frame identity, and time; immutable until stream completion	Frame invalid; no launch on size or geometry mismatch
Calibration	Channel and epoch validity; weight/norm/twiddle hashes; anchor, masks, rank, Q16 multiplier	Bundle rejected before data taking
CUDA stream	Input readiness event, stream-ordered output initialization, launch, completion event, and downstream wait	Frame invalid and stage health counter incremented
Fine/coarse outputs	Declared spans for power terms, diagnostics, completion/mask word, and optional debug taps	No truncation; size mismatch is configuration failure
Mask semantics	One aligned decision plus a separate validity bit; invalid is not encoded as clean	Downstream policy decides whether invalid data are retained, quarantined, or stop the run
Provenance	Source, binary, toolkit, schema, and configuration hashes; epoch identifier attached to monitoring record	Run cannot enter science mode without complete manifest

5. **Change control.** Treat any source, compiler, toolkit, geometry, twiddle, weight, schema, or calibration change as a new manifest. Code or geometry changes repeat the full verification suite; an epoch calibration change repeats the calibration and acceptance gates.

10.4 Monitoring: Tripwire, Sentinel, and Drift

The exact-marginal tripwire compares independently accumulated coarse and fine marginals as integers. It detects implementation and memory-path disagreement; it does not detect an astrophysical line at the wrong frequency. A separate schedule-agnostic sentinel compares persistent coarse excess with the fine-designated decision. Sustained coarse excess on frames the designated window does not reject is evidence for a moved anchor, an additional transmitter, or power outside the current census mask and triggers re-localization at the next epoch boundary.

This sentinel is required by a concrete legacy failure, not only by defensive design. In the earlier 23-channel survey, channel 33 appeared healthy inside the statistic — a near-0.5 mask fraction and no measured target-cell leak — while its extracted carriers lay in the skipped guard near -3.6 kHz with approximately 21 dB integrated SNR. The detector response there was about -17 dB. A strong off-nominal pilot could therefore be less visible than a weaker nominal one: “clean” and “blind” were indistinguishable from the target statistic alone. The same survey separated five likely instrument lines by rational receiver-clock coincidence, single-channel localization, and mask invariance, while retaining an unidentified sixth line rather than excising it by assumption. Appendix B records those diagnostics and their unresolved attribution. Operationally, the lesson is that a target statistic must be monitored by a broader spectral sentinel and a periodically refreshed line census.

Monitoring therefore has three layers: exact arithmetic health, statistical null/false-alarm health on non-designated bins, and environment drift. Dashboard rates are aggregated by channel, epoch, GPU, and calibration hash so an apparent broadcast change is not confused with a software rollout. The live integration is complete only when these records, the invalid-frame handling, and a failure injection exercise have been demonstrated on the actual `kotekan` buffer path.

10.5 Performance Acceptance

On an A100 benchmark GPU, the staged samples-to-powers path is reported at 2.14 ms, the fused path at 0.77 ms, and the fused path with decision epilogue at 0.90 ms for the $N = 16384$ geometry. These measurements demonstrate substantial compute margin relative to a 41.94-ms detector frame. They do not by themselves include framework scheduling, input readiness, inter-stage contention, or visibility-buffer synchronization. The live acceptance test therefore reports end-to-end latency percentiles and deadline misses under the full telescope load, with the kernel timing retained as a component rather than presented as the stage latency.

Chapter 11

Conclusions & Future Work

11.1 What This Work Establishes

This dissertation develops an instrument and a decision framework for a problem that is simultaneously statistical, computational, and cosmological. The engineering contribution is a narrow-line detector that uses the ATSC pilot as a continuously present, spectrally concentrated proxy; combines a fixed tone projection with local references and robust fine-axis normalization; and carries the complete decision through exact fixed-point arithmetic. Its digital contract is unusually strong: for exercised cases, independent implementations can be compared by integer equality rather than by a floating tolerance.

The scientific contribution is the refusal to stop at detectability. The archive products preserve the statistics needed to sweep masking policies, and the forecast translates each policy into lost exposure, surviving residual, BAO parameter bias, and integration-time cost. Equally important, the analysis marks where that chain is measured and where it is conditional: the standard fixes the pilot properties, the archive measures a scalar proxy in one coarse channel, and additional transfer measurements are required before that proxy becomes a validated visibility-domain contaminant across an entire 6-MHz allocation. Exact implementation evidence cannot substitute for those measurements.

The principal contributions can therefore be stated without claiming more than the evidence supports:

1. a standards-grounded pilot-frequency model and a nominal pilot-to-payload screening relation, separated from the received propagation transfer that remains to be measured;
2. a constraint-derived, exact-integer detector with explicit overflow, rounding, initialization, and verification contracts;
3. a restartable, bounded-storage archive survey that stores statistics and provenance rather than only historical mask bits;
4. empirical anchor, null, epoch, and correlation-time estimators with refusal semantics when a required population is absent; and
5. a two-tier BAO tolerance framework that can return recovery, excision, or measurement-limited outcomes and exposes the assumptions responsible for each.

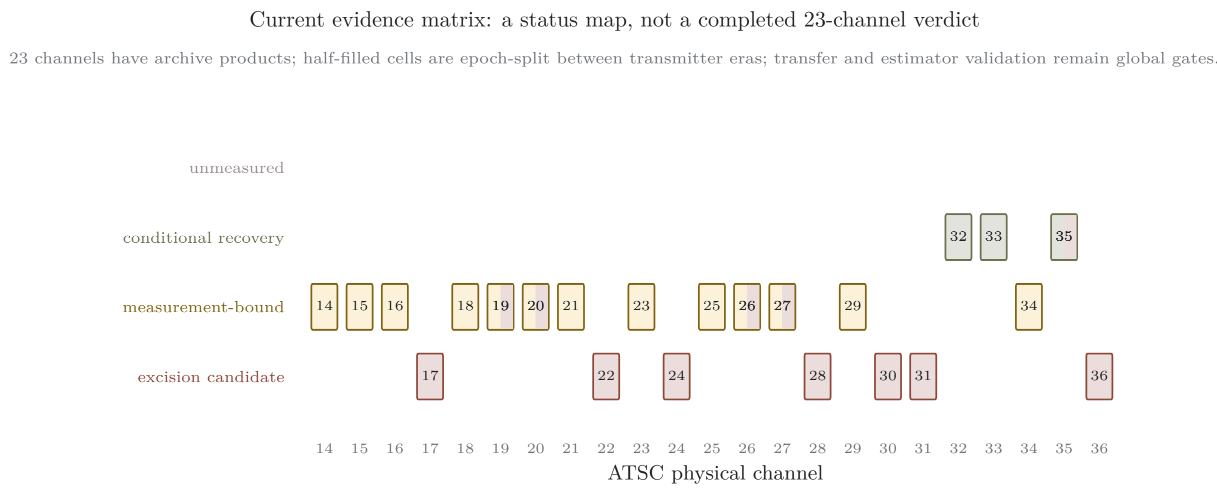

Figure 11.1: Current evidence map for the 23 ATSC allocations in the CHIME band, from the frozen August 2026 export: all 23 allocations carry archive products — the ten contiguous channels 27–36 with the full residual-chain evaluation, and channels 14–26 at the survey, epoch, and residual-chain level (Section 9.12). Categories are conditional on the pilot-to-allocation and visibility-transfer models and the per-bin Fisher estimator. Half-filled cells are epoch-split between transmitter eras: channel 35's early archive is forecast-feasible for the dilation tier while its post-switch-on environment is excision-shaped, and the dated sign-offs on channels 19, 20, 26, and 27 pair a measurement-bound off epoch with an occupancy-pinned on epoch. Channels 29 and 34 are controlled by measurements the snapshot archive cannot provide.

11.2 Current Channel-Level Synthesis

Under the current transmitter proxy, scalar residual chain, noise-shaped residual template, and per-redshift-bin Fisher estimator, the 23 measured channels divide as follows:

The thirteen lower-band allocations — channels 14–26 — were completed by the trawl in August 2026 under the same immutable schema and a shared kernel cohort, each complete against its archive holdings; the band-edge pair 14–15, last in the scan order, finished on 2026-08-19 and closed the survey. The caution against extrapolating from the ten-channel sample was borne out in the data: the new block contains two transmitter sign-offs resolved to the month (channels 19 and 26), a two-stage step-down (channel 20), an epoch-split persistent carrier whose anchor moves between its on-epochs (channel 17), a collection-curtailed archive on channel 24, and, on the band-edge pair, episodic faint carriers that pin the survey rule while releasing three-quarters of current-epoch frames at the working threshold (Section 9.12). Their survey- and epoch-level rows are reported there, and their residual-chain rows now stand beside the first block's (Table 9.8), computed by a released generator whose self-test reproduces the first block's published constants from raw products. Every screening status survives the chain unchanged; channel 21 emerges as the survey's best-floored coherence hostage (14,647 null frames, the archive's largest population) beside channel 29; channel 14 sets the survey's extremes of intra-day share (52.6%) and ground-filter weakness (2.7 dB); and no new-block channel reaches conditional recovery at the deployed basis under the coherence cap.

The growth-rate result is more conditional than the dilation result. In the present model, booking a BAO-preserving delay cut on both sides of the calculation can move the threshold-feasible channels inside the $f\sigma_8$ tolerance. That claim must be repeated with the collaboration's actual combined-bin estimator and with residual templates that reflect measured visibility structure. The appropriate

Table 11.1: Current conditional synthesis of channels 14–36. These are analysis statuses, not final operations commands. “Forecast-feasible” means that the recorded policy enters the present dilation-tolerance region; it does not waive the transfer and estimator validations listed in the final column.

Channel	Current status	Binding evidence or closing condition
14	measurement-limited; kept-and-masked	Episodic faint carrier; refused τ_c; measured −48.8 dB floor (1,151 nulls); inclusive-for-now keep (Sec. 9.12)
15	measurement-limited; kept-and-masked	Episodic faint carrier; refused τ_c; zero null frames, floor unmeasured; inclusive-for-now keep
16	measurement-limited	Trace with warm-season swell; refused τ_c and unmeasured floor
17	excision candidate; era-split	Persistent carrier, two on-epochs; occupancy-pinned; twenty null frames too few to stand on
18	measurement-limited	Trace, rising since 2024; refused τ_c; measured −41.3 dB floor
19	era-split	Sign-off Dec 2024; off-epoch floor −33.5 dB; off-epoch coherence refused, so the row is a bound
20	era-split	Step-down Sep 2022; off-epoch τ_c measured (167 min); conservative floor basis is the cheapest refinement
21	measurement-limited	Trace; the archive’s largest null population (14,647; floor −43.2 dB); cadence co-priority with 29
22	excision candidate	Persistent, steady; occupancy-pinned; no null population
23	measurement-limited	Trace; $\tau_c \leq 5$ min bounded; floor unmeasured — fine-stage null re-decision target
24	excision candidate	Persistent; collection-curtailed archive; τ_c measured (62 min); occupancy/cost wall
25	measurement-limited	Trace; refused τ_c and unmeasured floor
26	era-split	Sign-off Apr 2023; off-epoch floor −32.3 dB; off-epoch coherence refused, so the row is a bound
27	excision candidate (archive average); era-split	Transmitter off since the 2021–22 collection gap; current epoch is trace-level with a measured off-epoch floor (Sec. 9.12)
28	excision candidate	Near-continuous occupancy and inadequate verified null population
29	measurement-limited	Correlation-time refusal; contiguous cadence can change the verdict
30	excision candidate	Occupancy/cost wall; collection curtailed after Sep 2023, so the current state is unmonitored
31	excision candidate	Near-continuous occupancy despite measured correlation scale
32	forecast-feasible, current epoch	Post-2021 epoch; validate allocation and visibility transfer
33	forecast-feasible, conditional	Multi-structure environment; validate template and transfer
34	measurement-limited	Correlation-time and contiguous-scan requirement
35	epoch-split	Early epoch forecast-feasible; post-2021 switch-on is excision-shaped
36	excision candidate	Effectively always on with no defensible off-state floor

conclusion is not that the growth rate has already been recovered, but that a self-consistent filter configuration exists in the screening model and merits the closing analysis.

11.3 What Remains to Close the Verdict

The near program is fixed by the evidence matrix in Appendix A:

1. attach the immutable release ledger behind the now-complete 23-channel inventory (the trawl itself finished in August 2026; the exact discovered/eligible/processed/quarantined denominators of checklist item A remain to be exported);
2. run the specified cadence campaigns to measure or bound τ_c (checklist item C: half-day sessions

at $\sim$10-s cadence, channels 29 and 21 first — the survey's two best-floored coherence hostages — then 34); if they are not executed, the sidereal-day caps stand as permanent conservative bounds and certification shifts to the blocked holdout and per-channel jackknife of the final analysis;

3. measure simultaneous power across each 6-MHz allocation to infer the distribution of received pilot-to-shelf transfer by channel, epoch, and propagation state;

4. measure contaminated complex visibilities before and after the per-sidereal-day operation, preserving baseline and phase information, and separate coherent systematic residual from stochastic added variance;

5. repeat the tolerance calculation with the published combined multi-bin Fisher estimator and with localized, low-$k_{\|}$, wedge-like, sidereal-coherent, and empirical residual templates;

6. complete blocked, per-epoch holdout calibration and attach exact event, frame, storage, throughput, and reproduction denominators;

7. regenerate the standards-chain paired injection study through the current full-2048-input, two-stage pipeline, using a stationary-span waveform audit and reporting ideal-to-float, float-to-integer, and end-to-end crossing shifts with paired uncertainty; and

8. integrate the stage into the actual `kotekan` buffer path and pass alignment, failure-injection, monitoring, and end-to-end latency gates; and

9. schedule periodic spot-collection on channels excluded from baseband collection (channel 30 since September 2023): five transmitter sign-offs or station departures (channels 19, 20, 26, 27, 32) were recorded on collected channels during the survey span, and a channel the instrument does not sample can never report its own sign-off, leaving recoverable spectrum written off indefinitely.

Several forward engineering extensions remain valuable after those gates. A third, censored reference could protect the denominator from a discrete line; alternative reference weights could trade curvature rejection against variance; and the same exact-arithmetic and calibration architecture could target the bootstrap or pilot structures of OFDM broadcast standards. These are extensions of the detector family, not prerequisites for making the present evidence honest.

11.4 Closing Perspective

The strongest result of this work is not a promise that every television channel can be saved. It is a procedure that makes the answer auditable. The procedure starts from a feature the transmitter is required to emit, follows it through a bit-reproducible digital instrument, records enough information to change the policy without changing the past, and prices the remaining contamination in the same units as the cosmological measurement. Where a digital equality can be proved, the dissertation requires equality. Where the telescope can supply a measurement, it reports the measurement and its population. Where a transfer or cadence is missing, it carries a bound or a refusal rather than converting an assumption into a fact. That separation is what allows positive and negative channel verdicts to be scientifically useful rather than merely operational.

Appendix A

Evidence, Assumptions, and Closing Tests

The dissertation crosses several disciplinary boundaries, and the word "verified" does not mean the same thing in each of them. Table A.1 centralizes the claim boundary used throughout the revised text. It is not a substitute for the derivations in the body; it is a map from each major claim to the kind of evidence that supports it and to the experiment that would change its status. The distinction is intentional: bit equality can close an implementation question, whereas no amount of bit equality can validate an astrophysical transfer model.

Table A.1: Evidence matrix for the principal claims. “Conditional” means that the calculation is reproducible under the stated model but that a load-bearing transfer remains to be measured.

Claim	Status	Present evidence	Closing test or release record
Pilot frequency and nominal pilot-to-data power relation	specified	ATSC A/53 gives the symbol rate, pilot insertion, exact pilot-frequency relation, nominal 309-kHz placement, 1.25 bias, and approximately 11.3-dB pilot-to-data power ratio	Retain the cited standard edition and record the weight-generator convention in the release manifest.
Matched-filter projection for the declared known-tone model	analytic	Likelihood-ratio/matched-filter result under circular Gaussian noise and a known signal family	Report departures from the model separately; do not promote this result to global optimality of the complete calibrated detector.
Exact integer detector output	verified	Frozen weight bank and transform, golden vectors, independent reference implementations, concurrency emulation, and device equality tests	Release source commit, compiler/toolkit versions, weight/twiddle hashes, binary hash, exact test count, and edge-vector results.
Legacy paired injection loss	measured on superseded geometry	Standards-chain 8-VSB plus AWGN, 45,000 paired full-precision/packed trials, bootstrap crossing intervals, and a 200-trial CPU/GPU equality spot check	Regenerate with the present full-2048-input two-stage statistic, stationary-span waveform audit, empirical null, and forecast-selected rational threshold before quoting a current sensitivity or implementation-loss result.
Archive-scale reproduction	measured, incompletely reported	The current draft reports zero mismatches between independently built processing cohorts	Publish joined event and frame counts, join completeness, invalid/missing record counts, fields compared, and the immutable comparison manifest.
Triggered archive as a representation of transmitter occupancy	conditional	Eight calendar years of captures with per-event provenance, but a trigger-based rather than uniform sampling process	Quantify coverage by time of day, season, sidereal time, trigger class, and operations state; evaluate policies on blocked holdout intervals.
Pilot-to-shelf transfer across a 6-MHz allocation	conditional	Standard transmitter power relation and a receiver model; existing products observe the pilot-containing coarse channel only	Record simultaneous spectra across the allocation and infer the transfer distribution by channel, epoch, and propagation state.

Claim	Status	Present evidence	Closing test or release record
Scalar pilot-power proxy as visibility-domain residual after sidereal filtering	conditional	A nested temporal decomposition of the scalar proxy and an explicit coherence model	Measure contaminated complex visibilities before and after the per-sidereal-day operation, including baseline dependence and phase; fit separate coherent and stochastic residual terms.
Per-redshift-bin Fisher tolerance	modelled	Reproducible perturbation of the same instrument specification and clean code path used by the forecast branch	Repeat with the published combined multi-bin estimator and report changes in absolute errors, tolerances, policy ranking, and selected operating points.
Noise-shaped residual template	conditional	A transparent screening model $C_{\rm res} \propto C_N$	Repeat with frequency-localized, low-$k_\parallel$, wedge-like, sidereal-coherent, and empirically measured templates.
Channel status map (23 of 23 measured)	conditional	Archive products for all 23 channels (trawl completed August 2026); the full proxy and forecast chain for channels 27–36, and survey-, epoch-, and residual-chain results for 14–26 including dated sign-offs and transmitter-off floors	Complete the transfer validations, per-epoch calibration, contiguous scans, and the blocked holdout before issuing a final 23-channel operations matrix.

A compact visual version of the same boundary appears in Figure 1.1. The operational rule is simple: a claim can move from CONDITIONAL to MEASURED only when the named closing product is attached to the same immutable analysis bundle as the result it supports.

Appendix B

Legacy Validation Evidence and Reuse Boundaries

This appendix makes the useful legacy work part of the dissertation's canonical record without silently promoting it to the current detector geometry. The retained projects served three purposes: they exposed failure modes that shaped the present design, supplied measured implementation and survey diagnostics, and recorded enough source and compact data to audit how the claims were produced. Their numerical results are classified here as *measured on the recorded legacy geometry*, *analytically scaled*, or *historical prototype*. Only the first label is a measurement, and none of the three labels implies that the current two-stage operating point has already been validated.

B.1 Controlled-Injection Evidence

Section 6.7 integrates the principal result: the same standards-chain waveform and noise realizations were evaluated by matched full-precision and packed-`int4` paths. The source project preserves the 45-point sweep summary (15 SNR levels by three pilot offsets), waveform-audit JSON, weight-bank identity, CPU/GPU parity memo, decision records, producing commit, and the manuscript source. The large per-trial bundle was intentionally held outside the restored project, so the compact products can audit the reported crossings but cannot regenerate every trial from this directory alone.

Four distinctions from that experiment are retained as design requirements:

1. A paired floating-point/integer comparison isolates representation loss; comparing two independently generated sweeps does not.

2. An ideal-to-measured crossing difference includes waveform and convention error as well as implementation error and must not be labelled "fixed-point loss" without the paired control.

3. A nominal test waveform is not ground truth until its exact evaluated span has been checked independently. The startup transient found here changed the apparent pilot ratio by several tenths of a decibel.

4. Dimensionality must be stated with every sensitivity number. The measured curves used $R = 512$ rows per trial; the 13.6-dB improvement quoted for a full 2048-input frame was an analytic $1/\sqrt{R}$ scaling, not a regenerated deployment curve.

The earlier GPU-prototype manuscript is retained for a different reason. Its −0.309-dB mean and 0.59-dB maximum absolute shelf-equivalent bias summarize a vertical statistic residual over its tested range, whereas Table 6.1 reports horizontal detection-curve shifts at fixed operating probabilities. Its 14× real-time shortfall is likewise a result for its block definition, not for the fused kernel of Chapter 6. Keeping both records prevents a later reader from treating unlike metrics as a disagreement or a universal correction.

B.2 Sensitivity Geometry and Policy Lessons

The standalone sensitivity project developed a reusable artifact grammar: detector configuration, injection grid, trial arrays, detection-probability surface, required-SNR table, finite-trial support, figures, and a provenance summary. Its representative −21.699 dB/bin operating point belongs to the older $K = 128$, long-accumulation geometry and is not imported into the present result. What survives is the workflow: report the SNR convention, invert $P_\mathrm{d}(\mathrm{SNR}, P_\mathrm{FA})$ only where the trial count supports it, separate interpolation from measured grid points, and archive arrays rather than only figures.

Section 8.5 preserves the width-crossing coordinate and the exposure cost of tail cutting. These explain why a parameter-free mean crossing is a useful calibration health check and why a deeper left-tail cut cannot rescue signals far below one null width. They do not replace the present fine-stage rank estimator or the science-derived threshold. In particular, the legacy positive-excess rule accepts a null mask fraction near one half by construction; that quantity is a drift sentinel and retention coordinate, not a small false-alarm probability.

B.3 Blind Spots and Spectral Forensics

The earlier 23-channel survey produced several negative results worth retaining because they are easy to erase when only successful detections survive. Table B.1 records the most operationally important ones.

Two taxonomy distinctions also survive. A channel can be *separable but null-unusable*: an off-air minority proves that signal and null populations are distinct, yet the measured null is still too broad or contaminated to set an operating point. Or it can be *unseparable* inside the chosen statistic, as in the guard-region blind spot. The first state calls for a better null campaign; the second calls for a different observable. Neither is accurately described by a single "clean/dirty" label.

The old redshift-tiering note is retained only as geometry: refusing an allocation removes a narrow redshift slice, so conservative tiering costs depth at multiple points across the interval rather than moving one terminal redshift. Chapter 9 supersedes the old numerical forecast with the current Fisher and residual-chain conventions.

Table B.1: Legacy survey findings retained as design evidence. They describe the earlier 23-channel products and do not enlarge the present dissertation's ten-channel calibrated cohort.

Finding	Recorded evidence	Consequence retained here
Off-nominal blind spot	Channel 33 carried a guard-region line near -3.6 kHz at about 21 dB integrated SNR, where detector response was approximately -17 dB; eleven nearby translator/LPTV candidates lacked a tight source tolerance.	Pair the target statistic with a broad spectral sentinel, guard sweep, and epoch-specific anchor search; do not equate a clean target cell with a clean allocation.
Instrument tones	Five of six extra-cell lines lay within 9.5 Hz of rational receiver-clock fractions, were localized to one coarse channel, and changed by no more than 0.3 dB under the analytic mask. The unidentified sixth line was retained.	Require multiple independent attribution tests; unidentified means unexcised. Fixed-width notches must be checked for residual skirts.
Physical-invariant archive join	A join using day ± 1 plus integer frame count was 97.4% consistent on singleton days and classified 79.0% of 77,423 channel-event records; a day-only join mixed trigger classes.	When datasets share no identifier, validate a physical invariant before using the join and retain the unmatched population.
Recorded versus optimal subset	The recorded 16-channel subset contained 1548 common events; an unrestricted search later found 1829. The stacked headline changed from 3.74 to 3.53 MHz while per-channel and full-depth results were stable.	Report the recorded procedure and the post hoc optimum separately; robustness does not make the original selection preregistered.
Exact rational thresholds	Per-channel rational approximations used $Q \leq 2^{16}$, error at most 10^{-9}, zero decision disagreements over 339,196 frames, and a worst recorded cross-product width of 55 bits.	Calibration constants can remain data while the runtime decision stays exact; repeat the denominator and width audit for the current schema.

B.4 Canonical Evidence Directory

The dissertation source now contains one curated evidence directory rather than a sidecar notes archive. Under `evidence/legacy_projects/`:

- **Pilot-informed detector article** contains the principal manuscript, figures, compact survey/sweep products, and provenance memos.
- **Calibrated detector and GPU prototype** contains the earlier prototype manuscript, preview PDF, bibliography, and figure set.
- **Chapter 4 sensitivity project** contains the standalone chapter, compact NPZ/CSV/JSON products, generator, tables, and vector/raster previews.
- **Pilot-proxy findings ledger** contains the findings-ledger source, figure generators, tone table, and supporting figures.
- `evidence/reference_pdfs/` preserves the four reviewed PDFs as fixed reading copies, and `evidence/legacy_review.md` records the original keep/discard assessment.

The pilot-informed project's `provenance/BUNDLES_HELDOUT.md` names the remaining external artifacts: two large CANFAR results bundles, the archived 45,000-trial CPU records, and `all_spectra.npz`. Its hash file also records the generated ATSC waveform and weight-bank binaries, which are not present here. Their absence does not erase the compact evidence, but it prevents this dissertation package from claiming full raw-trial regeneration. The release manifest therefore distinguishes *retained source and compact products* from *externally held raw products*; closing the latter gap is a release task, not a prose edit.

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